



- Python C++

## 1.2

pybind11

- Python 2.7, 3.5+, PyPy/PyPy3 7.3
- lambda
- pybind11 C++11 (pybind11 uses C++11 move constructors and move assignment operators whenever possible to efficiently transfer custom data types.)
- Python buffer C++ Eigen NumPy
- pybind11 NumPy
- Python
- Boost.Python
- `constexpr`
- C++ Python pickle/unpickle

## 1.3

1. Clang/LLVM 3.3 (Apple Xcode's clang 5.0.0)
2. GCC 4.8
3. Microsoft Visual Studio 2015 Update 3
4. Intel classic C++ compiler 18 or newer (ICC 20.2 tested in CI)
5. Cygwin/GCC (previously tested on 2.5.1)
6. NVCC (CUDA 11.0 tested in CI)
7. NVIDIA PGI (20.9 tested in CI)

## 1.4 ☐☐

This project was created by Wenzel Jakob. Significant features and/or improvements to the code were contributed by Jonas Adler, Lori A. Burns, Sylvain Corlay, Eric Cousineau, Aaron Gokaslan, Ralf Grosse-Kunstleve, Trent Houlston, Axel Huebl, @hulucc, Yannick Jadoul, Sergey Lyskov Johan Mabilie, Tomasz Miąsko, Dean Moldovan, Ben Pritchard, Jason Rhineland, Boris Schäling, Pim Schellart, Henry Schreiner, Ivan Smirnov, Boris Staletic, and Patrick Stewart.

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## 1.5 ☐☐

See the [contributing guide](#) for information on building and contributing to pybind11.

## 1.6 License

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### 3. 安装

在 [pybind/pybind11 on GitHub](#) 上找到 pybind11 的源代码并克隆到本地。

#### 3.1 使用 Git 安装

在本地仓库中，使用 Git 安装 pybind11 子模块。



```
git submodule add -b stable ../../pybind/pybind11 extern/pybind11
git submodule update --init
```

在 `extern` 目录中，使用 Git 安装 pybind11 子模块。使用 `https` 或 `ssh` URL 克隆到本地。

在 `extern/pybind11/include` 目录中，使用 `Build System` 安装 pybind11。

#### 3.2 使用 PyPI 安装

使用 `pip` 安装 PyPI 上的 Pybind11。



```
pip install pybind11
```

在 `pybind11` 目录中，使用 `Python` 安装 pybind11。



```
pip install "pybind11[global]"
```

```
Python root /usr/local/include/pybind11  
/usr/local/share/cmake/pybind11  
pyproject.toml
```

### 3.3 conda-forge

You can use pybind11 with conda packaging via [conda-forge](#):



```
conda install -c conda-forge pybind11
```

### 3.4 vcpkg

Microsoft [vcpkg](#) pybind11

```
git clone https://github.com/Microsoft/vcpkg.git  
cd vcpkg  
./bootstrap-vcpkg.sh  
./vcpkg integrate install  
vcpkg install pybind11
```

### 3.5 brew

brew Homebrew on macOS, or Linuxbrew on Linux pybind11



```
brew install pybind11
```

## 3.6 其他位置

Other locations you can find pybind11 are [listed here](#); these are maintained by various packagers and the community.

# pybind11

## 1. 安装

### 1.1 安装

在python3-dev安装pybind11头文件include pybind11python3头文件  
cmake



```
set(PYTHON_TARGET_VER 3.6)
find_package(PythonInterp ${PYTHON_TARGET_VER} EXACT)
find_package(PythonLibs ${PYTHON_TARGET_VER} EXACT REQUIRED)

include_directories(pybind11_include_path)
include_directories(${PYTHON_INCLUDE_DIRS})
```

### 1.2 使用

```
#include <pybind11/pybind11.h>

int add(int i, int j) {
    return i + j;
}

PYBIND11_MODULE(example, m) {
    m.doc() = "pybind11 example plugin"; // optional module docstring
    m.def("add", &add, "A function which adds two numbers");
}
```

```

PYBIND11_MODULE example Python import
py::module_ m(pybind11::module_::def())

```

## 1.2.1

py::arg Python

```

m.def("add", &add, "A function which adds two numbers",
      py::arg("i"), py::arg("j"));

```

```

// regular notation
m.def("add1", &add, py::arg("i"), py::arg("j"));
// shorthand
using namespace pybind11::literals;
m.def("add2", &add, "i"_a, "j"_a);

```

Python



```

import example
example.add(i=1, j=2) #3L

```

## 1.2.2

pybind11 arg

```

m.def("add", &add, "A function which adds two numbers",
      py::arg("i") = 1, py::arg("j") = 2);

```

```
// regular notation
m.def("add1", &add, py::arg("i") = 1, py::arg("j") = 2);
// shorthand
m.def("add2", &add, "i"_a=1, "j"_a=2);
```

## 1.2.3 静态绑定

静态绑定（Static Binding）是指编译时就能确定调用的函数。

在 C++ 中，静态绑定通常通过函数重载（Function Overloading）实现。Python 解释器通过 C++ 的静态绑定机制来调用 Python 函数。

```
m.def("add", static_cast<int(*)>(int, int>(&add), "A function which adds
two int numbers");
m.def("add", static_cast<double(*)>(double, double>(&add), "A function
which adds two double numbers");
```

在 C++14 之前，静态绑定只能用于非静态成员函数。

```
m.def("add", py::overload_cast<int, int>(&add), "A function which adds two
int numbers");
m.def("add", py::overload_cast<double, double>(&add), "A function which
adds two double numbers");
```

在 C++14 中，静态绑定可以用于静态成员函数。以下代码展示了如何使用 `py::overload_cast` 来绑定静态成员函数：

## 1.3 属性

在 Python 中，属性（Attribute）是指对象的成员变量或方法。在 C++ 中，属性通常通过 `py::attr` 来定义。以下代码展示了如何使用 `py::attr` 来定义属性：

```
PYBIND11_MODULE(example, m) {
    m.attr("the_answer") = 42;
    py::object world = py::cast("World");
    m.attr("what") = world;
}
```

Python

```
```python
```

```
>>> import example
```

```
>>> example.the_answer
```

```
42
```

```
>>> example.what
```

```
'World'
```

## 1.4

C++ Pet

```
struct Pet {
    Pet(const std::string &name) : name(name) { }
    void setName(const std::string &name_) { name = name_; }
    const std::string &getName() const { return name; }

    std::string name;
};
```

```
#include <pybind11/pybind11.h>
namespace py = pybind11;

PYBIND11_MODULE(example, m) {
    py::class_<Pet>(m, "Pet")
        .def(py::init<const std::string &>())
        .def("setName", &Pet::setName)
        .def("getName", &Pet::getName)
        .def("__repr__",
            [](const Pet &a) {
                return "<example.Pet named '" + a.name + "'>";
            });
}
```

`class_` 对应 C++ `class` 或 `struct` 的 `init()` 函数，用于初始化类对象。

`print(p)` 输出 `<example.Pet named 'Molly'>`，这是 `__repr__` 方法返回的字符串。

Python 测试：



```
>>> import example
>>> p = example.Pet("Molly")
>>> print(p)
<example.Pet named 'Molly'>
>>> p.getName()
u'Molly'
>>> p.setName("Charly")
>>> p.getName()
u'Charly'
```

还可以使用 `class_::def_static` 来定义静态方法。

### 1.4.1 属性

使用 `class_::def_readwrite` 或 `class_::def_readonly` 来定义属性。

```
py::class_<Pet>(m, "Pet")
    .def(py::init<const std::string &>())
    .def_readwrite("name", &Pet::name)
    // ... remainder ...
```

Python



```
>>> p = example.Pet("Molly")
>>> p.name
u'Molly'
>>> p.name = "Charly"
>>> p.name
u'Charly'
```

`Pet::name` setter/getters

```
class Pet {
public:
    Pet(const std::string &name) : name(name) { }
    void setName(const std::string &name_) { name = name_; }
    const std::string &getName() const { return name; }
private:
    std::string name;
};
```

`class_::def_property()` (`class_::def_property_readonly()`)  
`setter`/`getter`

```
py::class_<Pet>(m, "Pet")
    .def(py::init<const std::string &>())
    .def_property("name", &Pet::getName, &Pet::setName)
    // ... remainder ...
```

`readonly`/`nullptr`

`class_::def_readwrite_static()`, `class_::def_readonly_static()`  
`class_::def_property_static()`, `class_::def_property_readonly_static()`



```
>>> p = example.Pet()
>>> p.name = "Charly" # OK, overwrite value in C++
>>> p.age = 2 # OK, dynamically add a new attribute
>>> p.__dict__ # just like a native Python class
{'age': 2}
```

\_\_\_\_\_ \_\_dict\_\_ \_\_\_\_\_  
 \_\_\_\_\_Python\_\_\_\_\_pybind11\_\_\_\_\_Python\_\_\_\_\_  
 \_\_\_\_\_

### 1.4.3

\_\_\_\_\_

```
struct Pet {
    Pet(const std::string &name, int age) : name(name), age(age) { }

    void set(int age_) { age = age_; }
    void set(const std::string &name_) { name = name_; }

    std::string name;
    int age;
};

// method 1
py::class_<Pet>(m, "Pet")
    .def(py::init<const std::string &, int>())
    .def("set", static_cast<void (Pet::*)(int)>(&Pet::set), "Set the pet's age")
    .def("set", static_cast<void (Pet::*)(const std::string &)>(&Pet::set), "Set the pet's name");

// method 2
py::class_<Pet>(m, "Pet")
    .def("set", py::overload_cast<int>(&Pet::set), "Set the pet's age")
    .def("set", py::overload_cast<const std::string &(&Pet::set), "Set the pet's name");
```

## 1.5 枚举

在C++中枚举类型

```
enum Flags {
    Read = 4,
    Write = 2,
    Execute = 1
};

py::enum_<Flags>(m, "Flags", py::arithmetic())
    .value("Read", Flags::Read)
    .value("Write", Flags::Write)
    .value("Execute", Flags::Execute)
    .export_values();
```

`enum_::export_values()` 在C++11中引入

在Python中，枚举类型 `__members__` 属性返回一个字典，其中键是枚举成员的名称，值是枚举成员的值。使用 `str(enum)` 可以获取枚举类型的字符串表示。

## 1.6 \*args 和 \*\*kwargs

Python中函数参数



```
def generic(*args, **kwargs):
    ... # do something with args and kwargs
```

在Python中，`pybind11` 库支持

```

void generic(py::args args, const py::kwargs& kwargs) {
    /// .. do something with args
    if (kwargs)
        /// .. do something with kwargs
}

/// Binding code
m.def("generic", &generic);

```

py::args 是 py::tuple 和 py::kwargs 的 py::dict

## 2. 非平凡返回

### 2.1 非平凡返回

Python C++ 非平凡返回类型 (no-trivial) 的函数，使用 `pybind11` 的 `model::def()` 或 `class_def()` 时，需要指定 `return_value_policy::automatic`。

非平凡返回类型的函数，在 C++ 中返回一个非平凡类型的对象，在 Python 中返回一个非平凡类型的对象。

```

/* Function declaration */
Data *get_data() { return _data; /* (pointer to a static data structure)
*/ }
...

/* Binding code */
m.def("get_data", &get_data); // <-- KABOOM, will cause crash when called
from Python

```

在 Python 中，`get_data()` 返回一个非平凡类型的对象。在 C++ 中，`get_data()` 返回一个非平凡类型的对象。在 Python 中，`return_value_policy::automatic` 是 `pybind11` 的默认值，返回一个非平凡类型的对象。

```
Python  _data Python pybind11 C++ operator delete()

```

```
return_value_policy::reference
```

```
m.def("get_data", &get_data, py::return_value_policy::reference);
```

```
pybind11
```

[illegible]

| 返回值                                                   | 策略                                                                                                                                                      |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>return_value_policy::automatic_reference</code> | <p>pybind11 默认使用 <code>return_value_policy::automatic_reference</code> 策略，该策略在 C++ 中返回的变量在 Python 中也是以引用方式返回的，即返回的变量在 C++ 中生命周期结束时，Python 中的变量也会失效。</p> |

pybind11 默认使用 `return_value_policy::automatic_reference` 策略，该策略在 C++ 中返回的变量在 Python 中也是以引用方式返回的，即返回的变量在 C++ 中生命周期结束时，Python 中的变量也会失效。

```
class_<MyClass>(m, "MyClass")
    .def_property("data", &MyClass::getData, &MyClass::setData,
        py::return_value_policy::copy);
```

pybind11 默认使用 `return_value_policy::automatic_reference` 策略，该策略在 C++ 中返回的变量在 Python 中也是以引用方式返回的，即返回的变量在 C++ 中生命周期结束时，Python 中的变量也会失效。

```
class_<MyClass>(m, "MyClass")
    .def_property("data"
        py::cpp_function(&MyClass::getData,
        py::return_value_policy::copy),
        py::cpp_function(&MyClass::setData)
    );
```

pybind11 默认使用 `return_value_policy::automatic_reference` 策略，该策略在 C++ 中返回的变量在 Python 中也是以引用方式返回的，即返回的变量在 C++ 中生命周期结束时，Python 中的变量也会失效。

pybind11

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## 2.2 保持存活

保持存活是指一个对象在垃圾回收之前，先被通知，以便它有机会清理资源。

### keep alive

C++ 保持存活 C++ 保持存活 `keep_alive<Nurse, Patient>` 保持存活  
Nurse 保持存活 Patient 保持存活 0 保持存活 1 保持存活 1 保持存活 this 保持存活 2 保持  
Nurse 保持存活 None 保持存活

nurse 保持存活 pybind11 保持存活 nurse 保持存活 nurse 保持存活 pybind11 保持  
保持存活

保持存活 "Could not cativate keep\_alive!" 保持存活 review  
保持存活

保持存活 list append 保持存活

```
py::class_<List>(m, "List").def("append", &List::append, py::keep_alive<1, 2>());
```

保持存活 1 this 保持存活 0 保持存活 void 保持存活  
保持存活

```
py::class_<Nurse>(m, "Nurse").def(py::init<Patient &>(), py::keep_alive<1, 2>());
```

Note: keep\_alive 在 Boost.Python 中 with\_custodian\_and\_ward 和  
with\_custodian\_and\_ward\_postcall 保持存活

### Call guard

call\_guard<T> 保持存活 T 保持存活 scope guard 保持存活

```
m.def("foo", foo, py::call_guard<T>());
```

□□□□□□□□

```
m.def("foo", [](args...) {
    T scope_guard;
    return foo(args...); // forwarded arguments
});
```

□□□□□□□□□□ `gil_scoped_release` □□□□□□□□□□□□

`call_guard` □□□□□□□□□□□□ `call_guard<T1, T2, T3 ...>` □□□□□□□□□□□□□□□□□□□□

See also: `test/test_call_policies.cpp` □□□□□□□□□□□□ `keep_alive` □  
`call_guard` □□□□

## 2.3 Keyword-only□□

Python3□□□keyword-only□□□□□□□□□□□□ \* □□□□□□□□



```
def f(a, *, b): # a can be positional or via keyword; b must be via
    keyword
    pass

f(a=1, b=2) # good
f(b=2, a=1) # good
f(1, b=2) # good
f(1, 2) # TypeError: f() takes 1 positional argument but 2 were given
```

pybind11□□□ `py::kw_only` □□□□□□□□□□□□

```
m.def("f", [](int a, int b) { /* ... */ },
    py::arg("a"), py::kw_only(), py::arg("b"));
```

py::args

## 2.4 Positional-only

python3.8 Positional-only pybind11 `py::pos_only()`

```
m.def("f", [](int a, int b) { /* ... */ },
      py::arg("a"), py::pos_only(), py::arg("b"));
```

`a` keyword-only

## 2.5 Non-converting

- `py::implicitly_convertible<A,B>()`
- 
- `std::complex<float>`
- Calling a function taking an Eigen matrix reference with a numpy array of the wrong type or of an incompatible data layout.

`py::arg` `.noconvert()`

```
m.def("floats_only", [](double f) { return 0.5 * f; },
      py::arg("f").noconvert());
m.def("floats_preferred", [](double f) { return 0.5 * f; }, py::arg("f"));
```

`TypeError`



```
>>> floats_preferred(4)
2.0
>>> floats_only(4)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: floats_only(): incompatible function arguments. The following
argument types are supported:
    1. (f: float) -> float

Invoked with: 4
```

```
PyObject* _a py::arg().noconvert()
```

### 3. 类型转换

#### 3.1 类型转换

```
PyObject*
```

```
struct Pet {
    Pet(const std::string &name) : name(name) { }
    std::string name;
};

struct Dog : Pet {
    Dog(const std::string &name) : Pet(name) { }
    std::string bark() const { return "woof!"; }
};
```

```
pybind11 1.10.0 C++ 11 类 class_ 2.0.0 类 class_
PyObject*
```

```
py::class_<Pet>(m, "Pet")
    .def(py::init<const std::string &>())
    .def_readwrite("name", &Pet::name);

// Method 1: template parameter:
py::class_<Dog, Pet /* ← specify C++ parent type */>(m, "Dog")
    .def(py::init<const std::string &>())
    .def("bark", &Dog::bark);

// Method 2: pass parent class_ object:
py::class_<Dog>(m, "Dog", pet /* ← specify Python parent type */)
    .def(py::init<const std::string &>())
    .def("bark", &Dog::bark);
```

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □



```
>>> p = example.Dog("Molly")
>>> p.name
u'Molly'
>>> p.bark()
u'woof!'
```

# Python

```
// 编译选项
m.def("pet_store", []() { return std::unique_ptr<Pet>(new Dog("Molly"));
});
```



```
>>> p = example.pet_store()
>>> type(p) # `Dog` instance behind `Pet` pointer
Pet        # no pointer downcasting for regular non-polymorphic types
>>> p.bark()
AttributeError: 'Pet' object has no attribute 'bark'
```

```
pet_store Dog Python Pet C++  
pybind11
```

```

struct PolymorphicPet {
    virtual ~PolymorphicPet() = default;
};

struct PolymorphicDog : PolymorphicPet {
    std::string bark() const { return "woof!"; }
};

// Same binding code
py::class_<PolymorphicPet>(m, "PolymorphicPet");
py::class_<PolymorphicDog, PolymorphicPet>(m, "PolymorphicDog")
    .def(py::init<>())
    .def("bark", &PolymorphicDog::bark);

// Again, return a base pointer to a derived instance
m.def("pet_store2", []() { return std::unique_ptr<PolymorphicPet>(new
PolymorphicDog); });

```



```

>>> p = example.pet_store2()
>>> type(p)
PolymorphicDog # automatically downcast
>>> p.bark()
u'woof!'

```

pybind11 is a C++ library that provides a simple and efficient way to integrate C++ code with Python. It allows you to write C++ code that can be called from Python, and vice versa, without the need for a complex build system or a deep understanding of C++ internals.

## 3.2 Python and C++

Python is a high-level, interpreted programming language that is easy to learn and use. It is widely used in a variety of applications, from web development to data science. C++ is a low-level, compiled programming language that is more powerful and flexible than Python, but it is also more difficult to learn and use.

```

class Animal {
public:
    virtual ~Animal() { }
    virtual std::string go(int n_times) = 0;
};

class Dog : public Animal {
public:
    std::string go(int n_times) override {
        std::string result;
        for (int i=0; i<n_times; ++i)
            result += "woof! ";
        return result;
    }
};

std::string call_go(Animal *animal) {
    return animal->go(3);
}

PYBIND11_MODULE(example, m) {
    py::class_<Animal>(m, "Animal")
        .def("go", &Animal::go);

    py::class_<Dog, Animal>(m, "Dog")
        .def(py::init<>());

    m.def("call_go", &call_go);
}

```

PythonAnimal“No constructor defined!”

Animal

```

class PyAnimal : public Animal {
public:
    /* Inherit the constructors */
    using Animal::Animal;

    /* Trampoline (need one for each virtual function) */
    std::string go(int n_times) override {
        PYBIND11_OVERRIDE_PURE(
            std::string, /* Return type */
            Animal,      /* Parent class */
            go,           /* Name of function in C++ (must match Python
name) */
            n_times      /* Argument(s) */
        );
    }
};

```

PYBIND11\_OVERRIDE\_PURE PYBIND11\_OVERRIDE  
 PYBIND11\_OVERRIDE\_PURE\_NAME PYBIND11\_OVERRIDE\_NAME  
 Python \_\_str\_\_

```

std::string toString() override {
    PYBIND11_OVERRIDE_NAME(
        std::string, // Return type (ret_type)
        Animal,      // Parent class (cname)
        "__str__",   // Name of method in Python (name)
        toString,    // Name of function in C++ (fn)
    );
}

```

Animal

```
PYBIND11_MODULE(example, m) {
    py::class_<Animal, PyAnimal /* <--- trampoline*/>(m, "Animal")
        .def(py::init<>())
        .def("go", &Animal::go);

    py::class_<Dog, Animal>(m, "Dog")
        .def(py::init<>());

    m.def("call_go", &call_go);
}
```

pybind11 的 `class_` 函数需要两个模板参数 `PyAnimal` 和 `PythonAnimal`

它们分别代表 C++ 的 `Animal` 类和 Python 的 `Animal` 类

```
py::class_<Animal, PyAnimal /* <--- trampoline*/>(m, "Animal");
    .def(py::init<>())
    .def("go", &PyAnimal::go); /* <--- THIS IS WRONG, use &Animal::go */
```

Python 的 `Animal::go` 函数应该定义为



```
from example import *
d = Dog()
call_go(d)      # u'woof! woof! woof! '
class Cat(Animal):
    def go(self, n_times):
        return "meow! " * n_times

c = Cat()
call_go(c)      # u'meow! meow! meow! '
```

Python 的 `__init__` 函数在 C++ 中应该定义为 `__init__`，而不是 `__init__`。在 pybind11 2.6 版本中，`TypeError` 错误



```
class Dachshund(Dog):
    def __init__(self, name):
        Dog.__init__(self)  # Without this, a TypeError is raised.
        self.name = name

    def bark(self):
        return "yap!"
```

```

class PyAnimal : public Animal {
public:
    using Animal::Animal; // Inherit constructors
    std::string go(int n_times) override {
PYBIND11_OVERRIDE_PURE(std::string, Animal, go, n_times); }
    std::string name() override { PYBIND11_OVERRIDE(std::string, Animal,
name, ); }
};
class PyDog : public Dog {
public:
    using Dog::Dog; // Inherit constructors
    std::string go(int n_times) override { PYBIND11_OVERRIDE(std::string,
Dog, go, n_times); }
    std::string name() override { PYBIND11_OVERRIDE(std::string, Dog,
name, ); }
    std::string bark() override { PYBIND11_OVERRIDE(std::string, Dog,
bark, ); }
};

```

name() bark()

pybind11

```

class Husky : public Dog {};
class PyHusky : public Husky {
public:
    using Husky::Husky; // Inherit constructors
    std::string go(int n_times) override {
PYBIND11_OVERRIDE_PURE(std::string, Husky, go, n_times); }
    std::string name() override { PYBIND11_OVERRIDE(std::string, Husky,
name, ); }
    std::string bark() override { PYBIND11_OVERRIDE(std::string, Husky,
bark, ); }
};

```



### 3.4

Pybind11 `std::unique_ptr`  
 Pybind11 `py::nodelete`  
 C++

```
/* ... definition ... */

class MyClass {
private:
    ~MyClass() { }
};

/* ... binding code ... */

py::class_<MyClass, std::unique_ptr<MyClass, py::nodelete>>(m, "MyClass")
    .def(py::init<>())
```

### 3.5

A B A B

```
py::class_<A>(m, "A")
    /// ... members ...

py::class_<B>(m, "B")
    .def(py::init<A>())
    /// ... members ...

m.def("func",
    [](const B &) { /* .... */ }
);
```

`func` A Python `func(B(a))` C++ `func(a)` A B

B A Python

```
py::implicitly_convertible<A, B>();
```

## 3.6 自定义

自定义 `Vector2` 数据类型

```
class Vector2 {  
public:  
    Vector2(float x, float y) : x(x), y(y) { }  
  
    Vector2 operator+(const Vector2 &v) const { return Vector2(x + v.x, y  
+ v.y); }  
    Vector2 operator*(float value) const { return Vector2(x * value, y *  
value); }  
    Vector2& operator+=(const Vector2 &v) { x += v.x; y += v.y; return  
*this; }  
    Vector2& operator*=(float v) { x *= v; y *= v; return *this; }  
  
    friend Vector2 operator*(float f, const Vector2 &v) {  
        return Vector2(f * v.x, f * v.y);  
    }  
  
    std::string toString() const {  
        return "[" + std::to_string(x) + ", " + std::to_string(y) + "];"  
    }  
private:  
    float x, y;  
};
```

自定义



```
#include <pybind11/operators.h>

PYBIND11_MODULE(example, m) {
    py::class_<Vector2>(m, "Vector2")
        .def(py::init<float, float>())
        .def(py::self + py::self)
        .def(py::self += py::self)
        .def(py::self *= float())
        .def(float() * py::self)
        .def(py::self * float())
        .def(-py::self)
        .def("__repr__", &Vector2::toString);
}
```

```
.def(py::self * float())
```

```
.def("__mul__", [](const Vector2 &a, float b) {
    return a * b;
}, py::is_operator())
```

### 3.7

Python `copy`

Python3 `pickle` `__copy__` `__deepcopy__`  
 Python2.7 `pybind11` `cPickle`

```
py::class_<Copyable>(m, "Copyable")
    .def("__copy__", [](const Copyable &self) {
        return Copyable(self);
    })
    .def("__deepcopy__", [](const Copyable &self, py::dict) {
        return Copyable(self);
    }, "memo"_a);
```

Note:

### 3.8

pybind11 holder pybind11 class\_

```
py::class_<MyType, BaseType1, BaseType2, BaseType3>(m, "MyType")
...
```

holder pybind11

Python C++ Python

pybind11 multiple\_inheritance

```
py::class_<MyType, BaseType2>(m, "MyType", py::multiple_inheritance());
```

### 3.9 protected

Python protected

```
class A {
protected:
    int foo() const { return 42; }
};
```

```
py::class_<A>(m, "A")
    .def("foo", &A::foo); // error: 'foo' is a protected member of 'A'
```

Python protected publicist pattern trampoline

```
class A {
protected:
    int foo() const { return 42; }
};

class Publicist : public A { // helper type for exposing protected
functions
public:
    using A::foo; // inherited with different access modifier
};

py::class_<A>(m, "A") // bind the primary class
    .def("foo", &Publicist::foo); // expose protected methods via the
publicist
```

&Publicist::foo &A::foo Publicist  
public

Python protected publicist pattern trampoline

```

class A {
public:
    virtual ~A() = default;

protected:
    virtual int foo() const { return 42; }
};

class Trampoline : public A {
public:
    int foo() const override { PYBIND11_OVERRIDE(int, A, foo, ); }
};

class Publicist : public A {
public:
    using A::foo;
};

py::class_<A, Trampoline>(m, "A") // <-- `Trampoline` here
    .def("foo", &Publicist::foo); // <-- `Publicist` here, not
    `Trampoline`!

```

### 3.10 如何final

C++11 如何final 如何final Python 如何final C++ 如何final

```

class IsFinal final {};

py::class_<IsFinal>(m, "IsFinal", py::is_final());

```

Python 如何final



```

class PyFinalChild(IsFinal):
    pass

TypeError: type 'IsFinal' is not an acceptable base type

```



| Exception thrown by C++                | Translated to Python exception type |
|----------------------------------------|-------------------------------------|
| <code>pybind11::buffer_error</code>    | <code>BufferError</code>            |
| <code>pybind11::import_error</code>    | <code>ImportError</code>            |
| <code>pybind11::attribute_error</code> | <code>AttributeError</code>         |
| Any other exception                    | <code>RuntimeError</code>           |

```

PythonPython
pybind11::error_already_set

```

```

Python handle::call() cast_error

```

## 4.2

```

pybind11pybind11 class
globalC++what()C++Python

```

```

py::register_exception<CppExp>(module, "PyExp");

```

```

PyExpPythonCppExpPyExp

```

```


```

```

py::register_local_exception<CppExp>(module, "PyExp");

```

```

handle

```

```

py::register_exception<CppExp>(module, "PyExp", PyExc_RuntimeError);
py::register_local_exception<CppExp>(module, "PyExp", PyExc_RuntimeError);

```

```

PyExpPyExpRuntimeError

```

```

PythonPythonStandard ExceptionsPyExc_Exception

```

```
py::register_exception_translator(translator) {}
py::register_local_exception_translator(translator) {}
void(std::exception_ptr) {}
```

### 9.3 C++ Python

C++ Python Python Python pybind11 Python  
 pybind11::error\_already\_set C++ Python  
 error\_already\_set Python Python C++

| Exception raised in Python | Thrown as C++ exception type |
|----------------------------|------------------------------|
| Any Python Exception       | pybind11::error_already_set  |

```
try {
    // open("missing.txt", "r")
    auto file = py::module_::import("io").attr("open")("missing.txt",
    "r");
    auto text = file.attr("read")();
    file.attr("close")();
} catch (py::error_already_set &e) {
    if (e.matches(PyExc_FileNotFoundError)) {
        py::print("missing.txt not found");
    } else if (e.matches(PyExc_PermissionError)) {
        py::print("missing.txt found but not accessible");
    } else {
        throw;
    }
}
```

C++ Python Python error\_already\_set.

```

try {
    py::eval("raise ValueError('The Ring')");
} catch (py::value_error &boromir) {
    // Boromir never gets the ring
    assert(false);
} catch (py::error_already_set &frodo) {
    // Frodo gets the ring
    py::print("I will take the ring");
}

try {
    // py::value_error is a request for pybind11 to raise a Python
    exception
    throw py::value_error("The ball");
} catch (py::error_already_set &cat) {
    // cat won't catch the ball since
    // py::value_error is not a Python exception
    assert(false);
} catch (py::value_error &dog) {
    // dog will catch the ball
    py::print("Run Spot run");
    throw; // Throw it again (pybind11 will raise ValueError)
}

```

## 9.4 Python C API

pybind11 wrappers Python C API Python C API  
 pybind11

Python C API Python `throw py::error_already_set();`  
 pybind11 Python `PyErr_SetString`

```

PyErr_SetString(PyExc_TypeError, "C API type error demo");
throw py::error_already_set();

// But it would be easier to simply...
throw py::type_error("pybind11 wrapper type error");

```

`PyErr_Clear`

PythonPython/pybind11

## 9.5 `unraisable`

PythonC++ `noexcept(true)` `std::terminate()` C++ Python `error_already_set` `error_already_set::discard_as_unraisable()` Python

`__del__` Python Python `unraisable` Python 3.8+ `system hook` `auditing event`

`noexcept` `try-catch` `error_already_set` `pybind11` Python Python `pybind11` C++ `noexcept` Python `discard_as_unraisable`

```
void nonthrowing_func() noexcept(true) {
    try {
        // ...
    } catch (py::error_already_set &eas) {
        // Discard the Python error using Python APIs, using the C++ magic
        // variable __func__. Python already knows the type and value and
of the
        // exception object.
        eas.discard_as_unraisable(__func__);
    } catch (const std::exception &e) {
        // Log and discard C++ exceptions.
        third_party::log(e);
    }
}
```



## 7.2 释放GIL

Python和C++的交互中，GIL的释放和获取是非常重要的。在Python的C++扩展中，我们通常使用`gil_scoped_release`和`gil_scoped_acquire`来管理GIL。在C++的Python扩展中，我们通常使用`PyGILState_STATE`来管理GIL。

```
class PyAnimal : public Animal {
public:
    /* Inherit the constructors */
    using Animal::Animal;

    /* Trampoline (need one for each virtual function) */
    std::string go(int n_times) {
        /* Acquire GIL before calling Python code */
        py::gil_scoped_acquire acquire;

        PYBIND11_OVERRIDE_PURE(
            std::string, /* Return type */
            Animal,      /* Parent class */
            go,           /* Name of function */
            n_times       /* Argument(s) */
        );
    }
};

PYBIND11_MODULE(example, m) {
    py::class_<Animal, PyAnimal> animal(m, "Animal");
    animal
        .def(py::init<>())
        .def("go", &Animal::go);

    py::class_<Dog>(m, "Dog", animal)
        .def(py::init<>());

    m.def("call_go", [](Animal *animal) -> std::string {
        /* Release GIL before calling into (potentially long-running) C++
        code */
        py::gil_scoped_release release;
        return call_go(animal);
    });
}
```

在C++的Python扩展中，我们通常使用`call_guard`和`call_go`来管理GIL。

```
m.def("call_go", &call_go, py::call_guard<py::gil_scoped_release>());
```

# mdbook-mermaid

## 简介Mermaid

Mermaid 是一个用于生成图表的标记语言，它使用简单的文本格式来描述图表结构，并可以生成各种类型的图表，如流程图、序列图、类图等。它支持多种图表类型，并且可以与 Markdown 文档结合使用，方便在文档中嵌入图表。

主要特点：

- 支持多种图表类型，如流程图、序列图、类图等。
- 可以与 Markdown 文档结合使用，方便在文档中嵌入图表。
- 支持 Gitbook 等文档生成工具。

## 安装Mermaid

安装

```
- npm install mermaid
- npm install mermaid
- npm install mermaid
```

安装完成后，可以在 Markdown 文档中使用 Mermaid 语法生成图表。

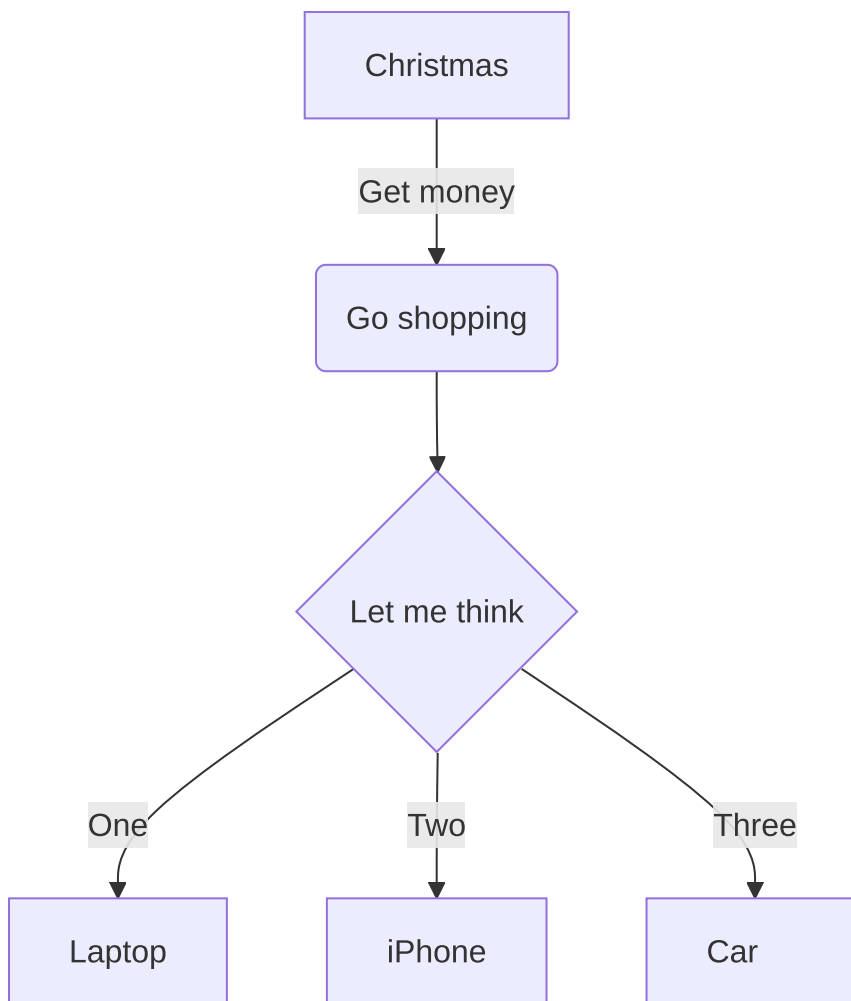
Mermaid 支持多种图表类型，如流程图、序列图、类图等。它使用简单的文本格式来描述图表结构，并可以生成各种类型的图表，如流程图、序列图、类图等。它支持多种图表类型，并且可以与 Markdown 文档结合使用，方便在文档中嵌入图表。

示例

```
graph TD
    A --> B
    B --> C
```

```
graph TD
  A[Christmas] -->|Get money| B[Go shopping]
  B --> C{Let me think}
  C -->|One| D[Laptop]
  C -->|Two| E[iPhone]
  C -->|Three| F[fa:fa-car Car]
```

□□



- □□□□: <https://github.com/mermaid-js/mermaid>
- □□□□: <https://mermaidjs.github.io/mermaid-live-editor/>
- □□□□: <https://mermaid-js.github.io/mermaid/#/flowchart>
- □□□□: <https://mermaid.nodejs.cn/syntax/flowchart.html>

# Mermaid□□□□□□

□□□□

□□□

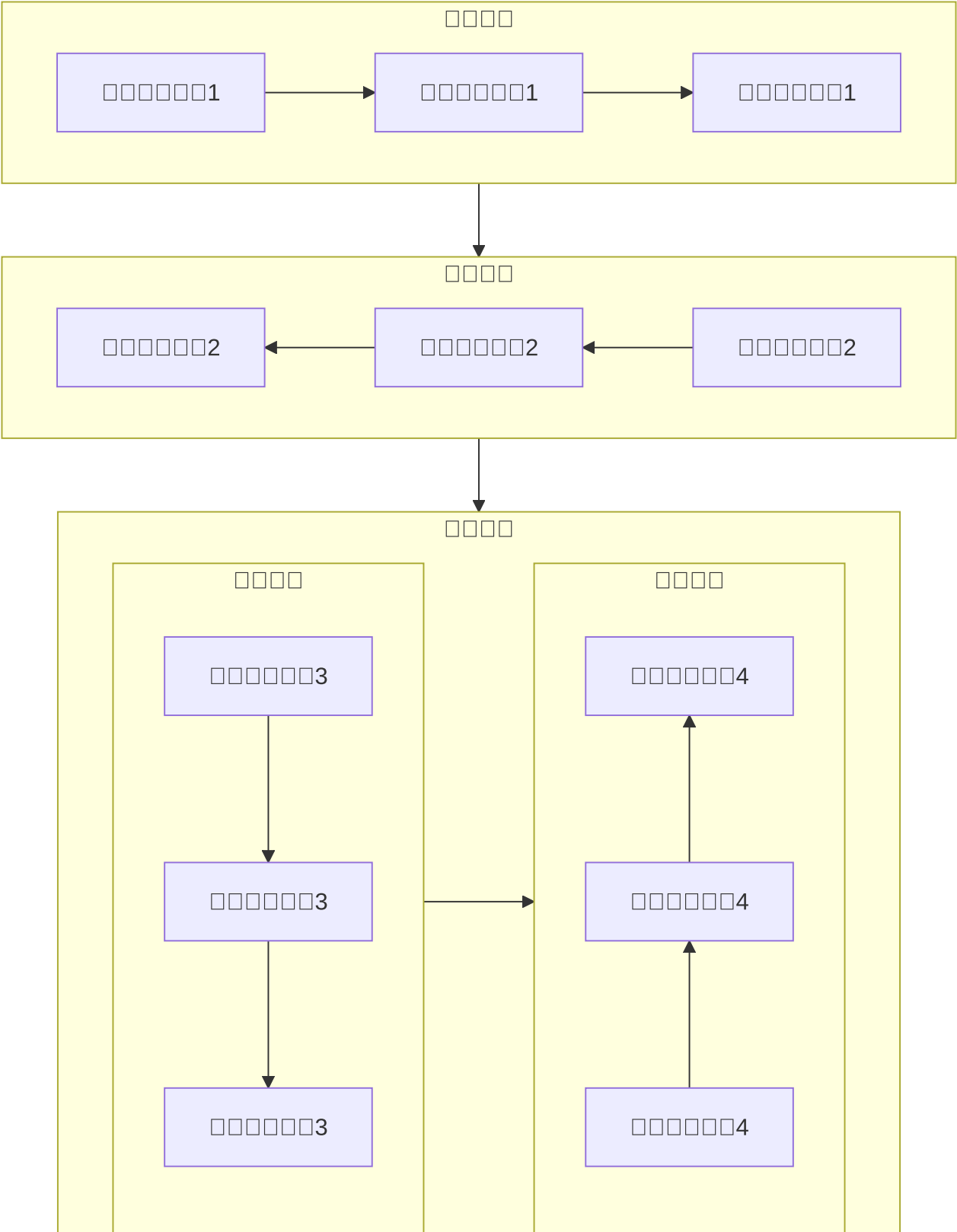
M↓

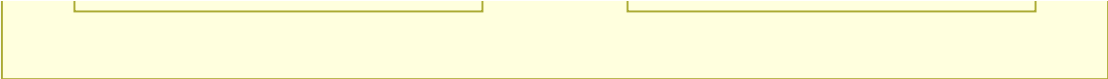
+ TB

+ BT

+ LR

+ RL





graph TD, direction TB; Start --> Stop; subgraph " " ; direction LR; subgraph " " ; direction TB; top[ ] --> bottom[ ]; end; left[ ] --> right[ ]; end; end;

graph TD; Start --> Stop;

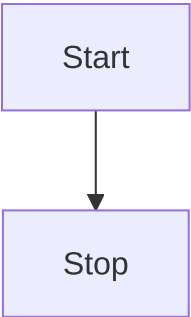
- TB

graph TD; Start --> Stop;

graph TD

```
graph TD; Start --> Stop;
```

graph TD



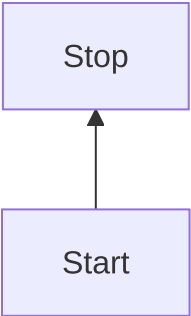
- BT

graph TD; Start --> Stop;

graph TD

```
graph BT
    Start --> Stop
```

□□



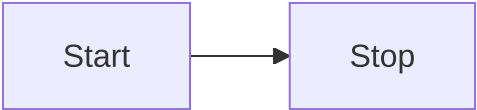
- LR

□□□□: from **L**eft to **R**ight

□□

```
graph LR
    Start --> Stop
```

□□



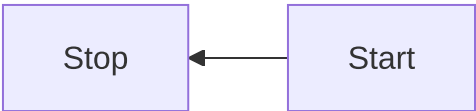
- RL

□□□□: from **R**ight to **L**eft

□□

```
graph RL
    Start --> Stop
```

□□



□□

□□□

M↓

- [ ]

+ [ ]

- [ [ ] ]

- [ ( ) ]

- [ { } ]

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▽File Handling

▵User Input

▭Multiple Documents

▭Process Automation

▭Paper Records

0000000,0000000000000000,000000000000,00000000,00
 [] 0000, () 0000, {} 0000
 (000 <> 000)00.

00: 00000000,000000.

00000

00000,0000000000,000000000 id 00,0000000000000000.

id 000000000000000000000000.

00

```
graph TD
  id
```

00



00000

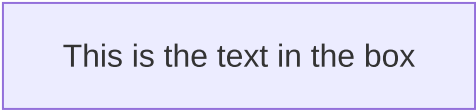
00000,000000,00000000 id 0000 <node shape> 00,0000000000000000.

id 0000000000,00000000000 <node shape> 0000,00 id 00000000000.

- 00

节点: [node description] , [] 描述, node description 描述.

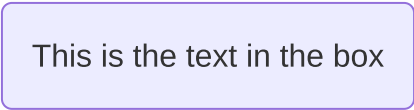
```
graph LR
    id1[This is the text in the box]
```



- 节点

节点: (node description) , () 描述, node description 描述.

```
graph LR
    id1(This is the text in the box)
```



- 节点

节点: ([node description]) , () 描述 [] 描述, 描述

`,node description` .

```
graph LR
    id1([This is the text in the box])
```

This is the text in the box

- 

: `[(node description)]` , `[ ]`  `( )` , `node description` .

```
graph LR
    id1[(Database)]
```



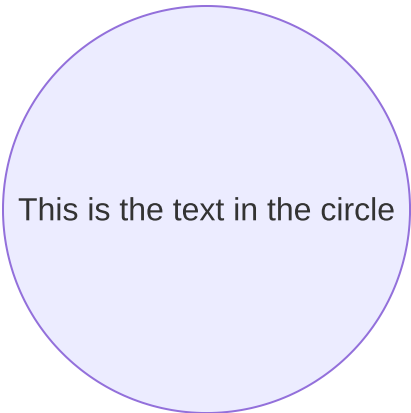
- 

: `((node description))` , `( )`  `( )` , `node description` .

❏

```
graph LR
    id1((This is the text in the circle))
```

❏



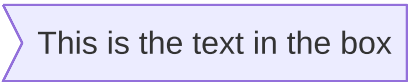
- ❶❷❸❹

```
❶❷❸❹: >node description] ,❶❷❸❹❶❷❸❹ > ,❶❷❸❹❶❷❸❹ ] ❶❷❸❹❶❷❸❹❶❷❸❹, node
description ❶❷❸❹❶❷❸❹.
```

❏

```
graph LR
    id1>This is the text in the box]
```

❏



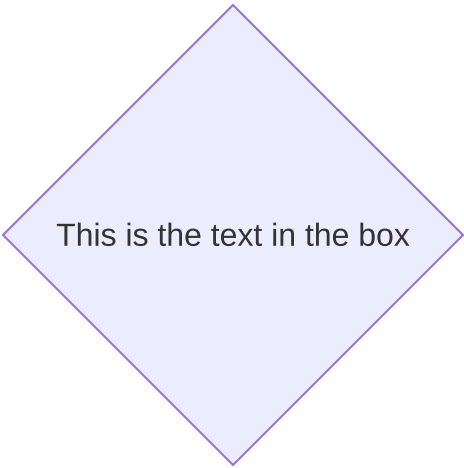
- ❶❷

节点: {node description} , {} 节点描述, node description 节点描述.

图

```
graph LR
  id1{This is the text in the box}
```

图



- 节点

节点: {{node description}} , {} 节点 {} 节点描述, node description 节点描述.

图

```
graph LR
  id1\\{\\{This is the text in the box\\}\\}
```

Gitbook 节点 {} 节点描述,节点描述,节点描述 \ 节点.

□□

This is the text in the box

- □□□□

```
□□□□: [/node description/] , [] □□□□ // □□□□□□□□□□□□, node
description □□□□□□□□.
```

□□

```
graph TD
  id1[/This is the text in the box/]
```

□□

This is the text in the box

- □□□□

```
□□□□: [\node description\] , [] □□□□ \\ □□□□□□□□□□□□, node
description □□□□□□□□.
```

□□

```
graph TD
  id1[\This is the text in the box\]
```

□□

This is the text in the box

- 

####: [/node description\] , [] / \ , node description .

```
graph TD
  A[/Christmas\]
```

Christmas

- 

####: [\node description/] , [] \ / , node description .

```
graph TD
  B[\Go shopping/]
```

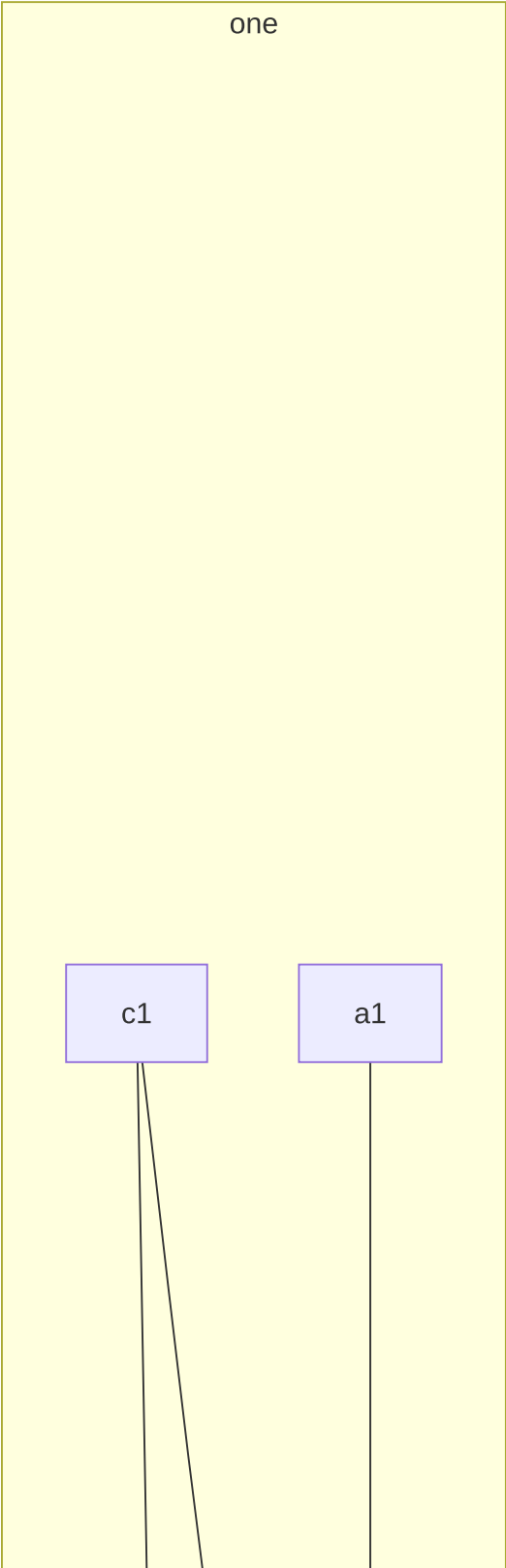
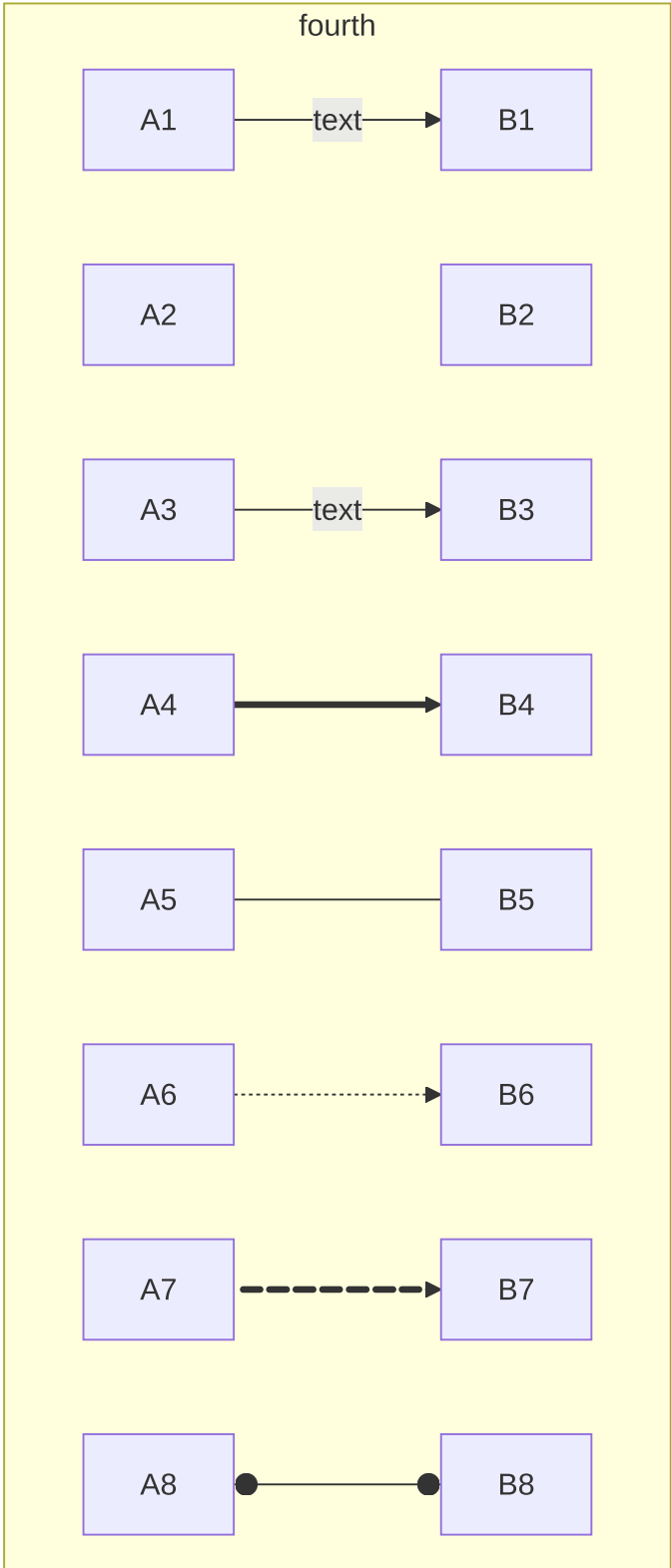
Go shopping

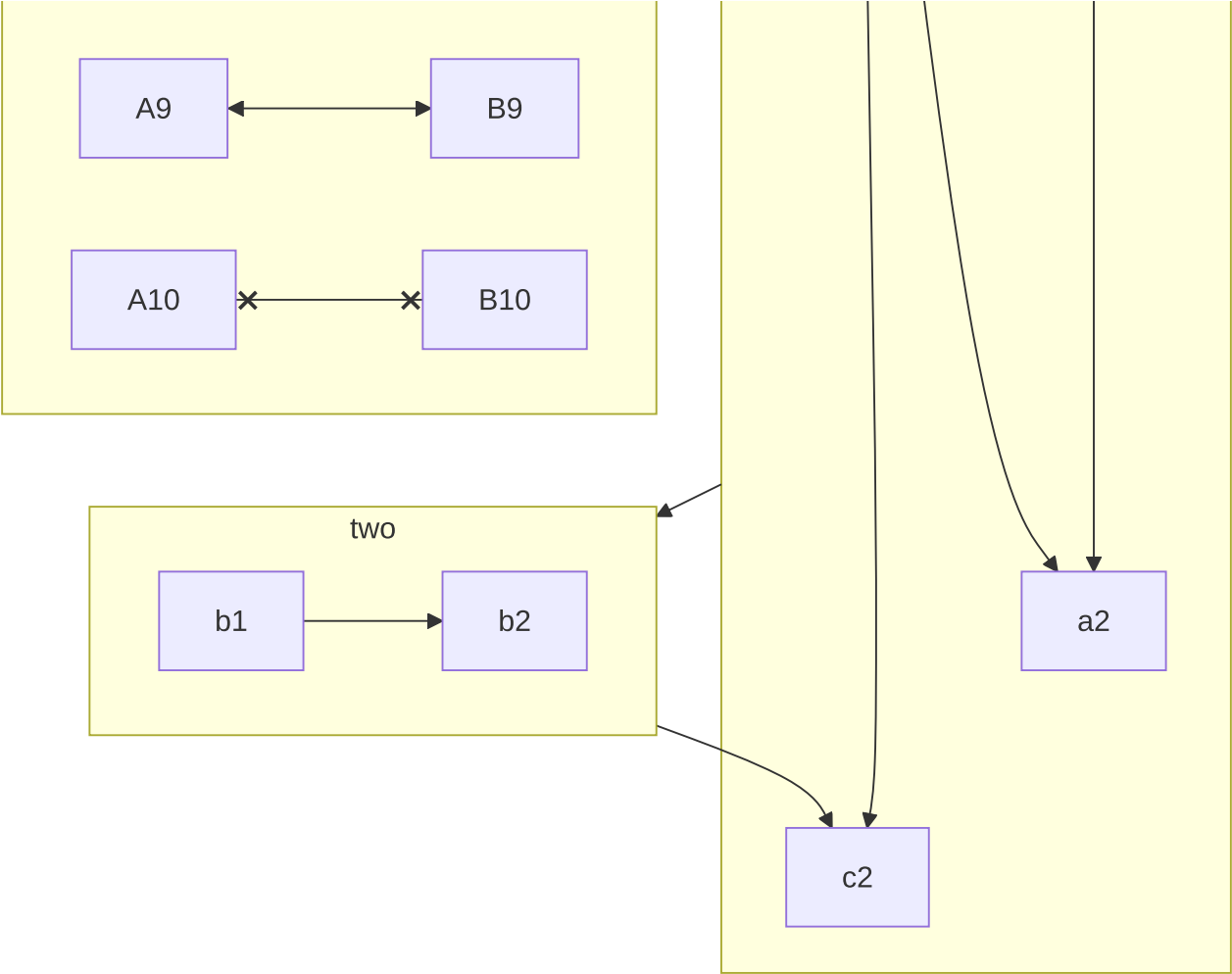
☐☐☐

☐☐☐

M↓

```
+ 00/00
- --
- -.
+ 000/000
- >
- -
+ 000/000
- 00
+ --0000
+ |0000|
- 00
+ -.0000
+ |0000|
+ 00
- ==
+ 0000
- -->
- ---
- -. ->
- .-
- 00000000
+ --0000-->
+ -->|0000|
- 00000000
+ --0000---
+ ---|0000|
- 00000000
+ -.0000-. ->
+ -. ->|0000|
- 00000000
+ -.0000-. -
+ -. -|0000|
- ==>
- ===
- 0000000000(2)
+ ==0000==>
+ ==>|0000|
- 0000000000(2)
+ ==0000===
+ ===|0000|
```





00000000,00000000000000000000,0000000000000000,00 -- 0000,00000000 -. -  
0000,0000000000 > ,00000000000 - .

00: 0000000,000000000,00000000000000000000,0000000000000000.

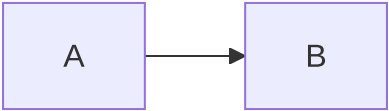
- 0000000000

0000: --> ,00 -- 0000, > 000000.

00

```
graph LR
  A-->B
```

🔗



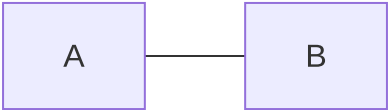
- 有向边

```
graph LR
  A --- B
  A -.- B
  A -.-> B
```

🔗

```
graph LR
  A --- B
```

🔗



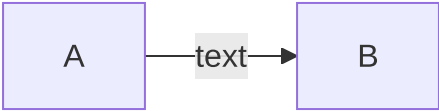
- 无向边

```
graph LR
  A -- connection line description --> B
  A --- text --> B
```

🔗

```
graph LR
  A -- text --> B
```

❏❏

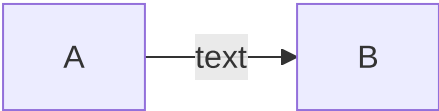


❏❏❏❏: |connection line description| ,❏❏ || ❏❏❏❏❏❏❏❏❏.

❏❏

```
graph LR
  A-->|text|B
```

❏❏



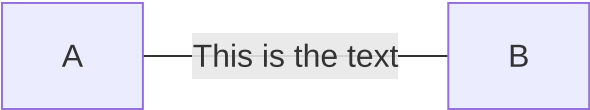
- ❏❏❏❏❏❏❏❏❏

❏❏❏❏: --connection line description ,❏❏❏❏❏ -- ❏❏❏❏❏❏❏❏❏,❏❏❏ --- ❏❏❏❏❏❏❏❏.

❏❏

```
graph LR
  A-- This is the text ---B
```

❏❏

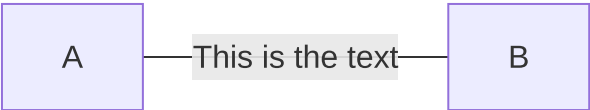


格式: |connection line description| , 逗号 || 分隔符.

图

```
graph LR
  A---|This is the text|B
```

图



- 格式

格式: -.connection line description.-> , 逗号 -. 分隔符, 逗号 .-> 图  
分隔符.

图

```
graph LR
  A-. text .-> B
```

图



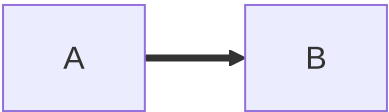
- 图的基本概念

```
graph LR
    A ==> B
```

图

```
graph LR
    A ==> B
```

图



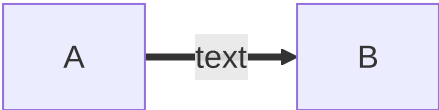
- 图的基本概念

```
graph LR
    A == connection line description ==> B
```

图

```
graph LR
    A == text ==> B
```

图



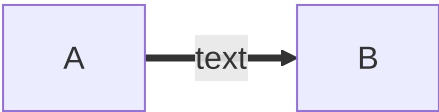
- 图的基本概念

连接: |connection line description| , 连接 || 连接描述.

连接

```
graph LR
  A ==>|text| B
```

连接



连接

连接



- + -->-->
- + &
- + ""
- + %%
- + subgraph



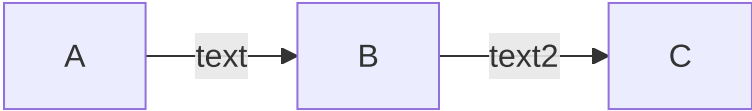
- 连接

连接

连接, A-->B-->C 连接 A-->B 连接 B-->C 连接.

```
graph LR
    A -- text --> B -- text2 --> C
```

□□



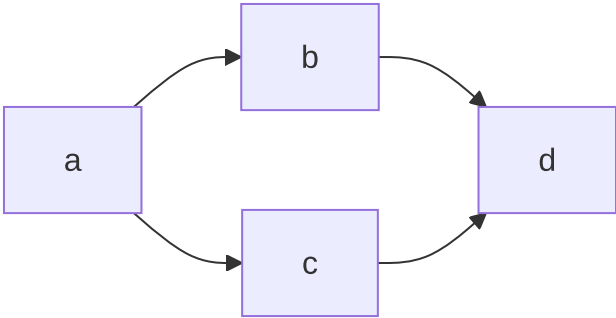
- □□□□□□□□

□□□□□□□□, A-->B & C □□□ A-->B □ A-->C □□.

□□

```
graph LR
    a --> b & c --> d
```

□□



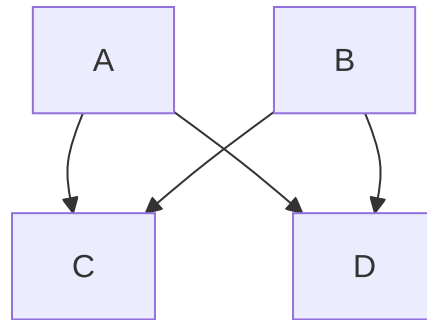
- □□□□□□□□

□□□□□□□□□□□□, A & B --> C & D □□□ A-->C , A-->D , B-->C □ B-->D □□□□□□.

□□

```
graph TB
    A & B --> C & D
```

11



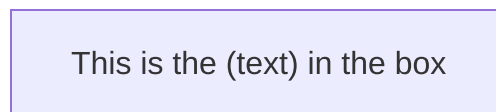
- □ □ □ □ □ □ □ □

□□□□□□□□□□□□□□□□ " " □□□□,□□□ □ □ ( ) □□ { } □□□□□□□.

11

```
graph LR
    id1["This is the (text) in the box"]
```

11



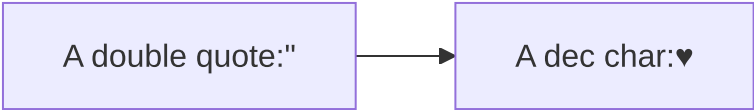
- [illegible]

□□ Html □□□□

11

```
graph LR
    A["A double quote:#quot;"] --> B["A dec char:#9829;"]
```

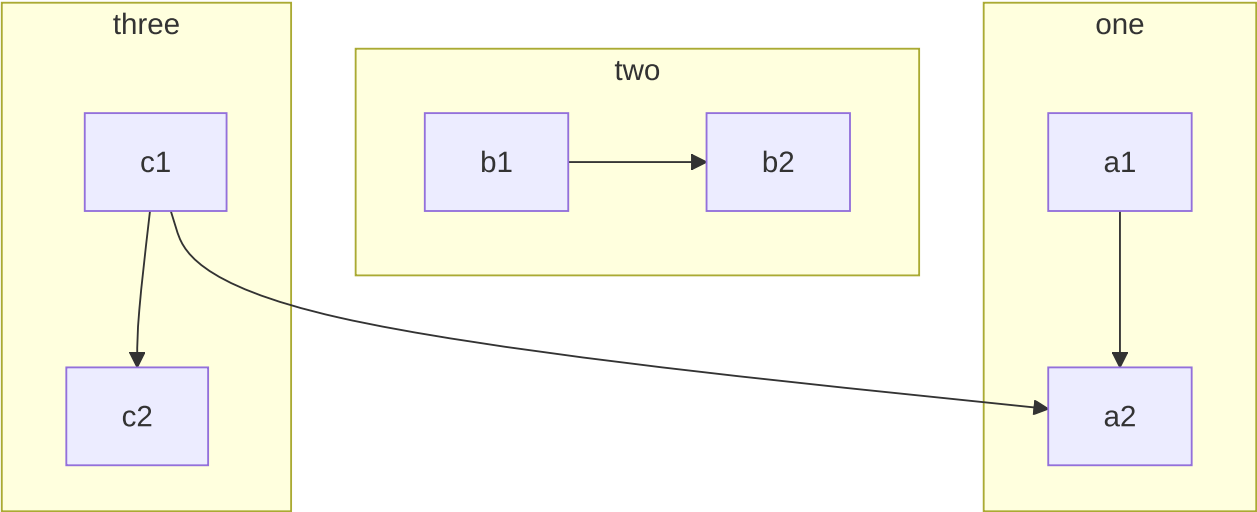
11



- 

```
subgraph title
  graph definition
end
```

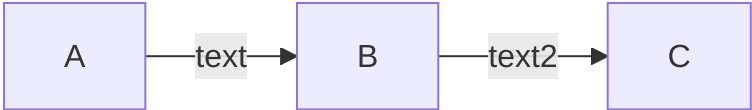
```
graph TB
  c1-->a2
  subgraph one
    a1-->a2
  end
  subgraph two
    b1-->b2
  end
  subgraph three
    c1-->c2
  end
```



- 图图图

图图 %% 图图图图图图图图.

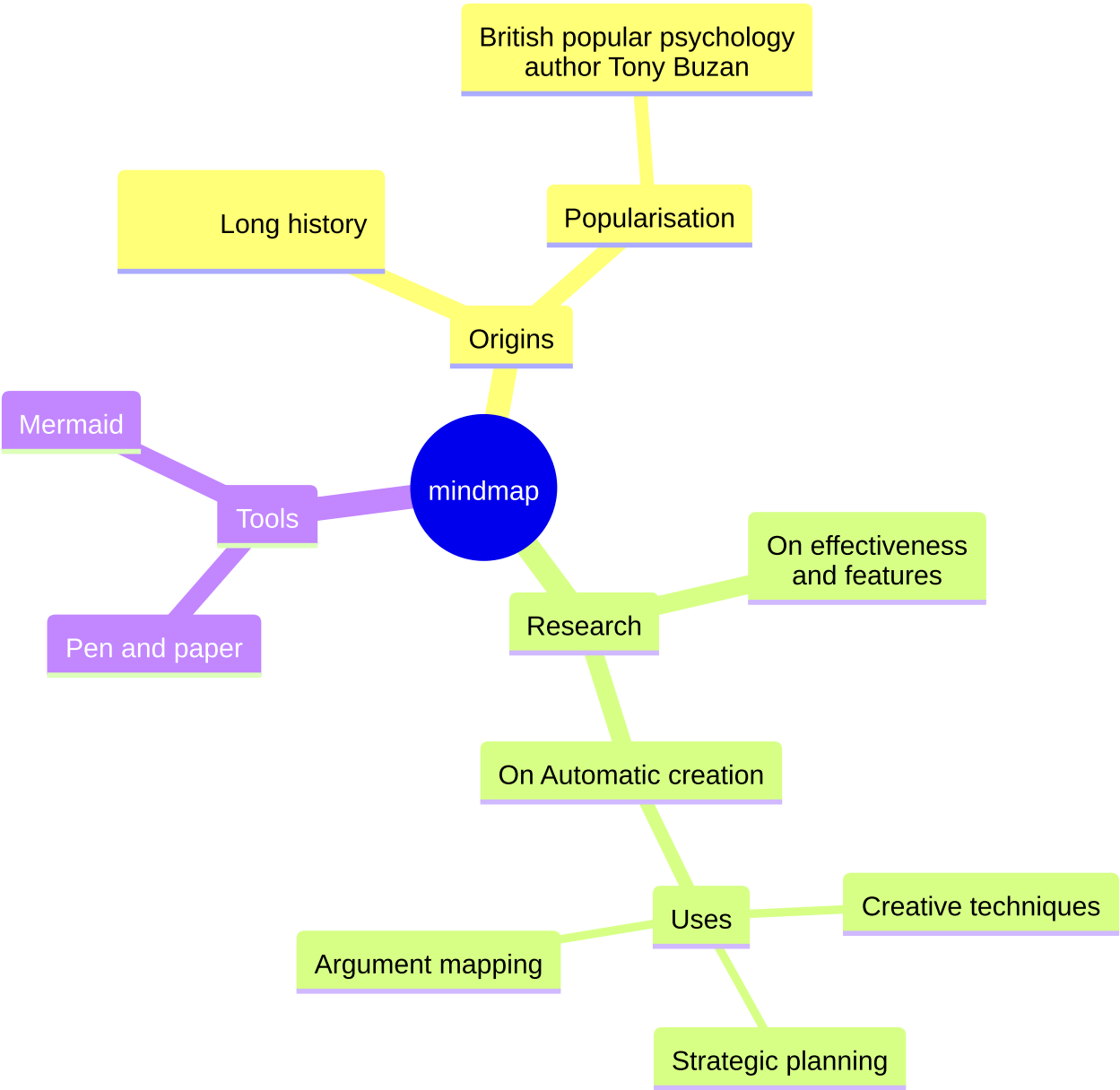
```
graph LR
%% this is a comment A -- text --> B{node}
  A -- text --> B -- text2 --> C
```



图图图图图图图图图图图图

图图

- 图图图图图图
- 图图图图图
- 图图图图



Mermaid 图生成器, 使用 Markdown 格式描述图结构, 生成 Mermaid 图代码。

图的基本概念

图是计算机科学中一种重要的数据结构，它由一组顶点和一组连接顶点的边组成。图可以用来表示各种现实世界中的关系，如社交网络、交通网络、计算机网络的拓扑结构等。

| 关键字      | 描述       | 说明        |
|----------|----------|-----------|
| graph    | graph    | 图的基本类型    |
| subgraph | subgraph | 子图        |
| top      | TB 或 BT  | 从上到下或从下到上 |
| bottom   | BT 或 TB  | 从下到上或从上到下 |
| left     | LR 或 RL  | 从左到右或从右到左 |
| right    | RL 或 LR  | 从右到左或从左到右 |

图的表示

图可以用邻接表、邻接矩阵、边集数组等多种方式表示。邻接表适用于稀疏图，邻接矩阵适用于稠密图。边集数组适用于无向图，邻接矩阵适用于有向图。

- 图的基本概念

| 符号 | 描述 | 说明     | 备注 |
|----|----|--------|----|
| [] | 图  | 图的基本类型 | 图  |
| () | 子图 | 子图     | 图  |
| {} | 图  | 图的基本类型 | 图  |
| <> | 图  | 图的基本类型 | 图  |
| -- | 图  | 图的基本类型 | 图  |
| -. | 图  | 图的基本类型 | 图  |
| == | 图  | 图的基本类型 | 图  |
| =: | 图  | 图的基本类型 | 图  |

| 符号  | 名称  | 描述    | 备注 |
|-----|-----|-------|----|
| >   | 大于  | 比较运算符 | 单目 |
| -   | 负号  | 一元运算符 | 单目 |
| xxx | 变量名 | 标识符   | 单目 |
| --  | 自减  | 二元运算符 | 双目 |
| -.  | 点分  | 标识符   | 单目 |
| ==  | 相等  | 比较运算符 | 双目 |
| =:  | 赋值  | 赋值运算符 | 双目 |

- 运算符

| 符号    | 名称  | 描述    | 备注       |
|-------|-----|-------|----------|
| [[]]  | 数组  | 数组访问  | 单目<br>[] |
| [()]  | 函数  | 函数调用  | 单目       |
| [{}]  | 字典  | 字典访问  | 单目<br>[] |
| (( )) | 圆括号 | 表达式   | 单目       |
| ([])  | 列表  | 列表构造  | 单目       |
| ({})  | 字典  | 字典构造  | 单目<br>[] |
| xxxx  | 常量  | 常量表达式 | 单目       |
| {[]}  | 集合  | 集合构造  | 单目<br>[] |
| {() } | 函数  | 函数构造  | 单目<br>[] |
| -->   | 右移  | 移位运算符 | 单目       |
| ---   | 左移  | 移位运算符 | 单目       |
| -.>   | 右移  | 移位运算符 | 单目<br>[] |

| 命令                                               | 参数         | 选项         | 说明       |
|--------------------------------------------------|------------|------------|----------|
| <code>-.-&gt;</code>                             | 连接描述       | 连接描述       | 连接       |
| <code>.-&gt;</code>                              | 连接描述       | 连接描述       | 连接       |
| <code>-.-</code>                                 | 连接描述       | 连接描述       | 连接       |
| <code>.-</code>                                  | 连接描述       | 连接描述       | 连接       |
| <code>==&gt;</code>                              | 连接描述       | 连接描述       | 连接       |
| <code>===</code>                                 | 连接描述       | 连接描述       | 连接       |
| <code>:=&gt;</code>                              | 连接描述       | 连接描述       | 连接<br>连接 |
| <code>:=:&gt;</code>                             | 连接描述       | 连接描述       | 连接<br>连接 |
| <code>:=</code>                                  | 连接描述       | 连接描述       | 连接<br>连接 |
| <code>xxx</code>                                 | 连接描述       | 连接描述<br>连接 | 连接       |
| <code>--connection line description-&gt;</code>  | 连接描述<br>连接 | 连接描述<br>连接 | 连接       |
| <code>-.connection line description.-&gt;</code> | 连接描述<br>连接 | 连接描述<br>连接 | 连接       |
| <code>--connection line description--</code>     | 连接描述<br>连接 | 连接描述<br>连接 | 连接       |
| <code>-.connection line description.-</code>     | 连接描述<br>连接 | 连接描述<br>连接 | 连接       |
| <code>==connection line description==&gt;</code> | 连接描述<br>连接 | 连接描述<br>连接 | 连接       |
| <code>:=connection line description:=&gt;</code> | 连接描述<br>连接 | 连接描述<br>连接 | 连接<br>连接 |
| <code>==connection line description==</code>     | 连接描述<br>连接 | 连接描述<br>连接 | 连接       |

| 节点                                               | 边                                 | 类                     | 样式                 |
|--------------------------------------------------|-----------------------------------|-----------------------|--------------------|
| <code>=:connection line<br/>description=:</code> | <code>graph LR; A --&gt; B</code> | <code>classDef</code> | <code>class</code> |

节点

节点是图论中的基本概念，用于表示事物或对象。在 Mermaid 中，节点通常用大写字母表示，如 A、B、C 等。节点之间通过边（Edge）连接，表示它们之间的关系。

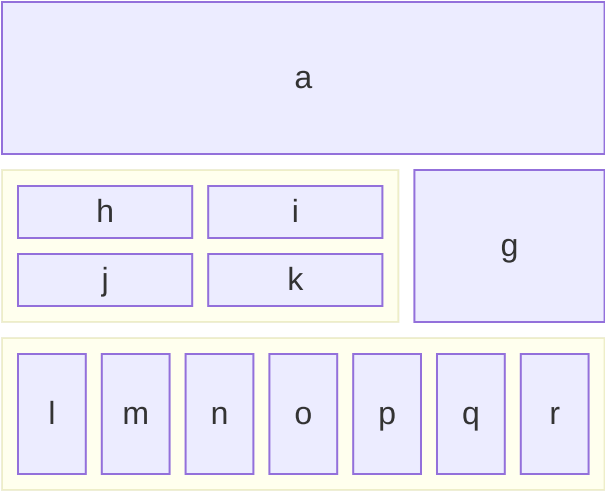
节点可以是简单的文本，也可以是带有样式的元素。例如，可以使用 `class` 关键字为节点指定类，从而应用预定义的样式。此外，还可以使用 `linkStyle` 关键字为边指定样式。

节点类：<https://mermaid-js.github.io/mermaid/#/flowchart?id=styling-and-classes>

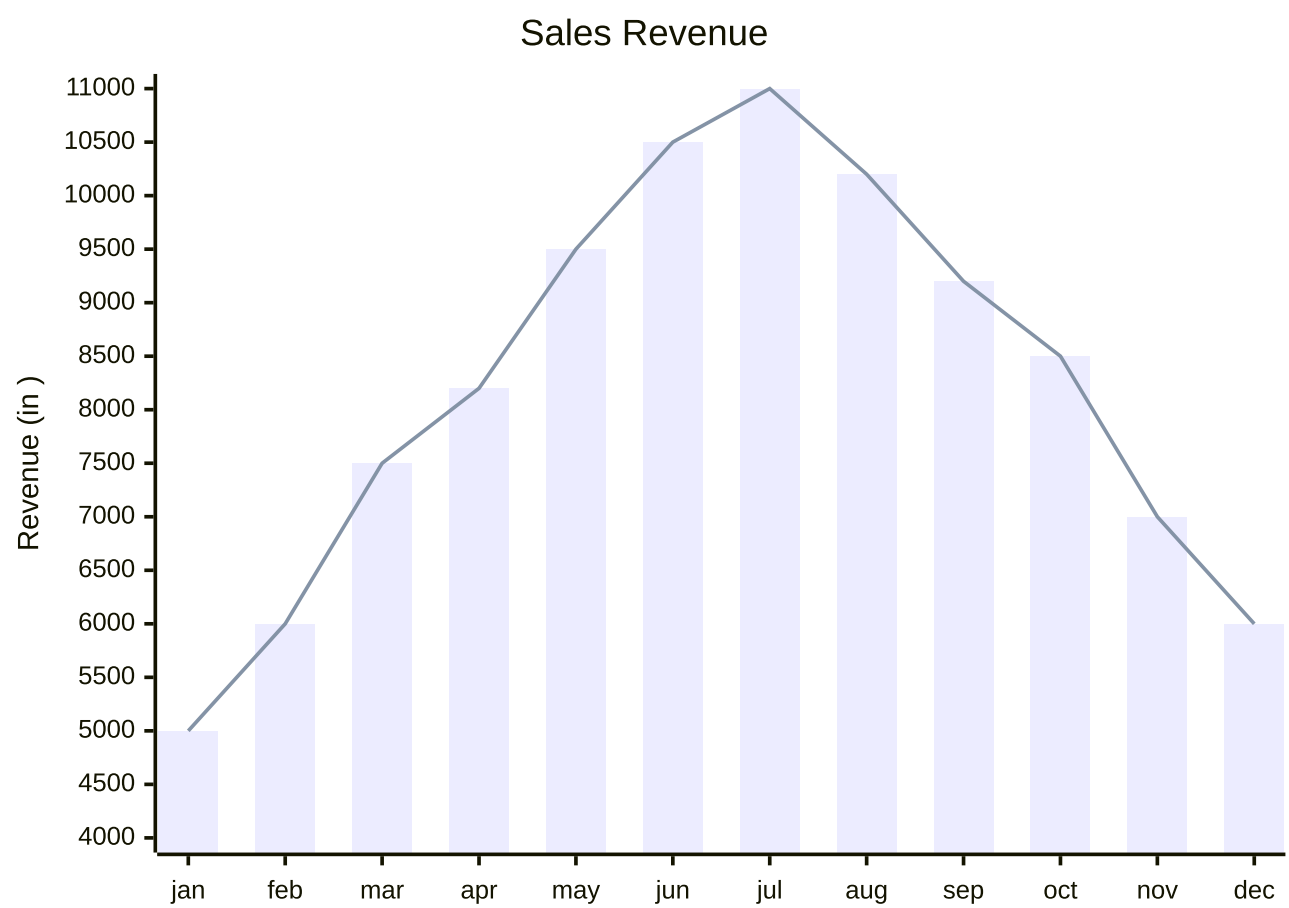
- 节点 Interaction : <https://mermaid-js.github.io/mermaid/#/flowchart?id=interaction>
- 节点 Styling and classes : <https://mermaid-js.github.io/mermaid/#/flowchart?id=interaction>
- 节点 Basic support for fontawesome: <https://mermaid-js.github.io/mermaid/#/flowchart?id=basic-support-for-fontawesome>
- 节点 <https://mermaid-js.github.io/mermaid/#/flowchart?id=graph-declarations-with-spaces-between-vertices-and-link-and-without-semicolon>

□□□

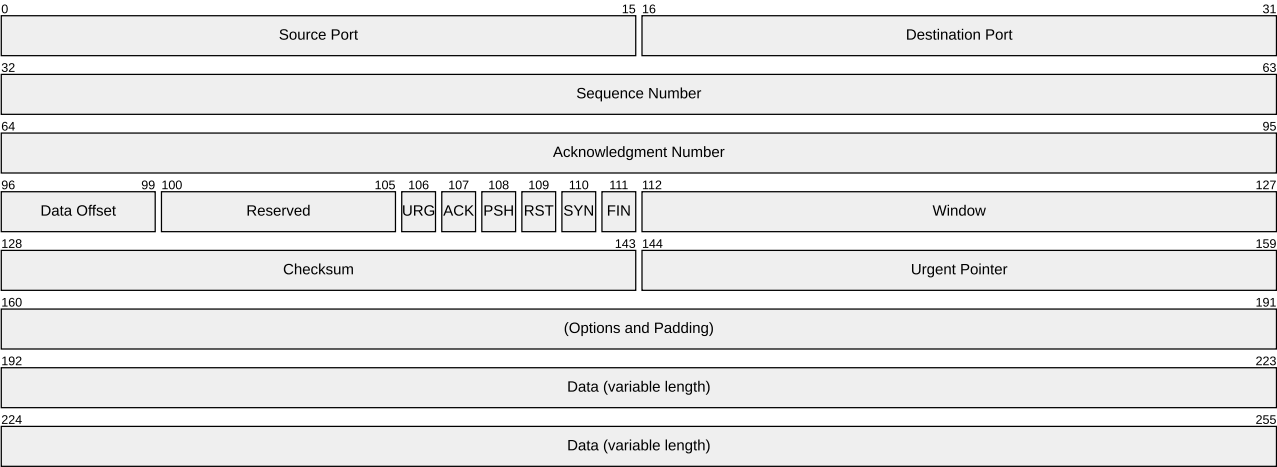
□□



XY



□□□



TCP Packet

# mdbook-admonish

The following admonishments are implemented by the [mdbook-admonish](#) plugin and are automatically themed to match Catppuccin.

## Directives

All supported directives are listed below.

`note`

### Note

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`abstract`, `summary`, `tldr`

### Abstract

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`info`, `todo`

 **Info**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

tip, hint, important

 **Tip**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

success, check, done

 **Success**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

question, help, faq

 **Question**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

warning, caution, attention

 **Warning**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

failure, fail, missing

 **Failure**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

danger, error

 **Danger**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

bug

 **Bug**

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

example

### Example

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

quote, cite

### ” Quote

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

## Bienvenue sur notre site de développement 3D !

Bienvenue sur notre site dédié au développement 3D. Ici, vous trouverez des ressources, des tutoriels et des informations utiles pour vous lancer dans le monde passionnant de la 3D.

### Info

A beautifully styled message.

### Un exemple

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Une note**

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Un warning**

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Collapsing note** 

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Le javascript c'est yolo préférez Typescript**

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

 **Referencing and dereferencing**

The opposite of *referencing* by using `&` is *dereferencing*, which is accomplished with the *dereference operator*, `*`.

## Bug

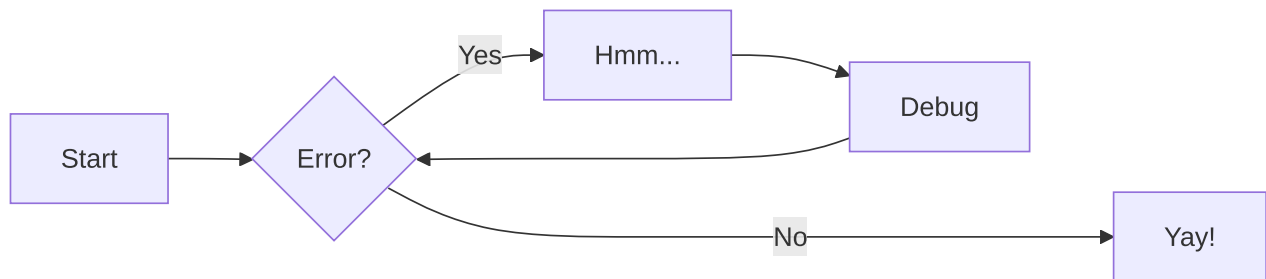
This syntax won't work in Python 3:



```
print "Hello, world!"
```

## À propos de nous

Nous sommes une équipe passionnée par la 3D et nous avons pour mission de partager nos connaissances avec la communauté. Vous trouverez ici des articles, des exemples de code et des démonstrations pour vous aider à démarrer votre voyage dans le développement 3D.



## Pour commencer

Si vous êtes nouveau dans le domaine de la 3D, ne vous inquiétez pas ! Notre page ["Getting Started"](#) vous guidera à travers les étapes essentielles pour démarrer rapidement.

## Restons en contact

N'hésitez pas à nous suivre sur les réseaux sociaux pour rester à jour avec nos dernières publications et annonces. Si vous avez des questions ou des commentaires, n'hésitez pas à nous contacter !

## Mizux



# mdbook\_mathjax

## Section 1

## Section 2

## Chapter 2

## Section 1

## Section 2

## Chapter 3

## Section 1

## Section 2

## Chapter 4

## Section 1

## Section 2

# Chapter 5

## Section 1

## Section 2



# Chapter 6

## Section 1

### Note

Highlights information that users should take into account, even when skimming.

### Tip

Optional information to help a user be more successful.

### Important

Crucial information necessary for users to succeed.

### Warning

Critical content demanding immediate user attention due to potential risks.

### Caution

Negative potential consequences of an action.

## Section 2

Here is an inline example,  $\pi(\theta)$ ,  
an equation,

$$\nabla f(x) \in \mathbb{R}^n,$$

and a regular \$ symbol.

Define  $f(x)$ :

$$f(x) = x^2$$

$$x \in \mathbb{R}$$

```
graph TD
  A[ Anyone ] -->|Can help | B( Go to github.com/yuzutech/kroki )
  B --> C{ How to contribute? }
  C --> D[ Reporting bugs ]
  C --> E[ Sharing ideas ]
  C --> F[ Advocating ]
```

□□□□

- □□□□□□□□Python: \\ □□□□ \\,□□□□□□□

```
def ex_gcd(a, b):
    if b == 0:
        return 1, 0, a
    else:
        x, y, g = ex_gcd(b, a % b)
        return y, x - (a // b) * y, g
```

```
N = 5
for n in range(1, N + 1):
    for m in range(1, N + 1):
        print(ex_gcd(n, m))
```

- □□□□□□□□

11...1

$x_1x_2...x_n$

$\vdots\vdots\ddots\vdots$

$x_1^{n-1}x_2^{n-2}...x_n^{n-1}$

=

$\prod_{1\leq j\leq i\leq n}(x_i-x_j)$

- 同位素示踪法



- 同位素示踪法

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$$

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\Psi(x,t)$$

- 同位素示踪法

$$\left\{\begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu_0 \left( \mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{array}\right.$$

Gauss’s law

Gauss’s law for magnetism

Maxwell–Faraday equation

Ampère’s circuital law

- 同位素示踪法

| $x$ | $y$ | $S$ |
|-----|-----|-----|
| 0   | 0   | 0   |
| 0   | 1   | 1   |
| 1   | 0   | 1   |
| 1   | 0   | 0   |

- 同位素示踪法

$$\mathbf{1}_{\mathbb{Q}}(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

- 同余式

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } a \equiv 0 \pmod{p} \\ 1 & \text{if } a \not\equiv 0 \pmod{p} \wedge \exists x : x^2 \equiv a \pmod{p} \\ -1 & \text{if } a \not\equiv 0 \pmod{p} \wedge \nexists x : x^2 \equiv a \pmod{p} \end{cases}$$

- 微分形式

$$\oint_{\Gamma} Pdx + Qdy + Rdz = \iiint_{\Sigma} \left( \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dydz \tag{1}$$

$$+ \left( \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dzdx \tag{2}$$

$$+ \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy \tag{3}$$

- 特殊函数

$$\Gamma(z) = \int_0^\infty t^{z-1} \mathrm{e}^{-t} dt$$

$$\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s}$$

- 逻辑

$\neg(P \vee Q) \iff \neg P \wedge \neg Q$

(propositional logic)

$\neg(P \wedge Q) \iff \neg P \vee \neg Q$

(4)

$(A \cup B)^c = A^c \cap B^c$

(set theory)

$(A \cap B)^c = A^c \cup B^c$

(5)

$\overline{A \cup B} = \overline{A} \cap \overline{B}$

(probability theory)

$\overline{A \cap B} = \overline{A} \cup \overline{B}$

(6)

- 全概率公式

$$P(A_i|B) = \frac{P(B|A_i) \cdot P(A_i)}{\sum_{j=1}^n P(B|A_j) \cdot P(A_j)}$$

- 泰勒级数

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \tag{7}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \tag{8}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \tag{9}$$

$$\ln(x+1) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{(n+1)!} \tag{10}$$

- 自然常数e的表达式

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

- 二项式定理

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

# mdbook-katex

HTML:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\Rightarrow x + iy = re^{i\theta}$$

Markdown (requires `mdbook-katex`):

$$\oint_C f(x, y) \, dA$$

Inspect element and use `Sources` tab (under `Debugger` on Firefox) to check that all CSS and fonts are properly loaded from GitHub pages instead of external CDN.

▼ Proof that  $e^{ix} = \cos x + i \sin x$

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \implies e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \\ \cos x &= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{(2m)!} = \sum_{m=0}^{\infty} \frac{(ix)^{2m}}{(2m)!} \\ \sin x &= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{(2s+1)!} = \sum_{s=0}^{\infty} \frac{(ix)^{2s+1}}{i(2s+1)!} \\ \cos x + i \sin x &= \sum_{l=0}^{\infty} \frac{(ix)^{2l}}{(2l)!} + \sum_{s=0}^{\infty} \frac{(ix)^{2s+1}}{(2s+1)!} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \\ &= e^{ix} \end{aligned}$$

Fourier Transform:

$$f(t) = \int_{-\infty}^{\infty} F(\omega) i^{4t\omega} d\omega$$
$$F(\omega) = \int_{-\infty}^{\infty} f(t) i^{-4t\omega} dt$$

Pauli Matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$
$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

# mdbook-whichlang



```
#include <stdio.h>

int main(void) {
    printf("Hello World\n");
}
```



```
#include <iostream>

int main()
{
    std::cout << "Hello World" << std::endl;
}
```



```
console.log("Hello World");
```



```
console.log("Hello World");
```



```
Console.log("Hello World")
```



```
h1 {  
  color: blue;  
}
```



```
set syntax=ruby
```



init.lua

```
print("Hello World")
```



```
print("Hello World")
```



```
echo "Hello World"
```



hello.zig

```
const std = @import("std");  
  
pub fn main() !void {  
    const stdout = std.io.getStdOut().writer();  
    try stdout.print("Hello, {s}!\n", .{"world"});  
}
```



```
package main

import "core:fmt"

main :: proc() {
    fmt.println("Hellope!")
}
```



hello.wat

```
(module
  (import "wasi_unstable" "fd_write"
    (func $fd_write (param i32 i32 i32 i32) (result i32))
  )

  (memory 1)
  (export "memory" (memory 0))

  (data (i32.const 0) "\08\00\00\00\0c\00\00\00Hello World\n")

  (func $main (export "_start")
    i32.const 1
    i32.const 0
    i32.const 1
    i32.const 20
    call $fd_write
    drop
  )
)
```

# mdbook-langtabs

## Some code

let's try with a simple example:



```
LogChannel(n"DEBUG", "hello world");
```



```
print("Hello, World!")
```



```
fn main() { println!("hello world"); }
```



```
print("Hello World")
```



```
#include <iostream>

int main() {
    std::cout << "Hello World!";
    return 0;
}
```



```
some:
  interesting:
    - property
```



```
{
  "some": { "interesting": ["property"] }
}
```

xml:

```
<some>
  <interesting>
    <property />
  </interesting>
</some>
```

contrary to code blocks, `inline` and `fenced` are left untouched.

it should also work when deeply nested:

1. outer:

1. inner:



```
GameInstance
  .GetStatusEffectSystem(this.GetGame())
  .ApplyStatusEffect(
    this.GetEntityID(),
    t"BaseStatusEffect.SplinterAddicted",
    this.GetRecordID(),
    this.GetEntityID());
```

# Hello World

Here's a simple "Hello World" example in different languages:



```
fn main() {  
    println!("Hello, World!");  
}
```



```
def main():  
    print("Hello, World!")  
  
if __name__ == "__main__":  
    main()
```



```
function main() {  
    console.log("Hello, World!");  
}  
  
main();
```



```
package main  
  
import "fmt"  
  
func main() {  
    fmt.Println("Hello, World!")  
}
```



```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

## Simple Function Example

Here's how you might define a function to calculate factorial in different languages:



```
fn factorial(n: u64) -> u64 {  
    match n {  
        0 | 1 => 1,  
        _ => n * factorial(n - 1)  
    }  
}  
  
fn main() {  
    println!("5! = {}", factorial(5)); // Outputs: 5! = 120  
}
```



```
def factorial(n):  
    if n <= 1:  
        return 1  
    return n * factorial(n - 1)  
  
print(f"5! = {factorial(5)}") # Outputs: 5! = 120
```



```
function factorial(n) {  
    if (n <= 1) return 1;  
    return n * factorial(n - 1);  
}  
  
console.log(`5! = ${factorial(5)}`); // Outputs: 5! = 120
```



```
package main  
  
import "fmt"  
  
func factorial(n uint64) uint64 {  
    if n <= 1 {  
        return 1  
    }  
    return n * factorial(n-1)  
}  
  
func main() {  
    fmt.Printf("5! = %d\n", factorial(5)) // Outputs: 5! = 120  
}
```



```
public class FactorialExample {  
    static long factorial(int n) {  
        if (n <= 1) return 1;  
        return n * factorial(n - 1);  
    }  
  
    public static void main(String[] args) {  
        System.out.println("5! = " + factorial(5)); // Outputs: 5! = 120  
    }  
}
```

# Data Structures Example

Here's how you might implement a simple stack in different languages:



```
struct Stack<T> {  
    items: Vec<T>,  
}  
  
impl<T> Stack<T> {  
    fn new() -> Self {  
        Stack { items: Vec::new() }  
    }  
  
    fn push(&mut self, item: T) {  
        self.items.push(item);  
    }  
  
    fn pop(&mut self) -> Option<T> {  
        self.items.pop()  
    }  
  
    fn is_empty(&self) -> bool {  
        self.items.is_empty()  
    }  
}  
  
fn main() {  
    let mut stack = Stack::new();  
    stack.push(1);  
    stack.push(2);  
    stack.push(3);  
  
    while let Some(item) = stack.pop() {  
        println!("Popped: {}", item);  
    }  
}
```



```
class Stack:
    def __init__(self):
        self.items = []

    def push(self, item):
        self.items.append(item)

    def pop(self):
        if not self.is_empty():
            return self.items.pop()
        return None

    def is_empty(self):
        return len(self.items) == 0

# Usage
stack = Stack()
stack.push(1)
stack.push(2)
stack.push(3)

while not stack.is_empty():
    print(f"Popped: {stack.pop()}")
```

JS

```
class Stack {
  constructor() {
    this.items = [];
  }

  push(item) {
    this.items.push(item);
  }

  pop() {
    if (!this.isEmpty()) {
      return this.items.pop();
    }
    return null;
  }

  isEmpty() {
    return this.items.length === 0;
  }
}

// Usage
const stack = new Stack();
stack.push(1);
stack.push(2);
stack.push(3);

while (!stack.isEmpty()) {
  console.log(`Popped: ${stack.pop()}`);
}
```



```
package main

import "fmt"

type Stack struct {
    items []int
}

func (s *Stack) Push(item int) {
    s.items = append(s.items, item)
}

func (s *Stack) Pop() (int, bool) {
    if s.IsEmpty() {
        return 0, false
    }

    index := len(s.items) - 1
    item := s.items[index]
    s.items = s.items[:index]
    return item, true
}

func (s *Stack) IsEmpty() bool {
    return len(s.items) == 0
}

func main() {
    stack := Stack{}
    stack.Push(1)
    stack.Push(2)
    stack.Push(3)

    for !stack.IsEmpty() {
        item, _ := stack.Pop()
        fmt.Printf("Popped: %d\n", item)
    }
}
```



```
import java.util.ArrayList;

public class StackExample {
    static class Stack<T> {
        private ArrayList<T> items = new ArrayList<>();

        public void push(T item) {
            items.add(item);
        }

        public T pop() {
            if (isEmpty()) {
                return null;
            }
            return items.remove(items.size() - 1);
        }

        public boolean isEmpty() {
            return items.isEmpty();
        }
    }

    public static void main(String[] args) {
        Stack<Integer> stack = new Stack<>();
        stack.push(1);
        stack.push(2);
        stack.push(3);

        while (!stack.isEmpty()) {
            System.out.println("Popped: " + stack.pop());
        }
    }
}
```

# kroki

## kroki-graphviz

```
digraph G {  
  a -> b [dir=both color="red:blue"]  
  c -> d [dir=none color="green:red;0.25:blue"]  
}
```

# kroki-plantuml

```
@startuml
clock    "Clock_0"    as C0 with period 50
clock    "Clock_1"    as C1 with period 50 pulse 15 offset 10
binary   "Binary"     as B
concise  "Concise"    as C
robust   "Robust"     as R
analog   "Analog"     as A

@0
C is Idle
R is Idle
A is 0

@100
B is high
C is Waiting
R is Processing
A is 3

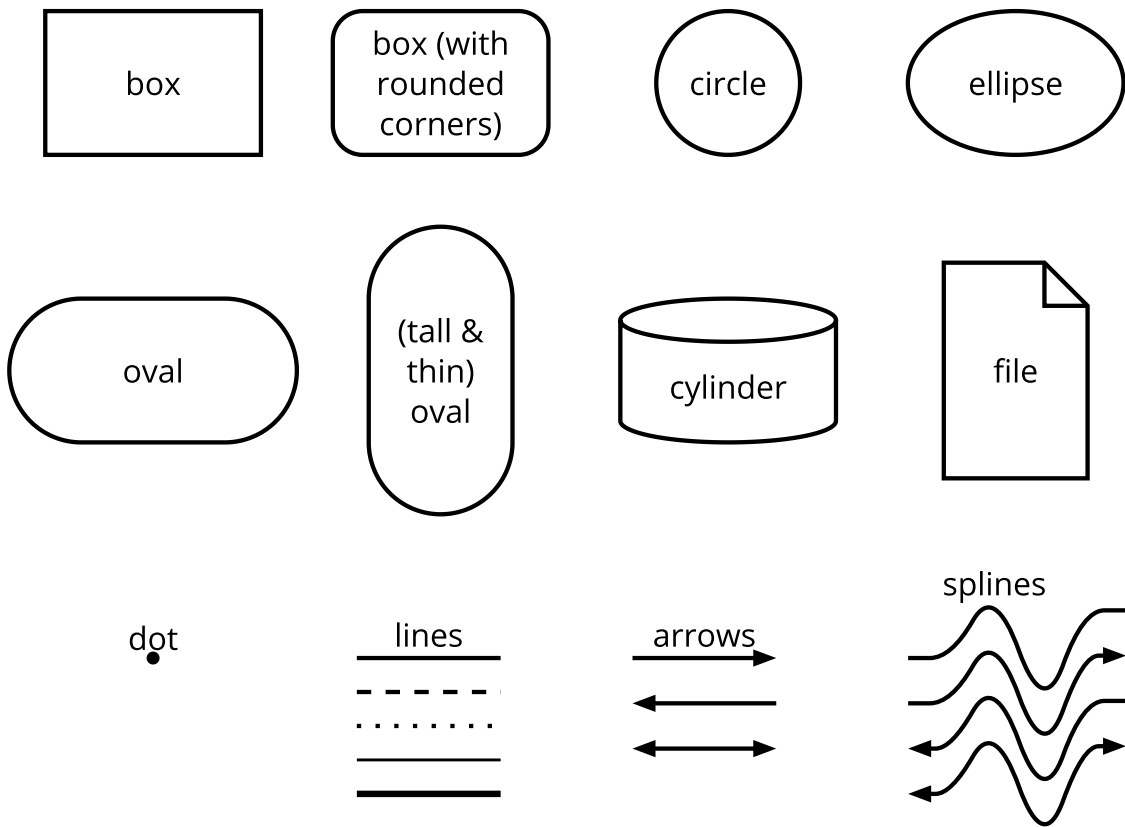
@300
R is Waiting
A is 1
@enduml
```

# kroki-symbolator

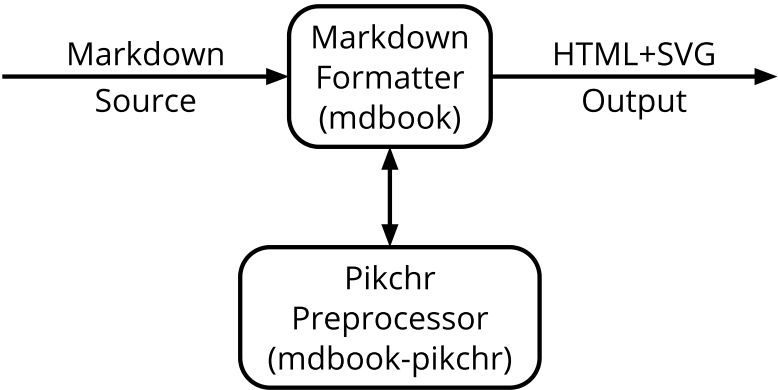
```
module demo_device #(
    //# {{}}
    parameter SIZE = 8,
    parameter RESET_ACTIVE_LEVEL = 1
) (
    //# {{clocks|Clocking}}
    input wire clock,
    //# {{control|Control signals}}
    input wire reset,
    input wire enable,
    //# {{data|Data ports}}
    input wire [SIZE-1:0] data_in,
    output wire [SIZE-1:0] data_out
);
    // ...
endmodule
```

# pikchr-demo1

## Examples Of Pikchr Objects

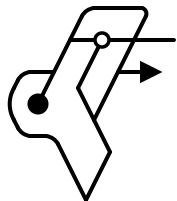


pikchr-demo2



# svgbob-demo

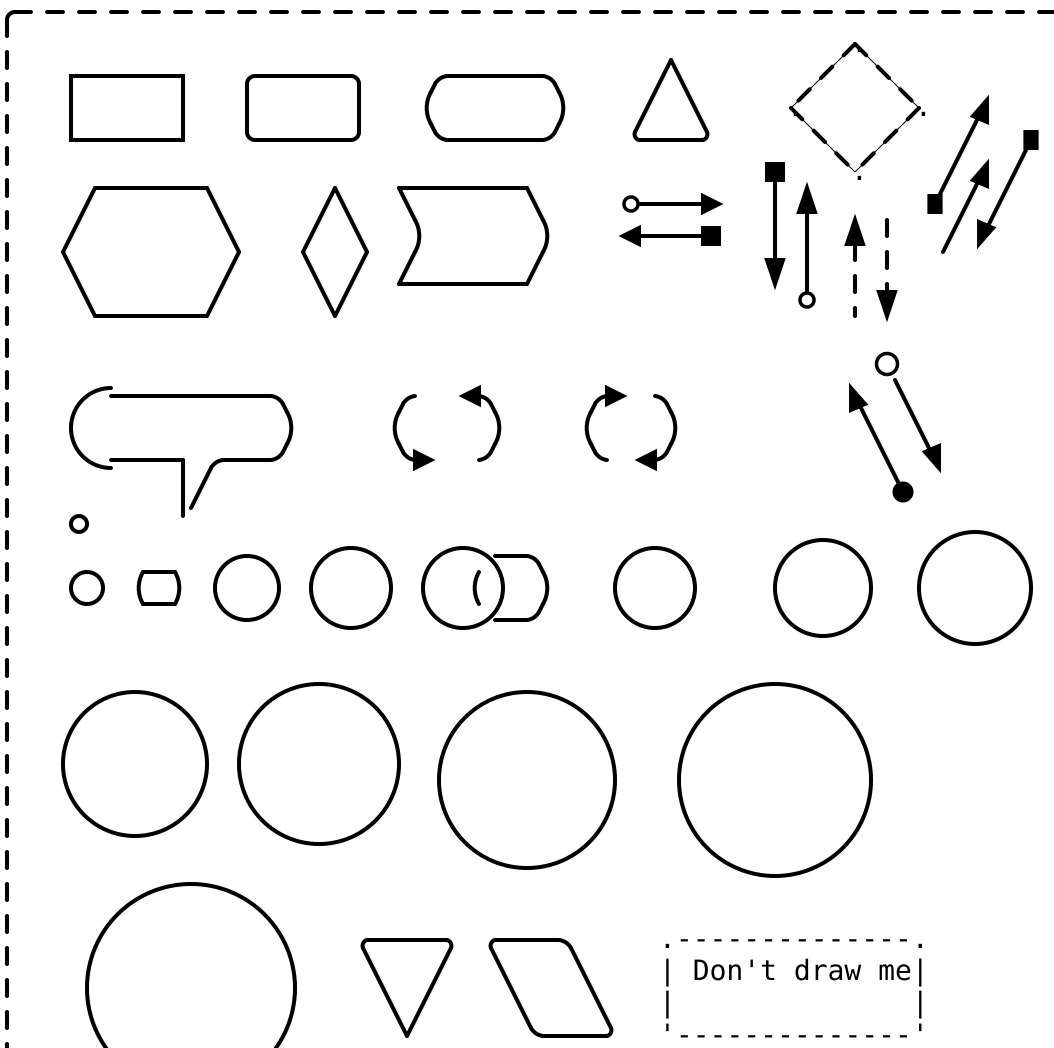
Svgbob is a diagramming model  
which uses a set of typing characters  
to approximate the intended shape.

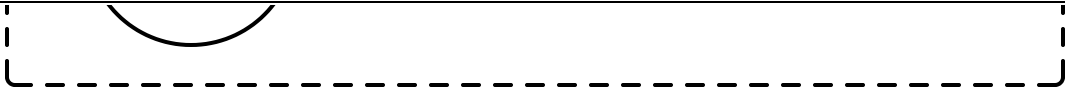


It uses a combination of characters  
which are readily available on your keyboards.

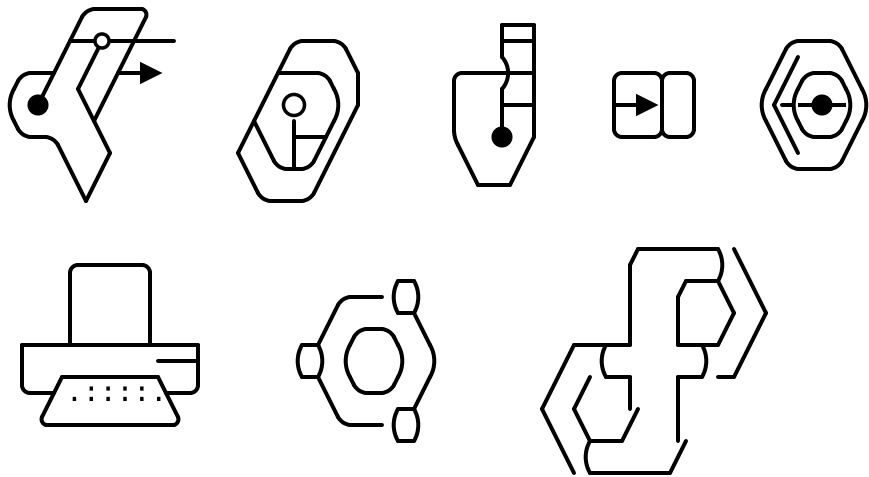
What can it do?

#### ➡ Basic shapes

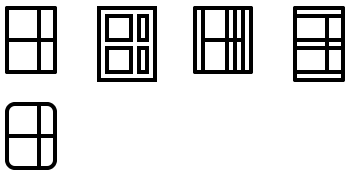




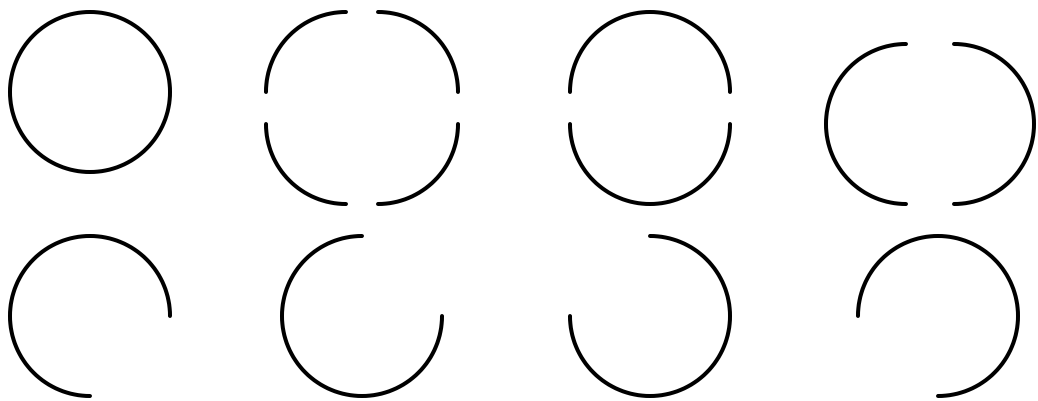
➡ Quick logo scribbles



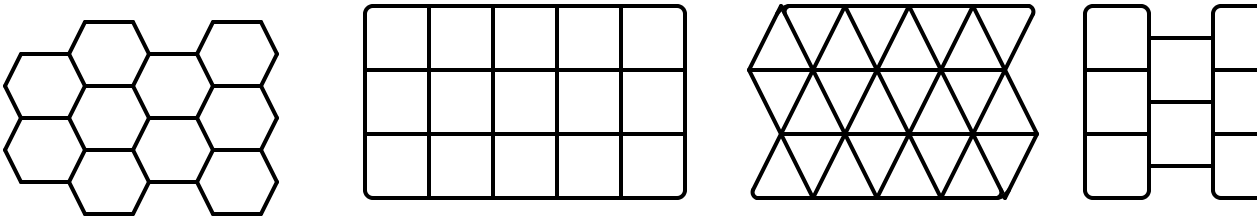
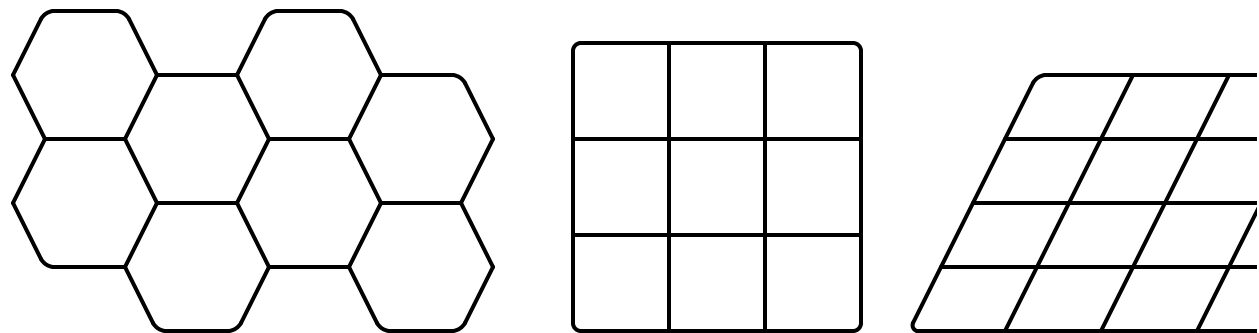
➡ Even unicode box drawing characters are supported



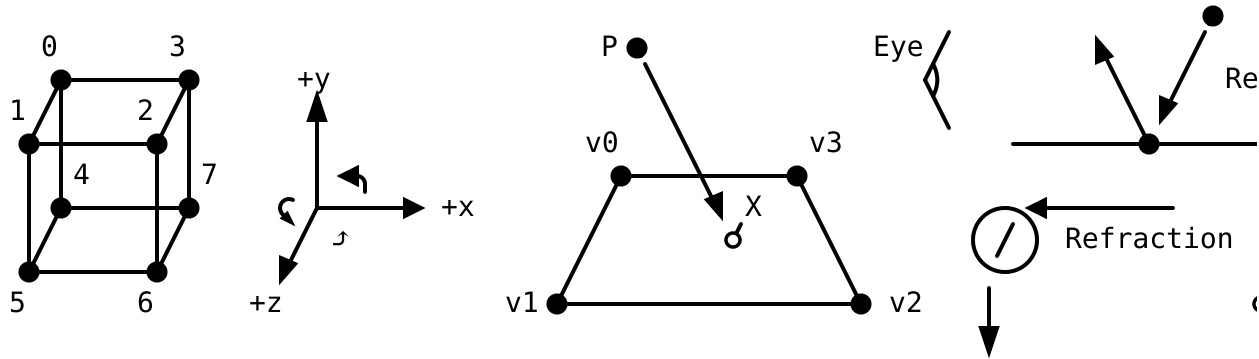
➡ Circle, quarter arcs, half circles, 3/4 quarter arcs



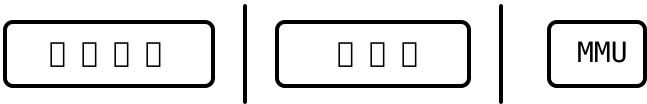
➡ Grids



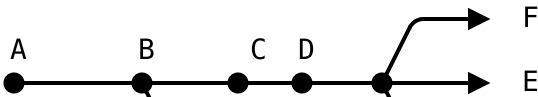
➤ Graphics Diagram

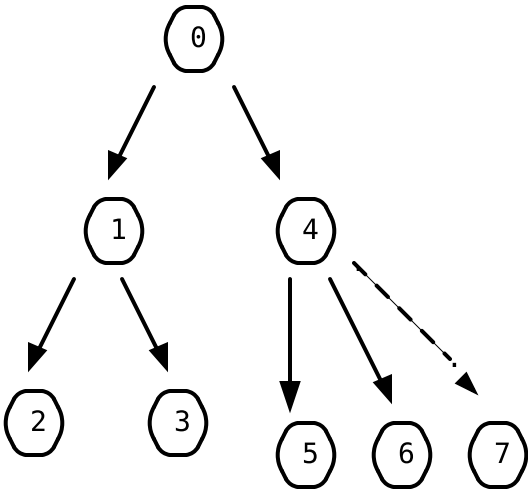
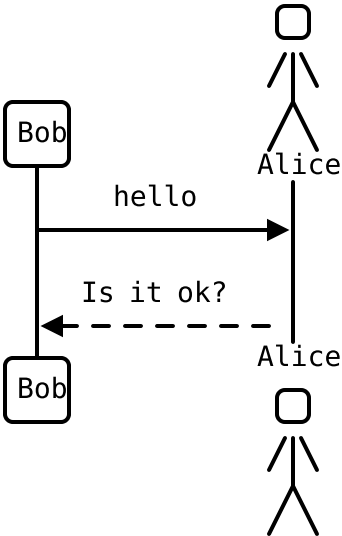
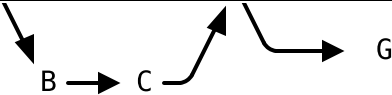


➤ CJK characters

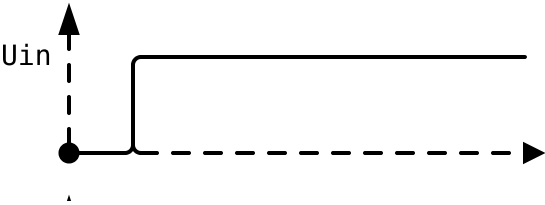


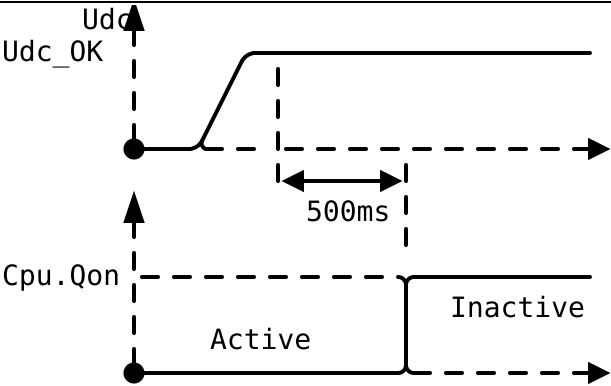
➤ Sequence Diagrams



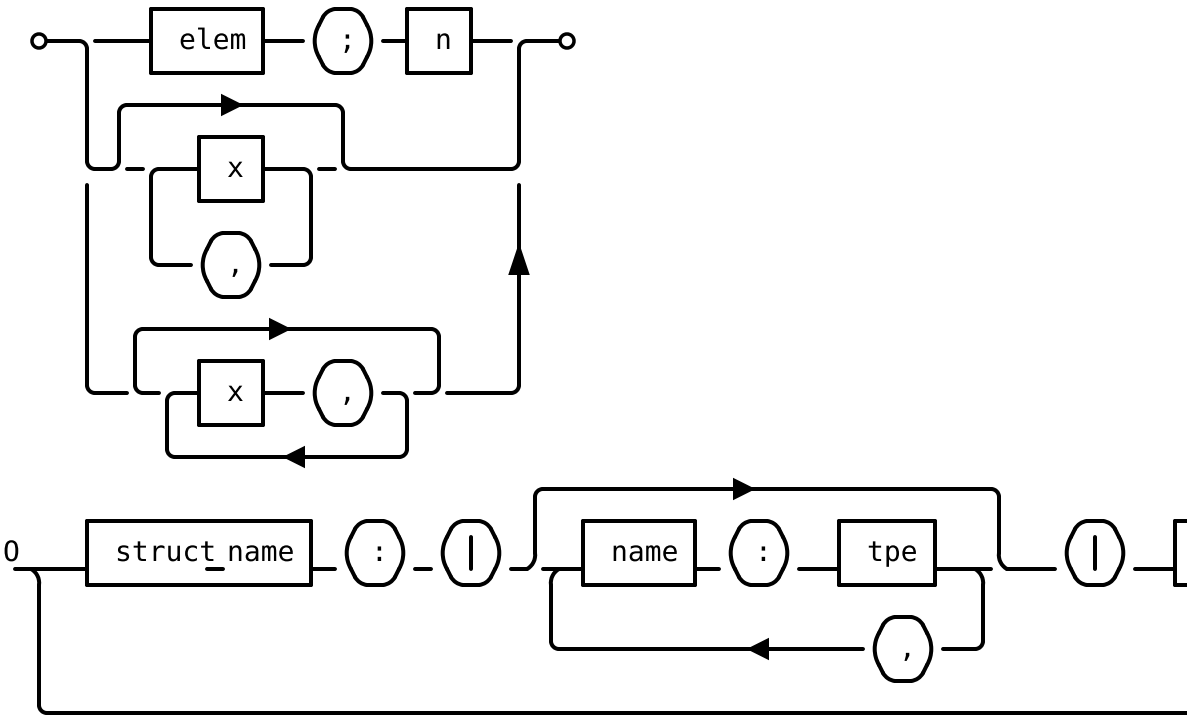


➡ Plot diagrams

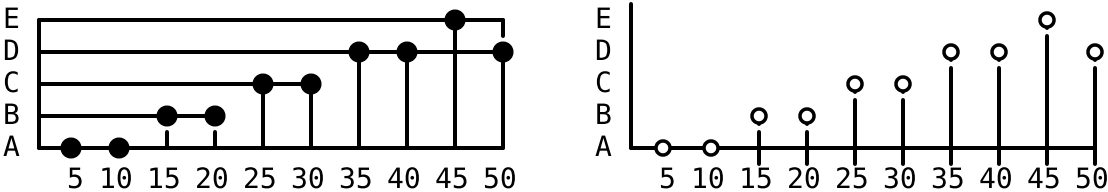


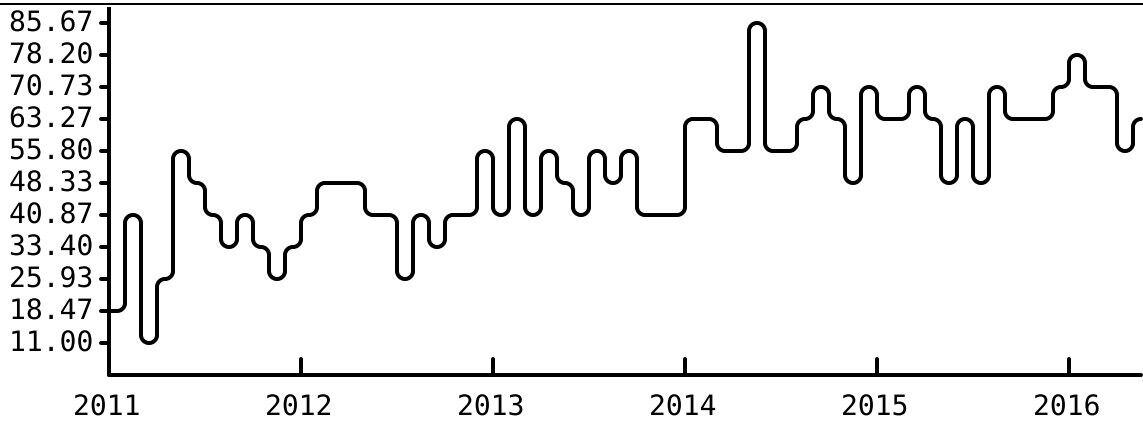


➤ Railroad diagrams

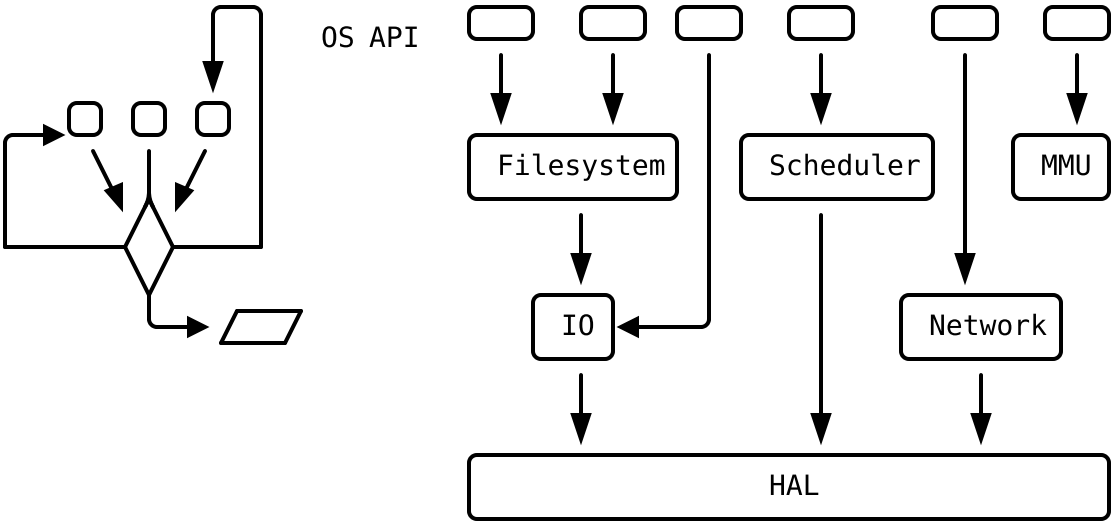


➤ Statistical charts

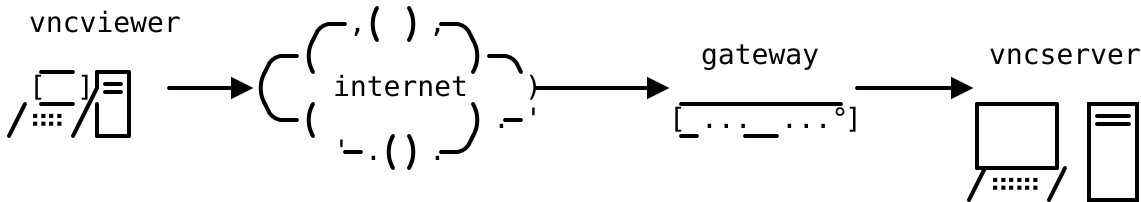


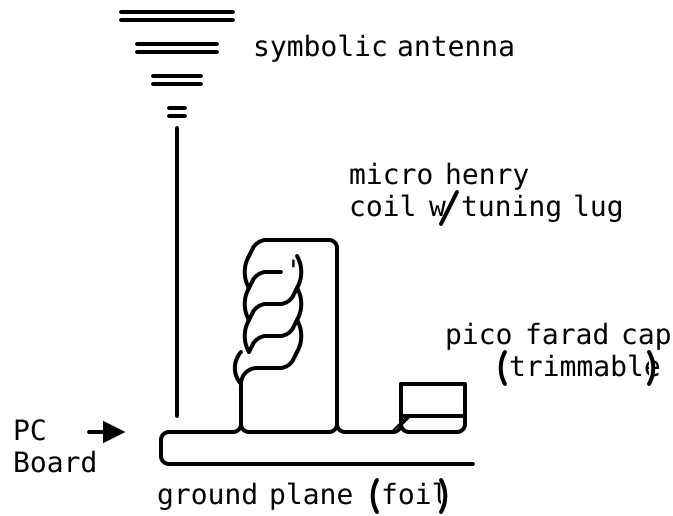
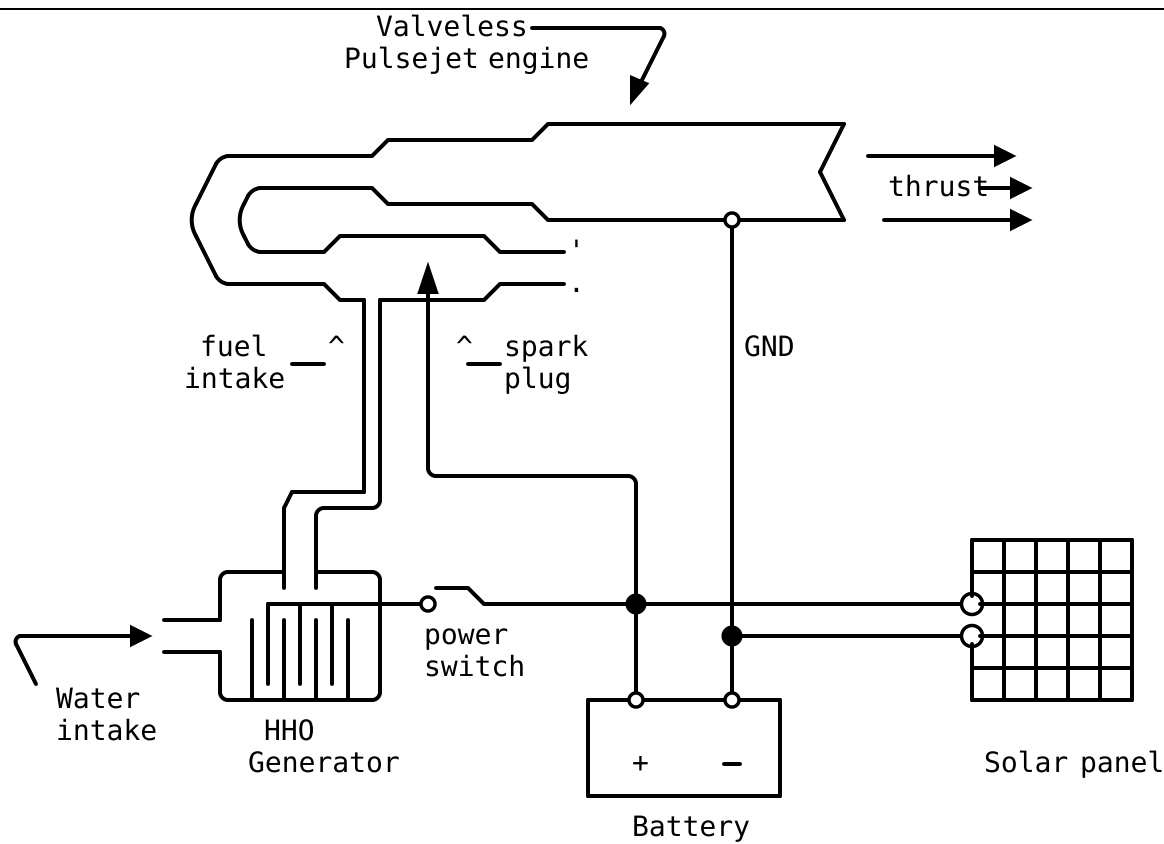


➤ Flow charts



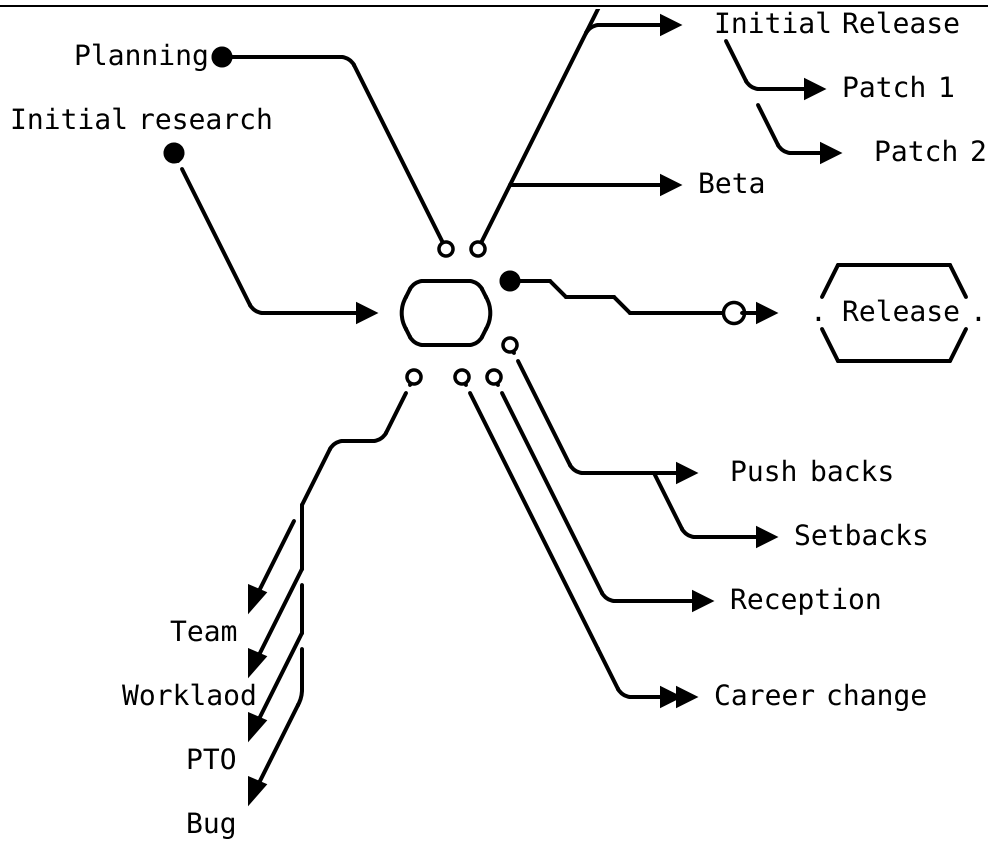
➤ Block diagrams



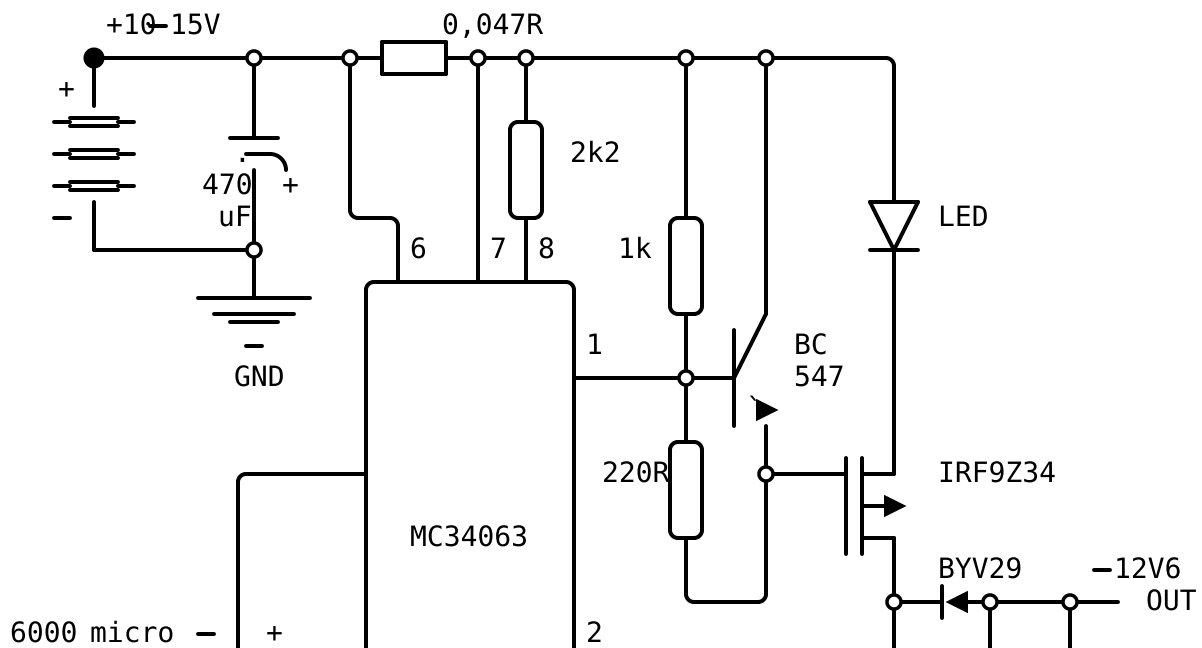


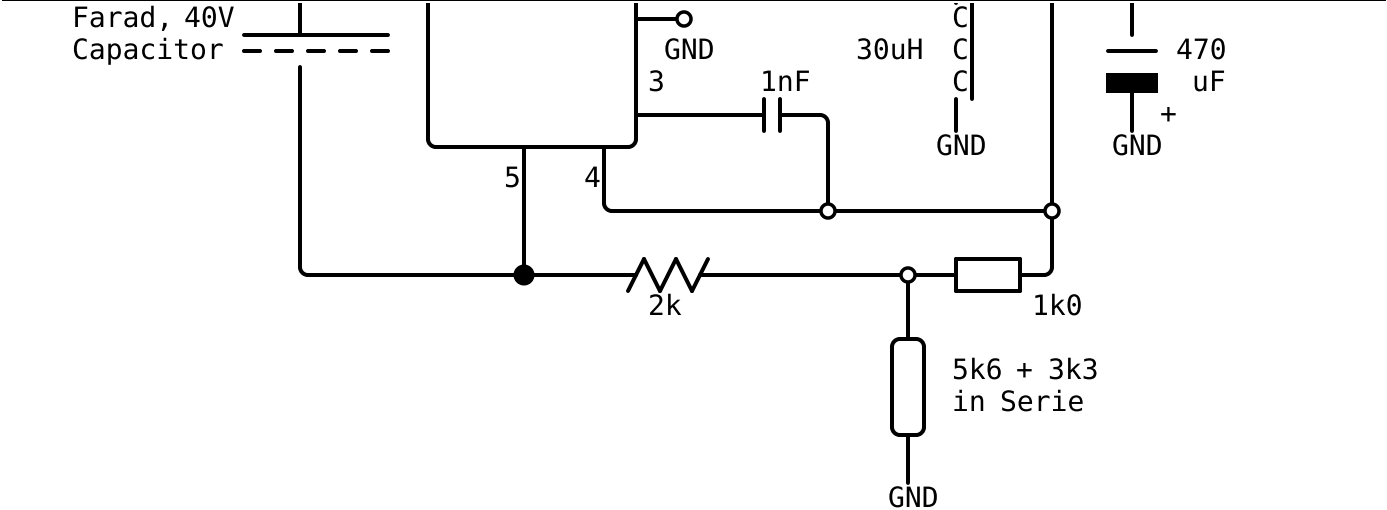
➡ Mindmaps

➡ Alpha

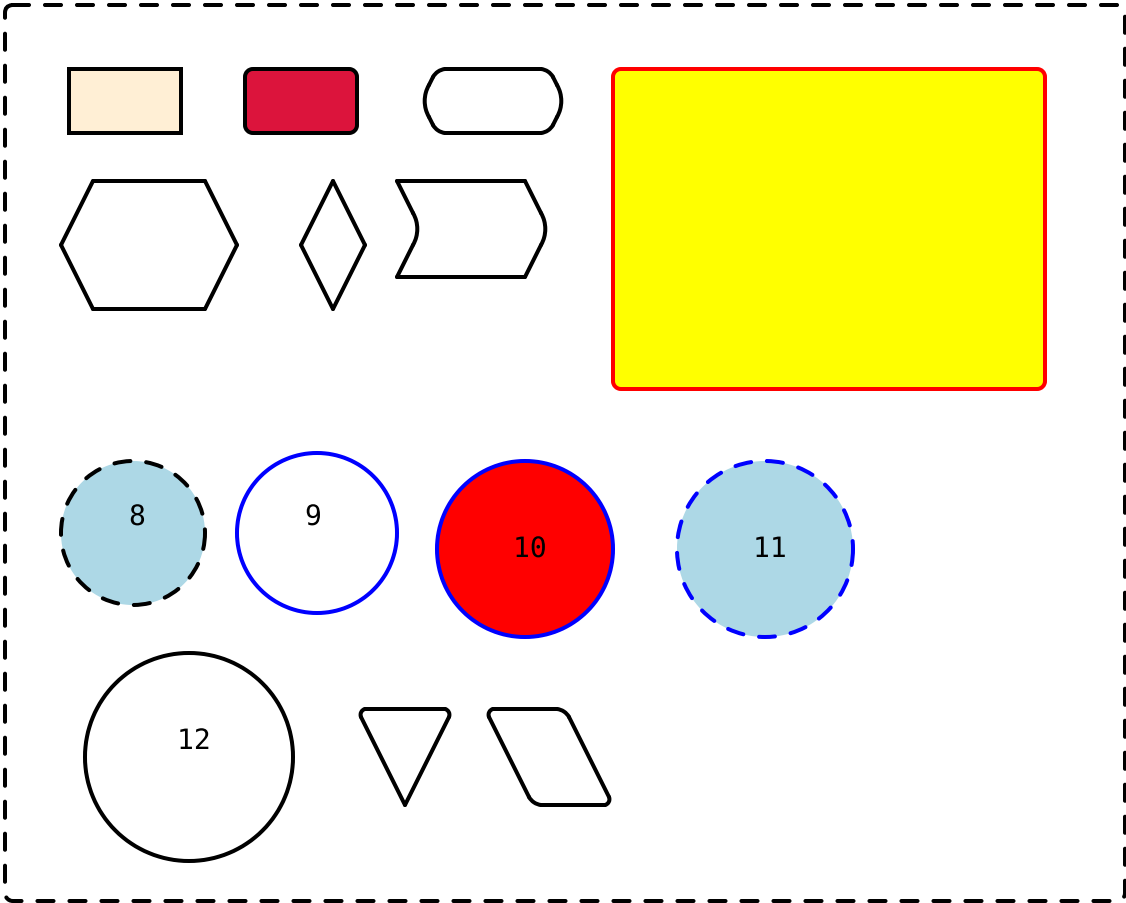


➡ It can do complex stuff such as circuit diagrams





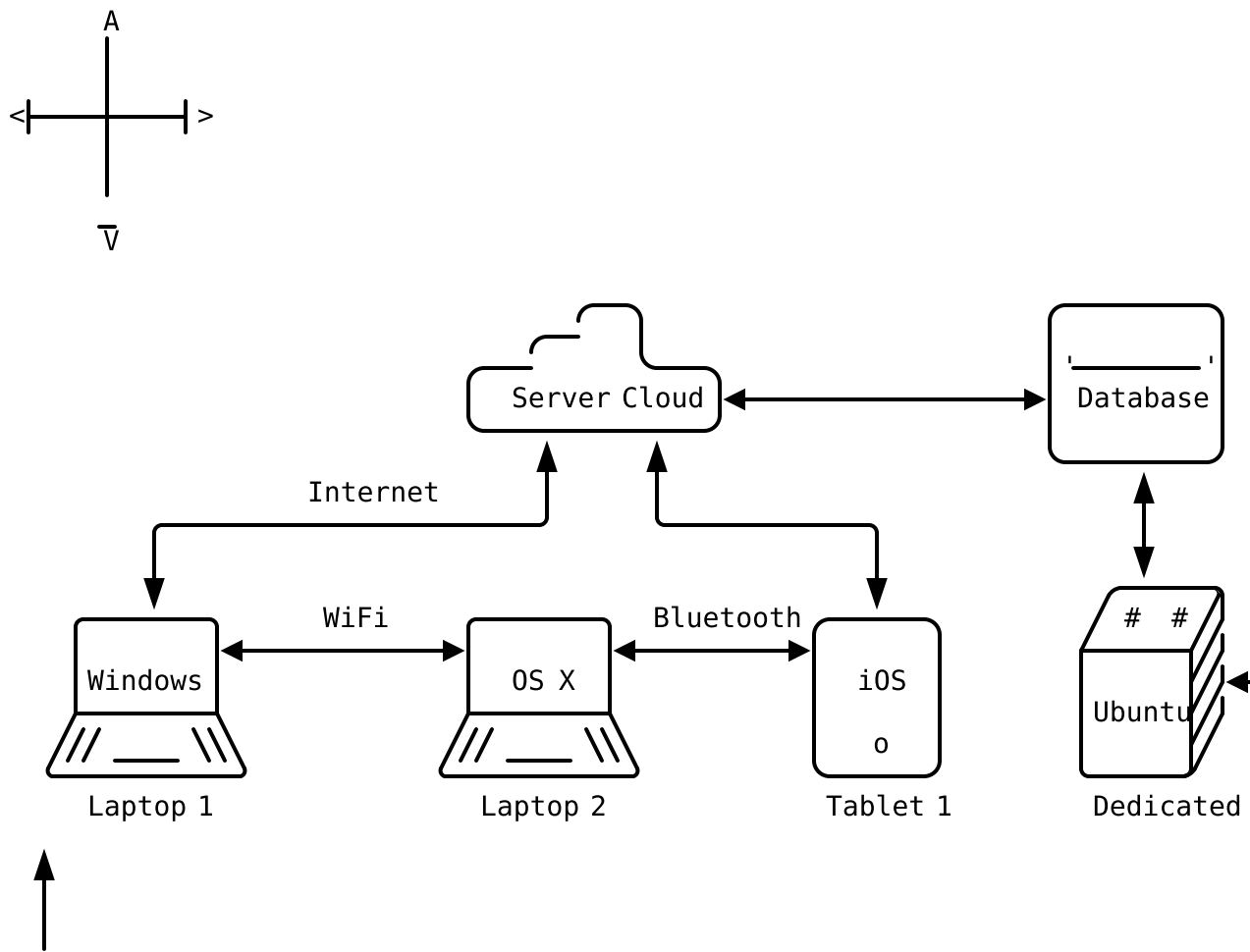
➔ Latest addition: Styling of tagged shapes



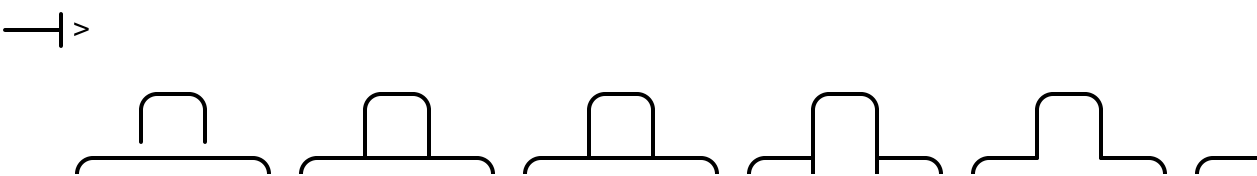
### Advantages:

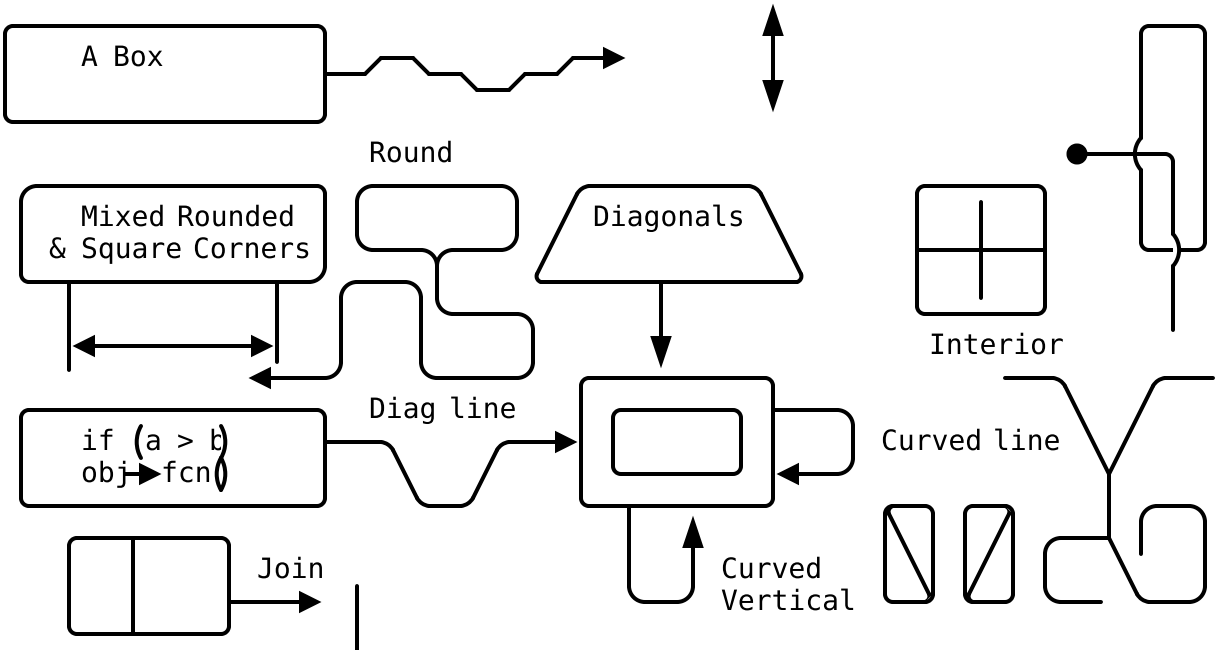
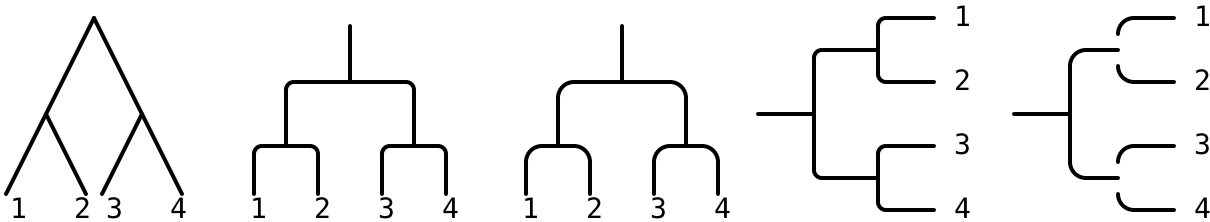
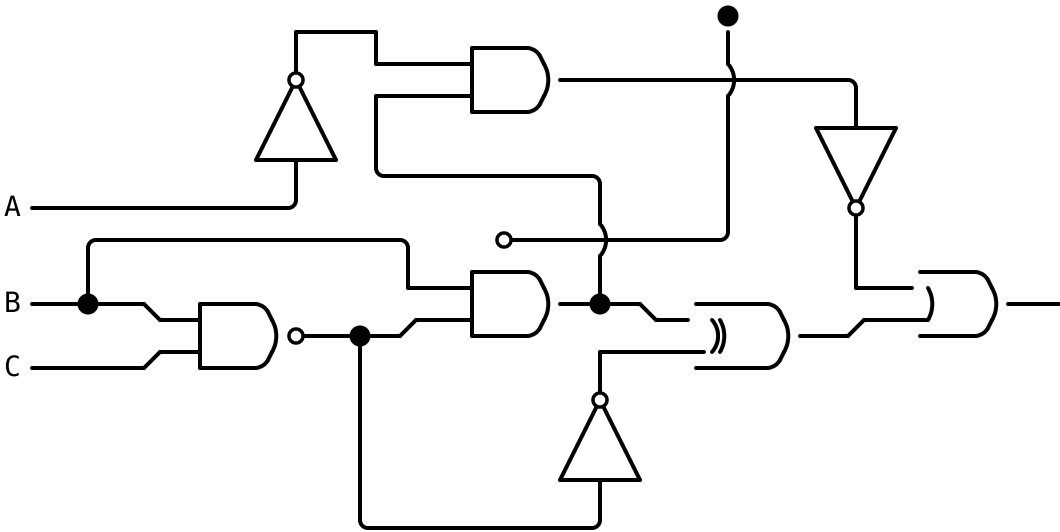
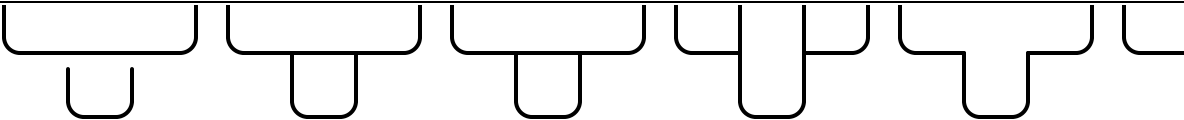
- ➊ Plain text format  
Ultimately portable, backward compatible and future proof.
- ➋ Degrades gracefully  
Even when not using a graphical renderer, it would still look good as text based diagrams. Paste the text in your source code.
- ➌ Easiest to use. Anyone knows how to edit text.

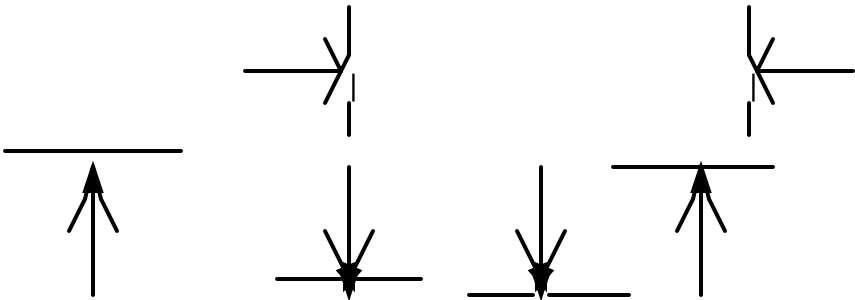
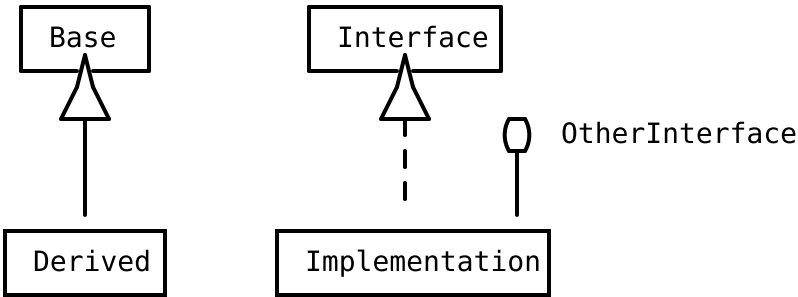
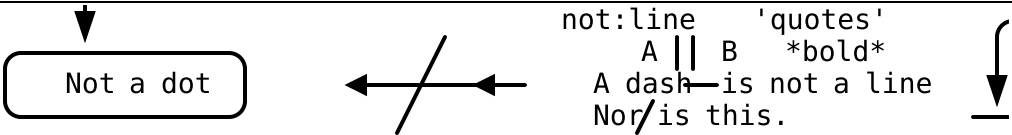
## svgbob-otherdemo



|       |       |       |     |       |         |
|-------|-------|-------|-----|-------|---------|
|       | $A_1$ | $A_2$ | ... | $A_j$ | ← Basis |
| $X_i$ | 16    | 4     | ... | 9     |         |
|       | 1     | -2    | ... | 10    |         |
|       | 8     | 52    | ... | 0     |         |
|       | 14    | 0     | ... | -1    |         |

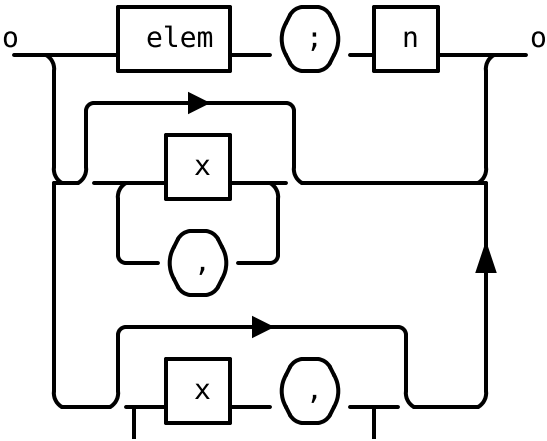


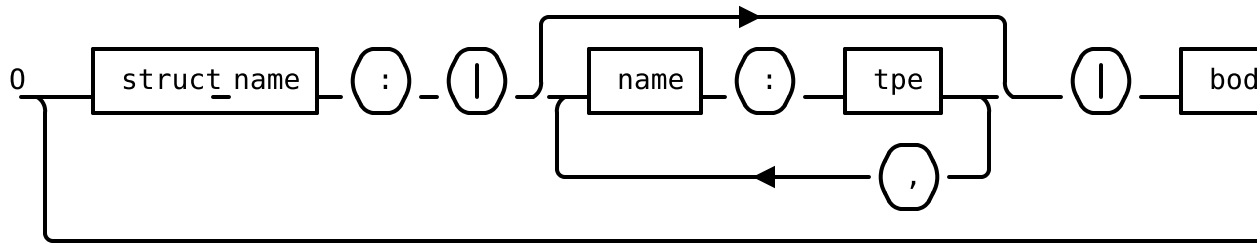




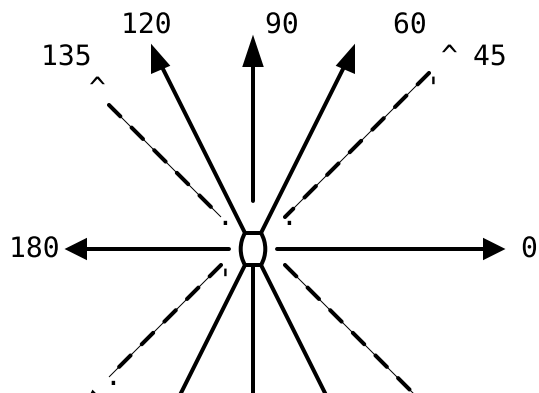
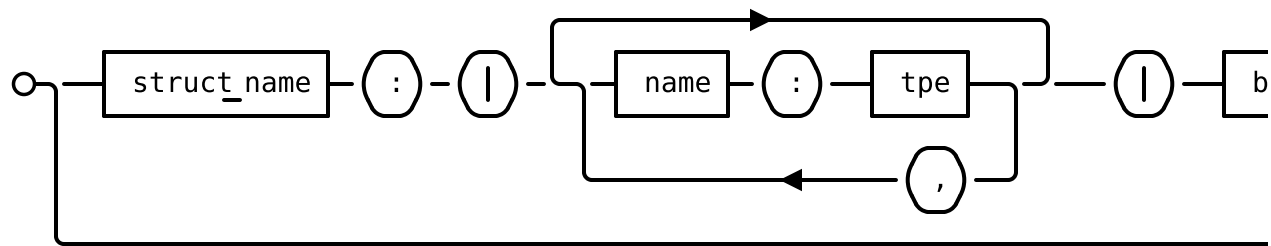
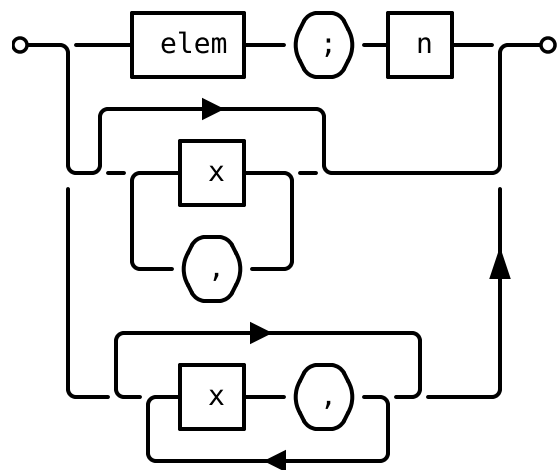
Poor man's rail road

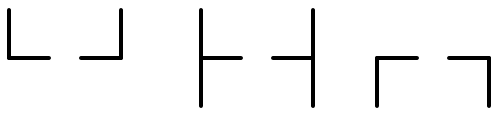
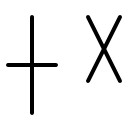
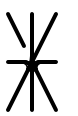
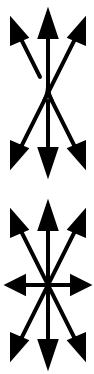
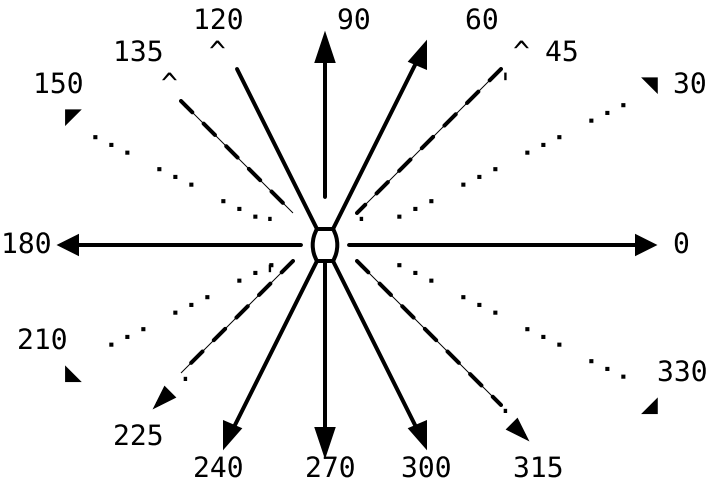
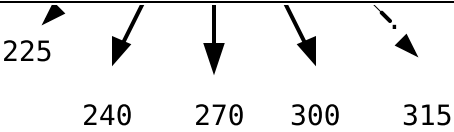
Style #1





Style #2





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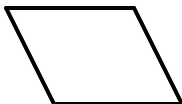
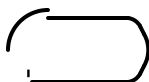
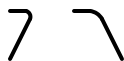
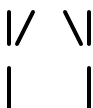
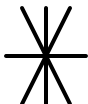
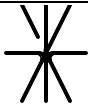
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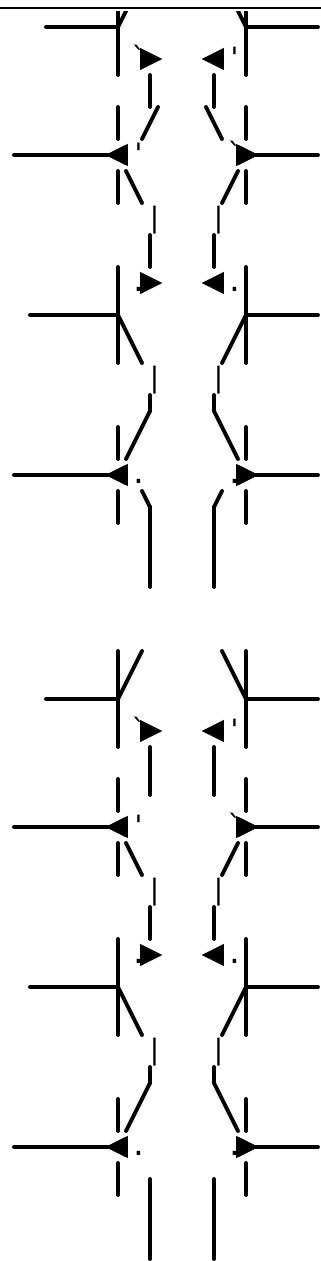
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...





# ECharts

## echarts-multi-function

linear

quadraticIn

quadraticOut

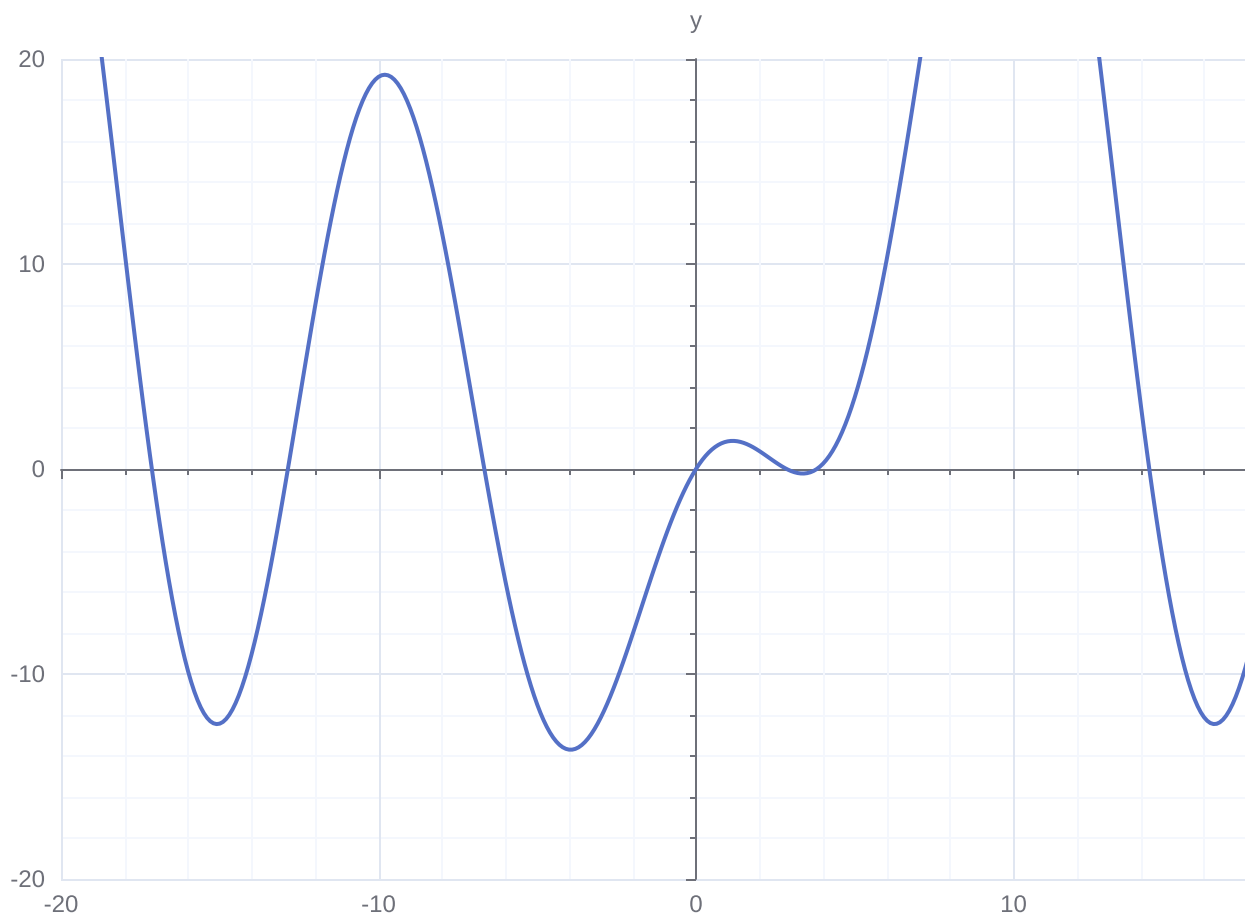
quadraticInOut

cubicIn

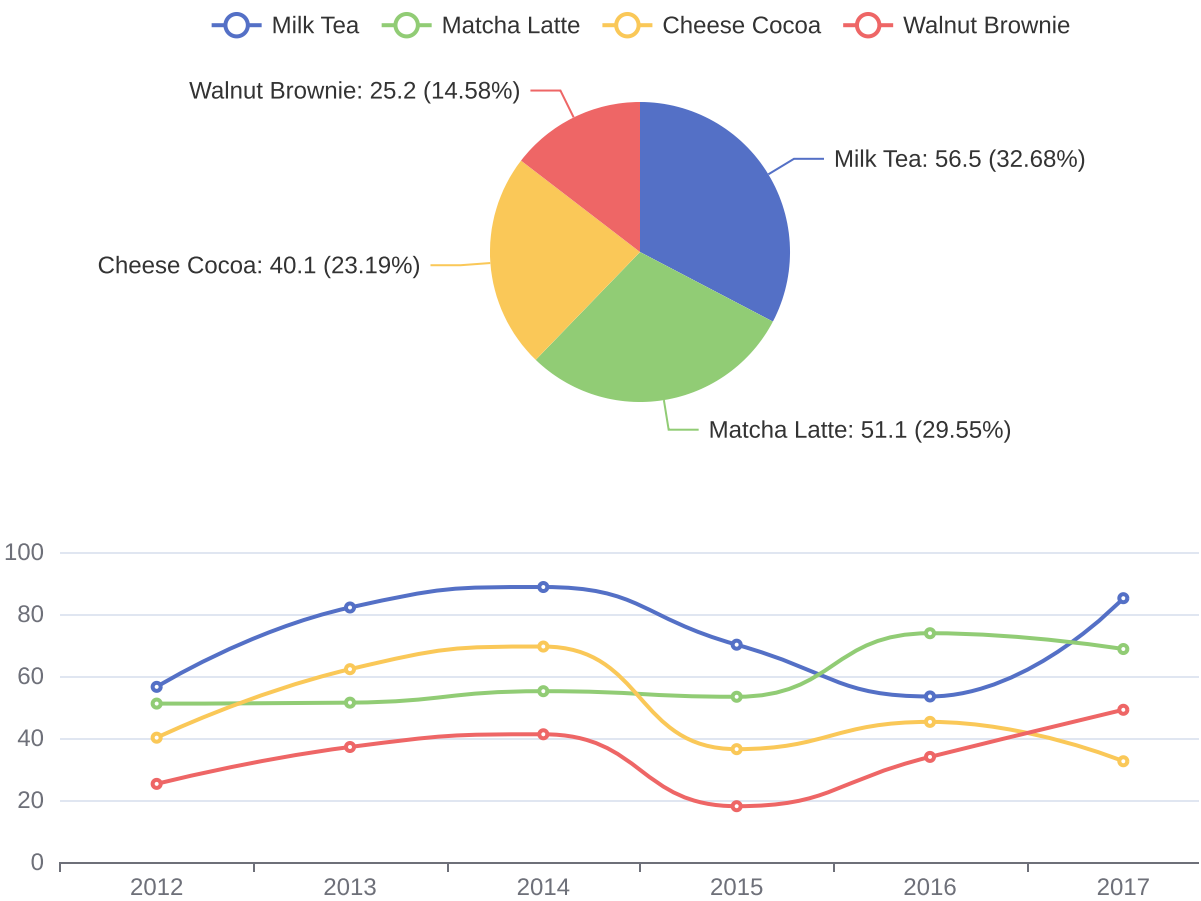


Different Easing Functions

echarts-function

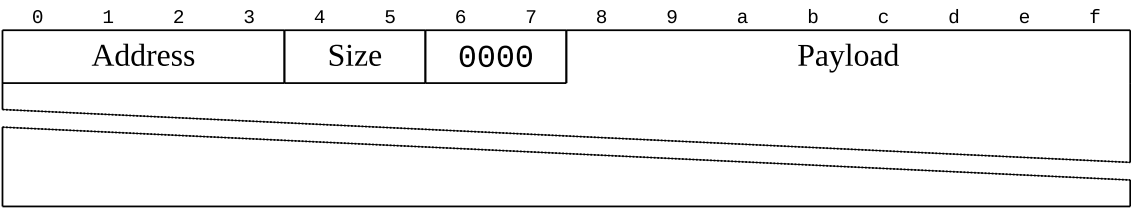


echarts-dataset

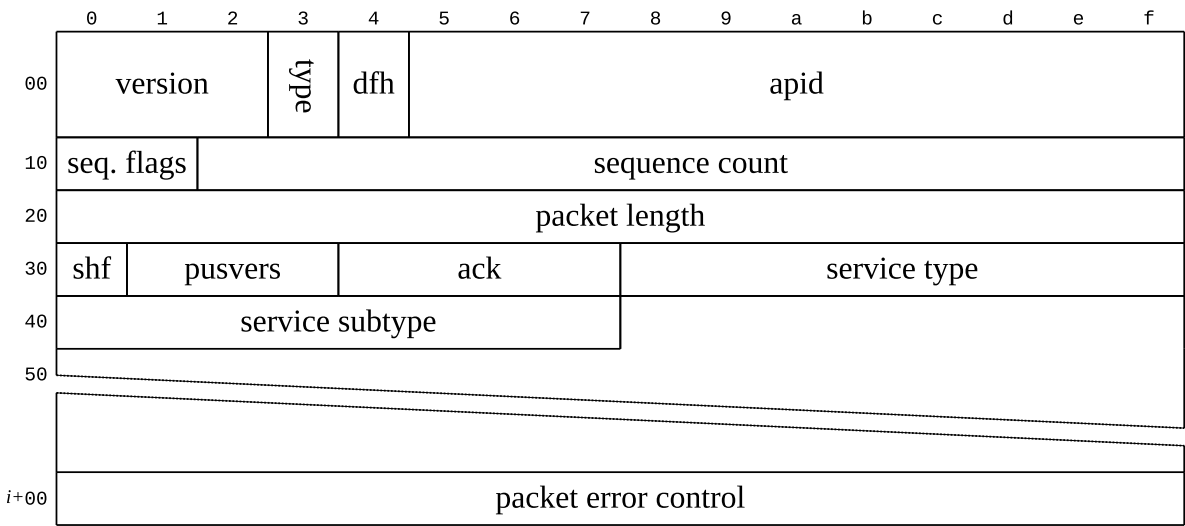


# Bytefield

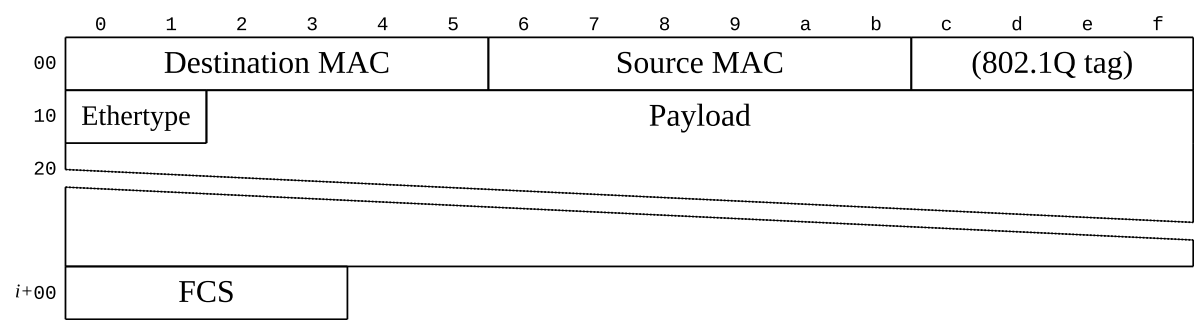
## Basic.



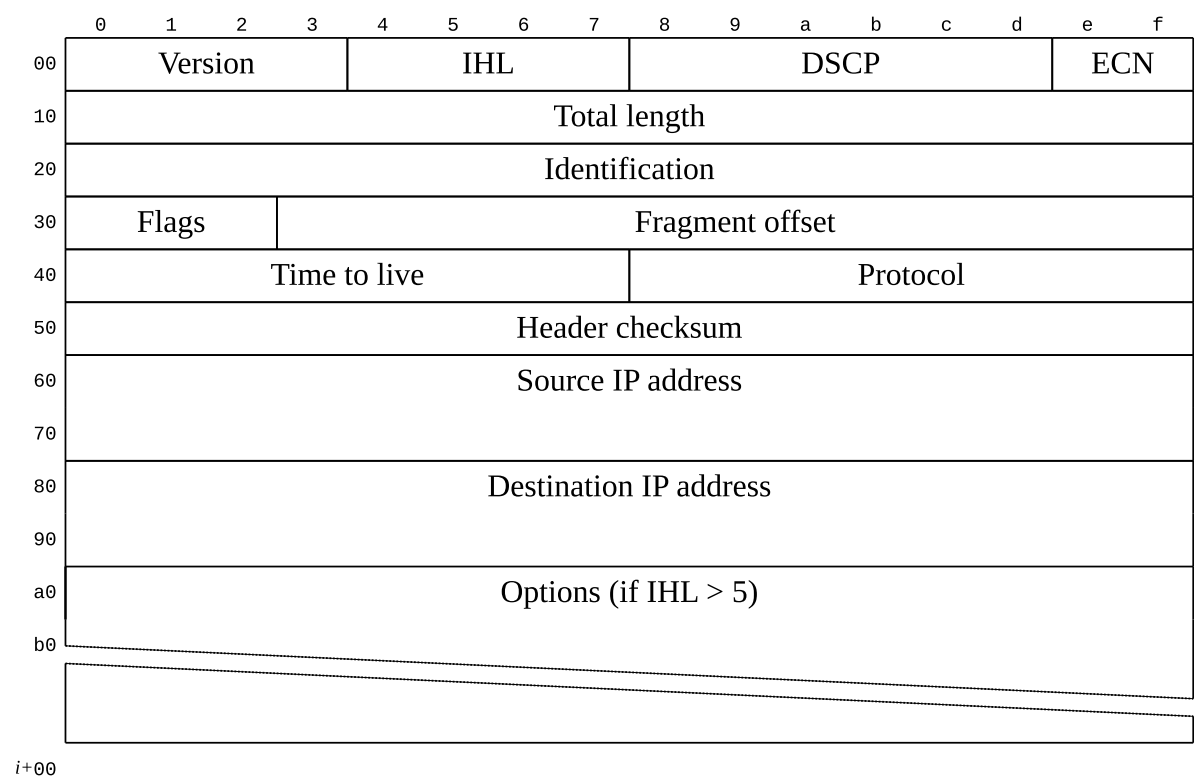
## ccsds.edn



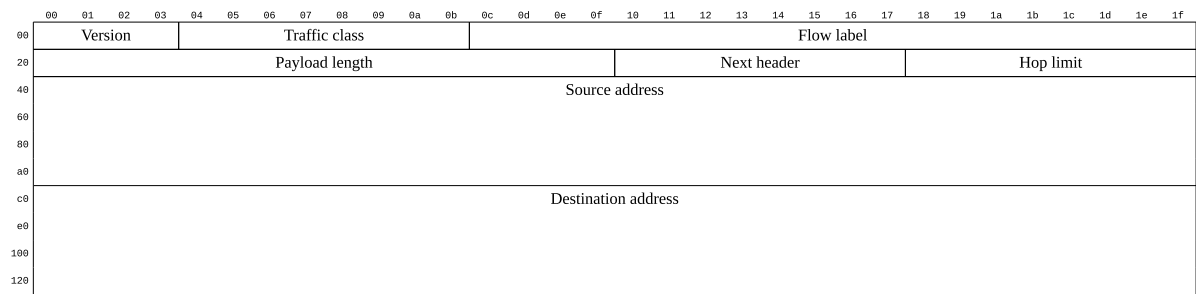
# ethernet.edn



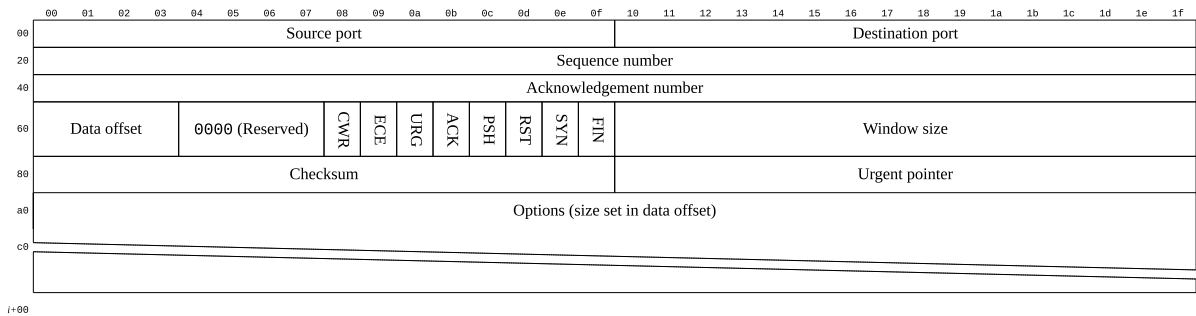
# ipv4.edn



# ipv6.edn



# tcp.edn



udp.edn

|             |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 00          | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 0a | 0b | 0c | 0d | 0e | 0f | 10               | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 1a | 1b | 1c | 1d | 1e | 1f |
| Source port |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | Destination port |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Length      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | Checksum         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

test.edn

|      | 0                   | 1                   | 2        | 3    | 4                        | 5  | 6                   | 7                  | 8  | 9    | a             | b                   | c                  | d  | e  | f    |  |
|------|---------------------|---------------------|----------|------|--------------------------|----|---------------------|--------------------|----|------|---------------|---------------------|--------------------|----|----|------|--|
|      | start               |                     |          | TxID |                          |    |                     |                    |    | type |               | args                |                    |    |    | tags |  |
| 00   | 11                  | 872349ae            |          |      |                          | 11 | TxID                |                    |    |      | 10            | 4702                |                    | 0f | 09 | 14   |  |
| 10   | 0000000c (12)       |                     |          |      | 06                       | 06 | 06                  | 03                 | 06 | 06   | 06            | 06                  | 03                 | 00 | 00 | 00   |  |
| 20   | 11                  | 00002104            |          |      |                          | 11 | 00000000            |                    |    |      | 11            | length <sub>1</sub> |                    |    |    | 14   |  |
| 30   | length <sub>1</sub> |                     |          |      | Cue and loop point bytes |    |                     |                    |    |      |               |                     |                    |    |    |      |  |
| 40   |                     |                     |          |      |                          |    |                     |                    |    |      |               |                     |                    |    |    |      |  |
|      |                     |                     |          |      |                          |    |                     |                    |    |      |               |                     |                    |    |    |      |  |
| i+00 |                     | 11                  | 00000036 |      |                          |    | 11                  | num <sub>hot</sub> |    |      |               | 11                  | num <sub>cue</sub> |    |    |      |  |
| i+10 | 11                  | length <sub>2</sub> |          |      |                          | 14 | length <sub>2</sub> |                    |    |      | Unknown bytes |                     |                    |    |    |      |  |
| i+20 |                     |                     |          |      |                          |    |                     |                    |    |      |               |                     |                    |    |    |      |  |
|      |                     |                     |          |      |                          |    |                     |                    |    |      |               |                     |                    |    |    |      |  |
| i+00 |                     |                     |          |      |                          |    |                     |                    |    |      |               |                     |                    |    |    |      |  |

# Latex Pictures



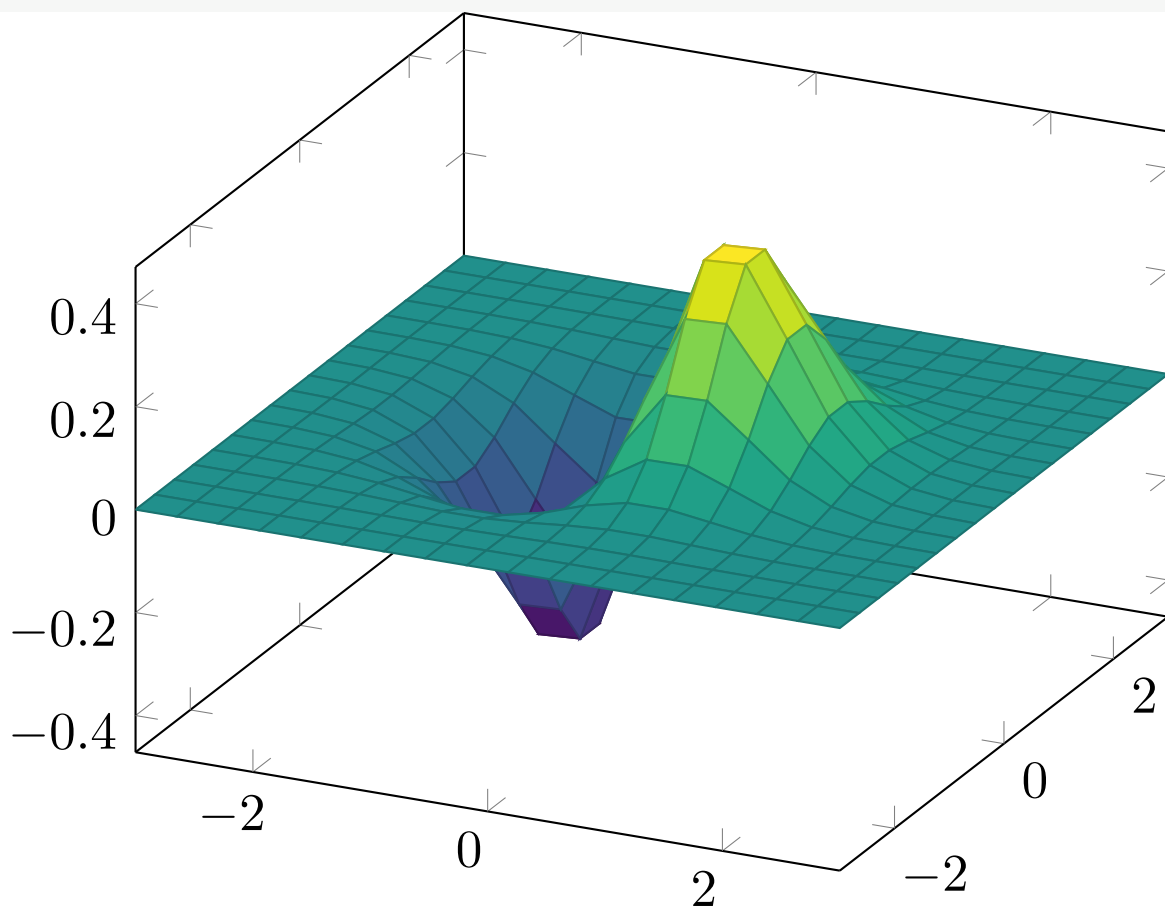
```
# node-tikzjax
chemfig
tikz-cd
circuitikz
pgfplots
array
amsmath
amstext
amsfonts
amssymb
tikz-3dplot
```

## tikz demo1

▼ demo1\_0.tex



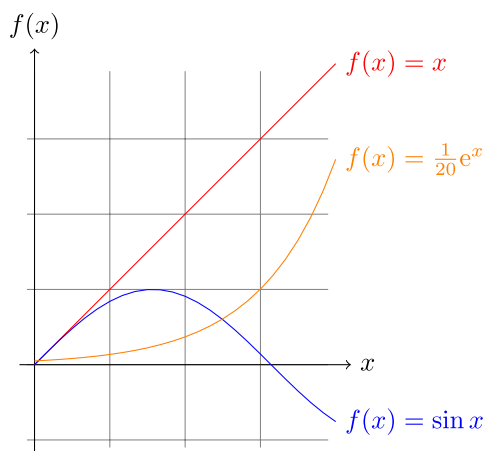
```
% demo1
\documentclass[tikz]{standalone}
\usepackage{pgfplots}
\pgfplotsset{compat=1.16}
\begin{document}
\begin{tikzpicture}[scale=2]
\begin{axis}[colormap/viridis]
\addplot3[
    surf,
    samples=18,
    domain=-3:3
]
{exp(-x^2-y^2)*x};
\end{axis}
\end{tikzpicture}
\end{document}
```



## tikz demo2

### ▼ demo2\_1.tex

```
% demo2
\documentclass[tikz]{standalone}
\begin{document}
\begin{tikzpicture}[domain=0:4]
\draw[very thin,color=gray] (-0.1,-1.1) grid (3.9,3.9);
\draw[->] (-0.2,0) -- (4.2,0) node[right] {$x$};
\draw[->] (0,-1.2) -- (0,4.2) node[above] {$f(x)$};
\draw[color=red] plot (\x,\x) node[right] {$f(x) = x$};
\draw[color=blue] plot (\x,{sin(\x r)}) node[right] {$f(x) = \sin x$};
\draw[color=orange] plot (\x,{0.05*exp(\x)}) node[right] {$f(x) = \frac{1}{20} e^x$};
\end{tikzpicture}
\end{document}
```

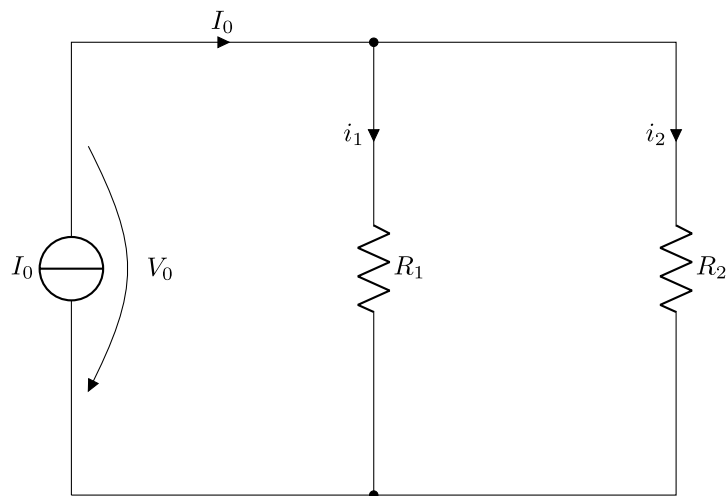


## tikz demo3

### ▼ demo3\_2.tex



```
% demo3
\documentclass[margin=10pt]{standalone}
\usepackage[siunitx]{circuitikz}
\begin{document}
\begin{tikzpicture}[scale=2]
\draw (0,0)
to[isource, l=$I_0$, v=$V_0$] (0,3)
to[short, -*, i=$I_0$] (2,3)
to[R=$R_1$, i>_=$i_1$] (2,0) -- (0,0);
\draw (2,3) -- (4,3)
to[R=$R_2$, i>_=$i_2$]
(4,0) to[short, -*] (2,0);
\end{tikzpicture}
\end{document}
```



## tikz demo4

▼ demo4\_3.tex



```

% demo4
% Poincaré Diagram: Classification of Phase Portraits in the  $(\det A, \text{Tr } A)$ -plane
% Author: Gernot Salzer, 22 Jan 2017
\documentclass[tikz,border=10pt]{standalone}
\usetikzlibrary{decorations.markings}
\tikzset
{
  every pin/.style = {pin edge = {<-}},      % pins are arrows from label to point
  > = stealth,                                % arrow tips look like stealth bombers
  flow/.style =      % everything marked as "flow" will be decorated with an arrow
  {
    decoration = {markings, mark=at position #1 with {\arrow{>}}},
    postaction = {decorate}
  },
  flow/.default = 0.5,                        % default position of the arrow is in the middle
  main/.style = {line width=1pt}              % thick lines for main graph
}
% \newtemplate[Scaling, default 0.18]{\NameOfTemplate}{Caption}{Code}
%
% Typesets Code and stores it in the box \NameOfTemplate.
% This way we avoid nested tikzpictures when inserting the templates into the
% main picture, since nesting is not guaranteed to work.
\newcommand\newtemplate[4][0.18]%
{
  \newsavebox#2%
  \savebox#2%
  {
    \begin{tabular}{@{}c@{}}
      \begin{tikzpicture}[scale=#1]
        #4
      \end{tikzpicture}
    \end{tabular}
    \templatecaption{#3}
  }
  \%
}
\newcommand\template[1]{\usebox{#1}}          % use the Code stored in box #1
\newcommand\templatecaption[1]{\sffamily\scriptsize#1} % typeset caption
\newcommand\Tr{\mathop{\mathrm{Tr}}}

```

```

\newtemplate\sink{sink}%
{\foreach \sx in {+,-} % for right/left half do:
  {\draw[flow] (\sx4,0) -- (0,0); % draw half of horizontal axis
  \draw[flow] (0,\sx4) -- (0,0); % draw half of vertical axis
  \foreach \sy in {+,-} % for upper/lower quadrant do:
    \foreach \a/\b in {2/1,3/0.44} % draw two half-parabolas
      \draw[flow,domain=\sx\a:0] plot (\x, {\sy\b*\x*\x});
  }
}

\newtemplate\source{source}%
{\foreach \sx in {+,-} % for right/left half do:
  {\draw[flow] (0,0) -- (\sx4,0); % draw half of horizontal axis
  \draw[flow] (0,0) -- (0,\sx4); % draw half of vertical axis
  \foreach \sy in {+,-} % for upper/lower quadrant do:
    \foreach \a/\b in {2/1,3/0.44} % draw two half-parabolas
      \draw[flow,domain=0:\sx\a] plot (\x, {\sy\b*\x*\x});
  }
}

\newtemplate\stablefp{line of stable fixed points}%
{\draw (-4,0) -- (4,0); % draw horizontal axis
\foreach \sy in {+,-} % for upper/lower half do:
  {\draw[flow] (0,\sy4) -- (0,0); % draw half of vertical axis
  \foreach \x in {-3,-2,-1,1,2,3} % draw six vertical half-lines
    \draw[flow] (\x,\sy3) -- (\x,0);
  }
}

\newtemplate\unstablefp{line of unstable fixed points}%
{\draw (-4,0) -- (4,0); % draw horizontal axis
\foreach \sy in {+,-} % for upper/lower half do:
  {\draw[flow] (0,0) -- (0,\sy4); % draw half of vertical axis
  \foreach \x in {-3,-2,-1,1,2,3} % draw six vertical half-lines
    \draw[flow] (\x,0) -- (\x,\sy3);
  }
}

\newtemplate\spiralsink{spiral sink}%
{\draw (-4,0) -- (4,0); % draw horizontal axis
\draw (0,-4) -- (0,4); % draw vertical axis
\draw [samples=100,smooth,domain=27:7] % draw spiral
  plot ({\x r}:{0.005*\x*\x}); % Using "flow" here gives
"Dimension
  \def\x{26} % too large", so we draw
a tiny
  \draw[->] ({\x r}:{0.005*\x*\x}) -- +(0.01,-0.01);% tangent for the
arrow.
}

```

```

\newtemplate\spiralsource{spiral source}%
{\draw (-4,0) -- (4,0);           % draw horizontal axis
 \draw (0,-4) -- (0,4);           % draw vertical axis
 \draw [samples=100,smooth,domain=10:28] % draw spiral
      plot ({-\x r}:{0.005*\x*\x}); % Using "flow" here gives
"Dimension
 \def\x{27.5}                     % too large", so we draw
a tiny
 \draw[<-] ({-\x r}:{0.005*\x*\x}) -- +(0.01,-0.008);% tangent for the
arrow.
}
\newtemplate[0.15]\centre{center}% British spelling since \center is in
use
{\draw (-4,0) -- (4,0);           % draw horizontal axis
 \draw (0,-4) -- (0,4);           % draw vertical axis
 \foreach \r in {1,2,3}           % draw three circles
      \draw[flow=0.63] (\r,0) arc (0:-360:\r cm);
}
\newtemplate\saddle{saddle}%
{\foreach \sx in {+,-}           % for right/left half do:
 {\draw[flow] (\sx4,0) -- (0,0); % draw half of horizontal axis
 \draw[flow] (0,0) -- (0,\sx4); % draw half of vertical axis
 \foreach \sy in {+,-}           % for upper/lower quadrant do:
      \foreach \a/\b/\c/\d in {2.8/0.3/0.7/0.6, 3.9/0.4/1.3/1.1}
          \draw[flow] (\sx\a,\sy\b) % draw two bent lines
              .. controls (\sx\c,\sy\d) and (\sx\d,\sy\c)
              .. (\sx\b,\sy\a);
      }
 }
}
\newtemplate\degensink{degenerate sink}%
{\draw (0,-4) -- (0,4);           % draw vertical axis
 \foreach \s in {+,-}           % for upper/lower half do:
 {\draw[flow] (\s4,0) -- (0,0); % draw half of horizontal axis
 \foreach \a/\b/\c/\d in {3.5/4/1.5/1, 2.5/2/1/0.8}
      \draw[flow] (\s-3.5,\s\a) % draw two bent lines
          .. controls (\s\b,\s\c) and (\s\b,\s\d)
          .. (0,0);
      }
 }
}
\newtemplate\degensource{degenerate source}%
{\draw (0,-4) -- (0,4);           % draw vertical axis
 \foreach \s in {+,-}           % for upper/lower half do:
 {\draw[flow] (0,0) -- (\s4,0); % draw half of horizontal axis
 \foreach \a/\b/\c/\d in {3.5/4/1.5/1, 2.5/2/1/0.8}
      \draw[flow] (0,0) % draw two bent lines

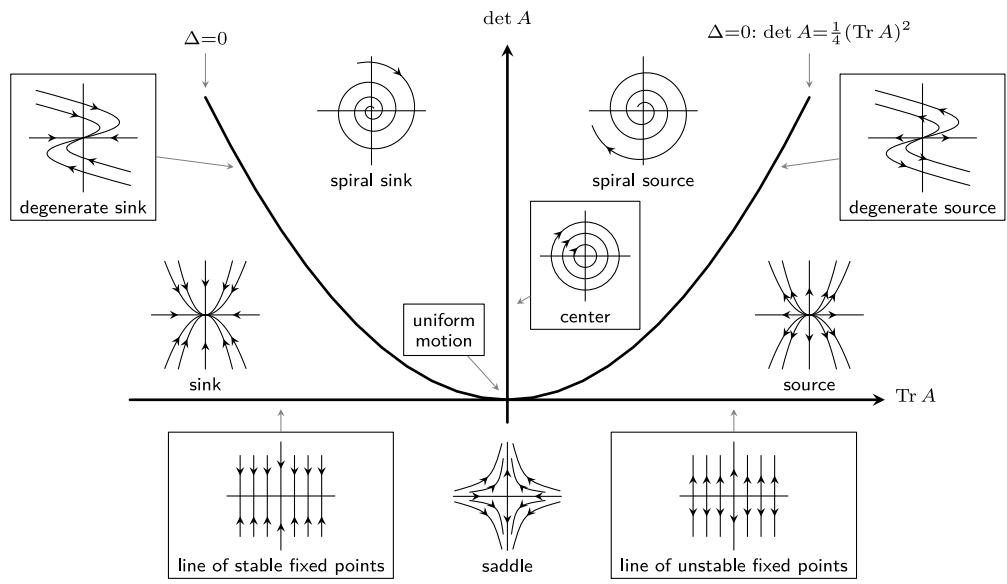
```

```

    .. controls (\s\b,\s\d) and (\s\b,\s\c)
    .. (\s-3.5,\s\ a);
  }
}
\begin{document}
\begin{tikzpicture}[line cap=round,line join=round]
  % MAIN DIAGRAM
  \draw [main,->] (0,-0.3) -- (0,4.7) %
vertical axis
  node [label={[above]\scriptstyle\det A$}] {}
  node [label={[above,yshift=0.8cm]%
    {\sffamily\large Poincar'\e Diagram: Classification of Phase
    Portraits in the $(\det A,\Tr A)$-plane}}] {}
  \draw [main,->] (-5,0) -- (5,0) %
horizontal axis
  node [label={[right,yshift=-0.5ex]\scriptstyle\Tr A$}] {};
  \draw [main, domain=-4:4] plot (\x, {0.25*\x*\x}); % main
graph
  \node at (-4,4) [pin={[above]\scriptstyle\Delta=0$}] {};
  \node at ( 4,4) [pin={[above,align=left]%
    {\scriptstyle\Delta=0:\;\det A=\frac{1}{4}(\Tr A)^2$}}] {};
  % TEMPLATES describing areas
  \node at ( 0 , -1.4) {\template\saddle};
  \node at (-4 , 1 ) {\template\sink};
  \node at ( 4 , 1 ) {\template\source};
  \node at (-1.8, 3.7) {\template\spiralsink};
  \node at ( 1.8, 3.7) {\template\spiralsource};
  % TEMPLATES labeling lines and points
  \node at ( 0 , 1.2) [pin={[draw,right,xshift=0.3cm]%
    \template\centre}] {};
  \node at (-3 , 0 ) [pin={[draw,below,yshift=-1cm]%
    \template\stablefp}] {};
  \node at ( 3 , 0 ) [pin={[draw,below,yshift=-1cm]%
    \template\unstablefp}] {};
  \node at (-3.5,{0.25*3.5*3.5}) [pin=
  {[draw,left,xshift=-1.15cm,yshift=-0.3cm]%
    \template\degensink}] {};
  \node at ( 3.5,{0.25*3.5*3.5}) [pin=
  {[draw,right,xshift=0.9cm,yshift=-0.3cm]%
    \template\degensource}] {};
  \node at ( 0 , 0 ) [pin={[draw,above left,align=center,xshift=-0.3cm]%
    \templatecaption{uniform}\[-1ex]\templatecaption{motion}}] {};
\end{tikzpicture}
\end{document}

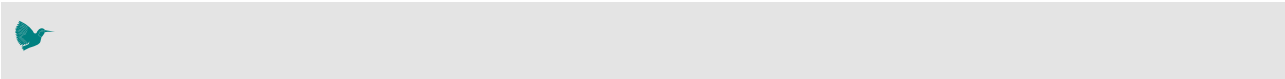
```

Poincaré Diagram: Classification of Phase Portraits in the  $(\det A, \text{Tr } A)$ -plane



tikz demo5

▼ demo5\_4.tex



```

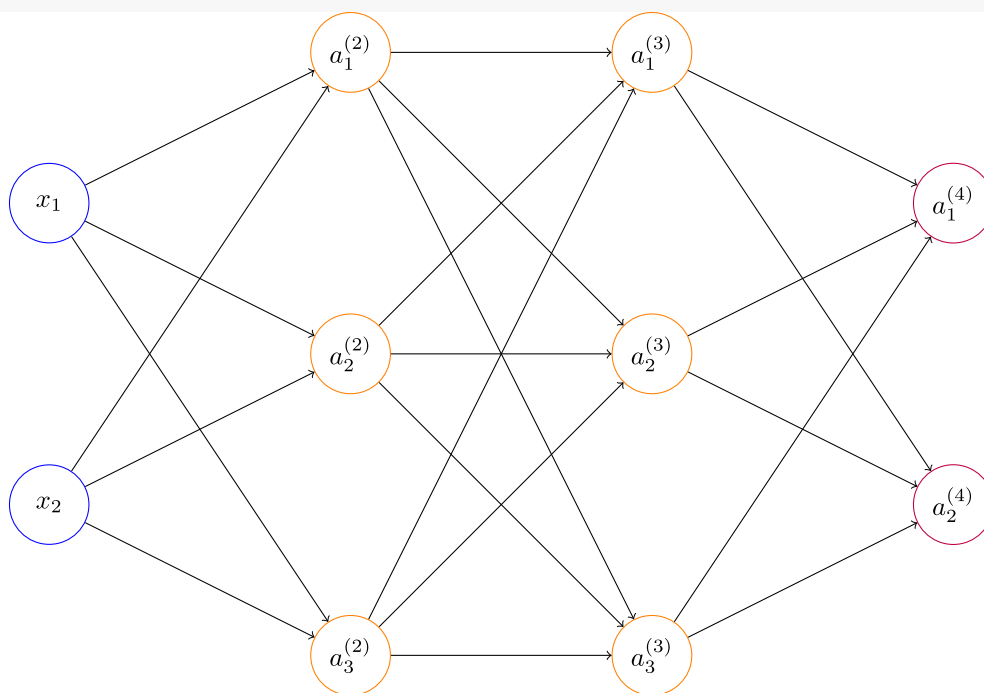
% demo5
\documentclass[tikz]{standalone}
\begin{document}
\begin{tikzpicture}[scale=2]
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=blue] (1) at(0,2){ $x_1$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=blue] (2) at(0,0){ $x_2$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (3) at(2,-1){ $a_3^{(2)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (4) at(2,1){ $a_2^{(2)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (5) at(2,3){ $a_1^{(2)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (6) at(4,-1){ $a_3^{(3)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (7) at(4,1){ $a_2^{(3)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (8) at(4,3){ $a_1^{(3)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=purple] (9) at(6,2){ $a_1^{(4)}$ };
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=purple] (10) at(6,0){ $a_2^{(4)}$ };
\draw[->] (1) --(3);
\draw[->] (1) --(4);
\draw[->] (1) --(5);
\draw[->] (2) --(3);
\draw[->] (2) --(4);
\draw[->] (2) --(5);
\draw[->] (3) --(6);
\draw[->] (3) --(7);
\draw[->] (3) --(8);

```

```

\draw[->] (4) --(6);
\draw[->] (4) --(7);
\draw[->] (4) --(8);
\draw[->] (5) --(6);
\draw[->] (5) --(7);
\draw[->] (5) --(8);
\draw[->] (6) --(9);
\draw[->] (6) --(10);
\draw[->] (7) --(9);
\draw[->] (7) --(10);
\draw[->] (8) --(9);
\draw[->] (8) --(10);
\end{tikzpicture}
\end{document}

```



## tikz demo6

▼ demo6\_5.tex



```

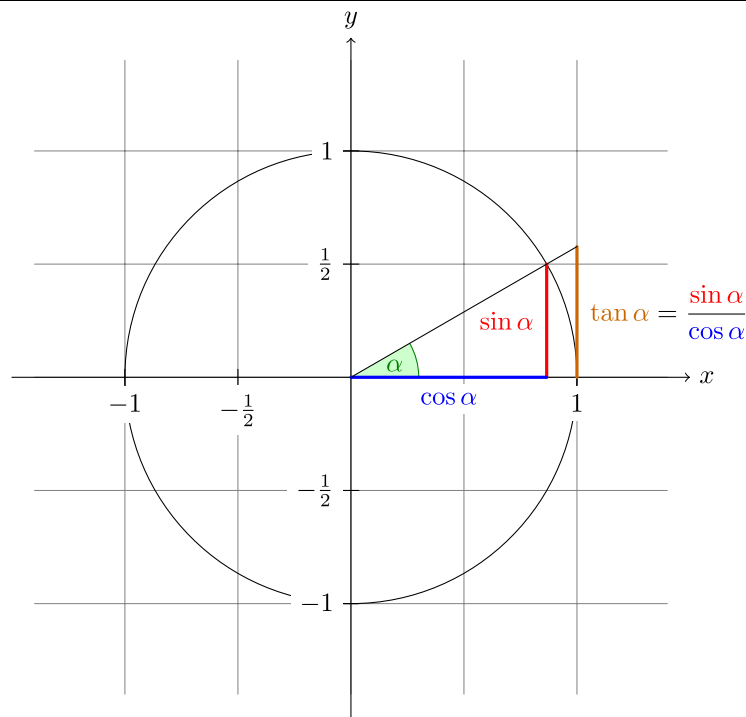
% demo6
% Author: Till Tantau
% Source: The PGF/TikZ manual
\documentclass[tikz]{standalone}
\begin{document}
\pagestyle{empty}
\begin{tikzpicture}[scale=3,cap=round]
% Local definitions
\def\costhirty{0.8660256}
% Colors
\colorlet{anglecolor}{green!50!black}
\colorlet{sincolor}{red}
\colorlet{tancolor}{orange!80!black}
\colorlet{coscolor}{blue}
% Styles
\tikzstyle{axes}=[]
\tikzstyle{important line}=[very thick]
\tikzstyle{information text}=[rounded corners,fill=red!10,inner sep=1ex]
% The graphic
\draw[style=help lines,step=0.5cm] (-1.4,-1.4) grid (1.4,1.4);
\draw (0,0) circle (1cm);
\begin{scope}[style=axes]
\draw[->] (-1.5,0) -- (1.5,0) node[right] {$x$};
\draw[->] (0,-1.5) -- (0,1.5) node[above] {$y$};
\foreach \x/\xtext in {-1, -.5/-\frac{1}{2}, 1}
\draw[xshift=\x cm] (0pt,1pt) -- (0pt,-1pt) node[below,fill=white]
{$\xtext$};
\foreach \y/\ytext in {-1, -.5/-\frac{1}{2}, .5/\frac{1}{2}, 1}
\draw[yshift=\y cm] (1pt,0pt) -- (-1pt,0pt) node[left,fill=white]
{$\ytext$};
\end{scope}
\filldraw[fill=green!20,draw=anglecolor] (0,0) -- (3mm,0pt) arc(0:30:3mm);
\draw (15:2mm) node[anglecolor] {$\alpha$};
\draw[style=important line,sincolor]
(30:1cm) -- node[left=1pt,fill=white] {$\sin \alpha$} +(0,-.5);
\draw[style=important line,coscolor]
(0,0) -- node[below=2pt,fill=white] {$\cos \alpha$} (\costhirty,0);
\draw[style=important line,tancolor] (1,0) --
node [right=1pt,fill=white]
{
$\displaystyle \tan \alpha \color{black}=$
\frac{\color{sincolor}\sin \alpha}{\color{coscolor}\cos \alpha}$
} (intersection of 0,0--30:1cm and 1,0--1,1) coordinate (t);

```

```

\draw (0,0) -- (t);
\draw[xshift=1.85cm] node [right,text width=6cm,style=information text]
{
  The {\color{anglecolor} angle  $\alpha$ } is  $30^\circ$  in the
  example ( $\pi/6$  in radians). The {\color{sincolor}sine} of
     $\alpha$ , which is the height of the red line, is
  \[
    {\color{sincolor} \sin \alpha} = 1/2.
  \]
  By the Theorem of Pythagoras we have  ${\color{coscolor} \cos^2 \alpha}$ 
+
     ${\color{sincolor} \sin^2 \alpha} = 1$ . Thus the length of the blue
    line, which is the {\color{coscolor}cosine} of  $\alpha$ , must be
  \[
    {\color{coscolor} \cos \alpha} = \sqrt{1 - 1/4} = \textstyle
    \frac{1}{2} \sqrt{3}.
  \]
  This shows that {\color{tancolor}  $\tan \alpha$ }, which is the
  height of the orange line, is
  \[
    {\color{tancolor} \tan \alpha} = \frac{{\color{sincolor} \sin
      \alpha}}{{\color{coscolor} \cos \alpha}} = 1/\sqrt{3}.
  \]
};
\end{tikzpicture}
\end{document}

```



The **angle**  $\alpha$  is  $30^\circ$  in the example ( $\pi/6$  in radians). The **sine of**  $\alpha$ , which is the height of the red line, is

$$\sin \alpha = 1/2.$$

By the Theorem of Pythagoras we have  $\cos^2 \alpha + \sin^2 \alpha = 1$ . Thus the length of the blue line, which is the **cosine of**  $\alpha$ , must be

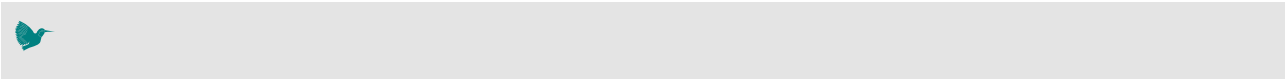
$$\cos \alpha = \sqrt{1 - 1/4} = \frac{1}{2}\sqrt{3}.$$

This shows that **tan**  $\alpha$ , which is the height of the orange line, is

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = 1/\sqrt{3}.$$

# tikz demo7

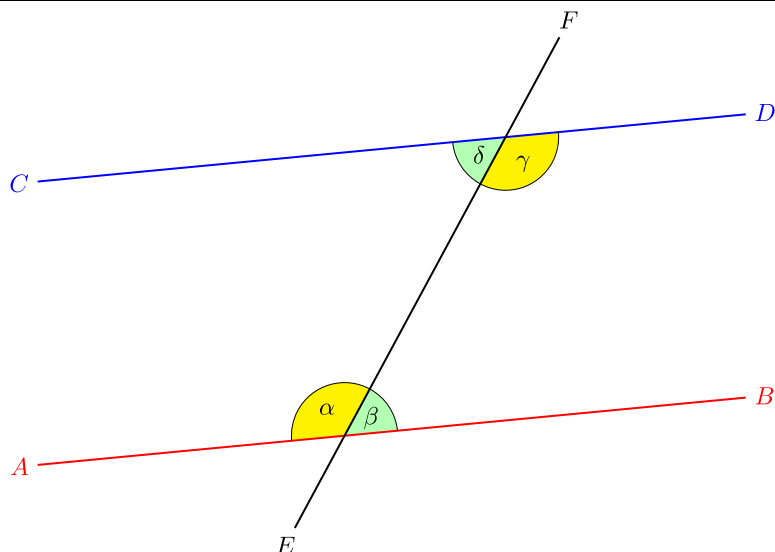
▼ demo7\_6.tex



```

% demo7
\documentclass[tikz]{standalone}
\usetikzlibrary {angles,calc,quotes}
\begin{document}
\begin{tikzpicture}[angle radius=.75cm, scale=2]
\node (A) at (-2,0) [red,left] {$A$};
\node (B) at ( 3,.5) [red,right] {$B$};
\node (C) at (-2,2) [blue,left] {$C$};
\node (D) at ( 3,2.5) [blue,right] {$D$};
\node (E) at (60:-5mm) [below] {$E$};
\node (F) at (60:3.5cm) [above] {$F$};
\coordinate (X) at (intersection cs:first line={(A)--(B)}, second line=
{(E)--(F)});
\coordinate (Y) at (intersection cs:first line={(C)--(D)}, second line=
{(E)--(F)});
\path
  (A) edge [red, thick] (B)
  (C) edge [blue, thick] (D)
  (E) edge [thick] (F)
  pic ["$\alpha$", draw, fill=yellow] {angle = F--X--A}
  pic ["$\beta$", draw, fill=green!30] {angle = B--X--F}
  pic ["$\gamma$", draw, fill=yellow] {angle = E--Y--D}
  pic ["$\delta$", draw, fill=green!30] {angle = C--Y--E};
\node at ($ (D)!.5!(B) $) [right=1cm,text width=6cm,rounded
corners,fill=red!20,inner sep=1ex]
{
  When we assume that $\color{red}AB$ and $\color{blue}CD$ are
  parallel, i.\,e., $\color{red}AB \mathbin{\parallel} \color{blue}CD$,
  then $\alpha = \gamma$ and $\beta = \delta$.
};
\end{tikzpicture}
\end{document}

```



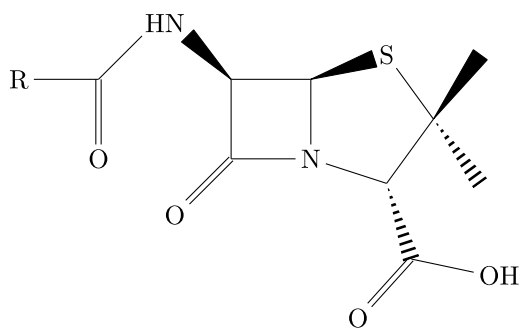
When we assume that  $AB$  and  $CD$  are parallel, i. e.,  $AB \parallel CD$ , then  $\alpha = \gamma$  and  $\beta = \delta$ .

## tikz demo8

### ▼ demo8\_7.tex

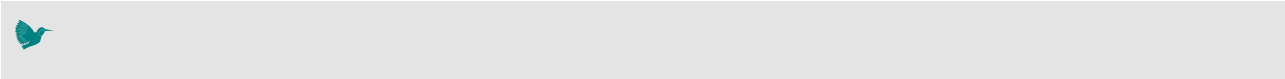


```
% demo8
\documentclass[tikz]{standalone}
\usepackage{chemfig}
\begin{document}
\chemfig{[: -90]HN(-[: -45](-[: -45]R)=[: +45]O)>[: +45]*4(-(=O)-N*5(-(<[: -60]O)-[: +60]OH)-(<[: +0])(<[: -108])-S>--)}
\end{document}
```



# tikz demo9

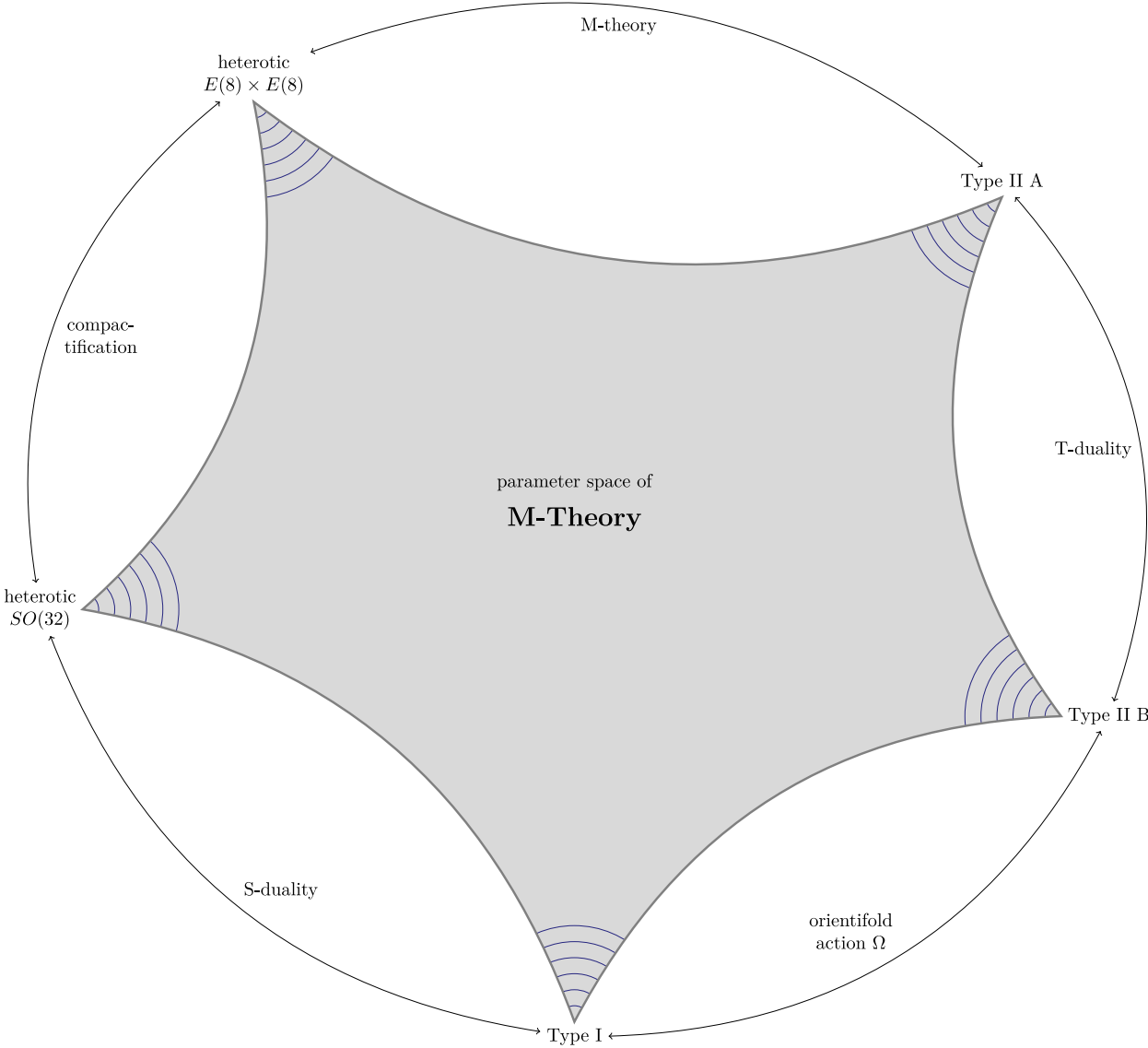
▼ samples\_8.tex



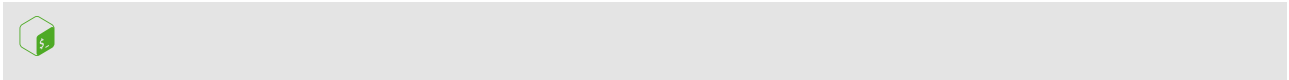
```

\documentclass[tikz]{standalone}
\definecolor{DarkBlue}{rgb}{0,0,0.5} % 深蓝色
\begin{document}
\begin{tikzpicture}[scale=2]
  \node (so32) [align=center] at (-5,-1) {heterotic\\$SO(32)$};
  \node (e8e8) [align=center] at (-3,4) {heterotic\\$E(8) \times E(8)$};
  \node (tiia) [align=center] at (4,3) {Type II A};
  \node (tiib) [align=center] at (5,-2) {Type II B};
  \node (ti) [align=center] at (0,-5) {Type I};
  \draw[bend left,<->] (so32) to node [below right,align=center] {compact-\\tification} (e8e8);
  \draw[bend left,<->] (e8e8) to node [below left] {M-theory} (tiia);
  \draw[bend left,<->] (tiia) to node [below left] {T-duality} (tiib);
  \draw[bend left,<->] (tiib) to node [above left,align=center]
{orientifold\\action $\Omega$} (ti);
  \draw[bend left,<->] (ti) to node [above right] {S-duality} (so32);
  \begin{scope}
    % 深蓝色
    \clip (so32.east) to [bend right] (e8e8.south)
      to [bend right] (tiia.south)
      to [bend right] (tiib.west)
      to [bend right] (ti.north)
      to [bend right] cycle; % 深蓝色
    \foreach \c in {so32.east,e8e8.south,tiia.south,tiib.west,ti.north}{%
      \foreach \r in {1,...,6}{%
        \draw[DarkBlue] (\c) circle (\r*0.15cm); % 深蓝色
      }
    }
  \end{scope}
  % 深蓝色
  \draw[bend right,very thick,gray,fill,fill opacity=0.3]
    (so32.east) to [bend right] (e8e8.south)
    to [bend right] (tiia.south)
    to [bend right] (tiib.west)
    to [bend right] (ti.north)
    to [bend right] cycle; % 深蓝色
  \node (mth) [align=center] at (0,0) {parameter space of\\[2ex]{\Large
\textbf{M-Theory}}}};
\end{tikzpicture}
\end{document}

```



TexNote.



```
# latex      c/c++/rust
\#           ==> \\\#
\$           ==> \\\$
\%           ==> \\%\
\&          ==> \\&
\_          ==> \\_\
\{           ==> \\{\
\}           ==> \\}\
\~{}        ==> \\~\{\}
\^{}        ==> \\^\{\}
\           ==> \\
\<          ==> \\<
\>          ==> \\>
```

# Latex Document.

## Latex demo

test in  
margin  
here

### L<sup>A</sup>T<sub>E</sub>X.js Showcase

made with ♥ by Michael Brade

2017–2021

#### Abstract

This document will show most of the features of L<sup>A</sup>T<sub>E</sub>X.js while at the same time being a gentle introduction to L<sup>A</sup>T<sub>E</sub>X. In the appendix, the API as well as the format of custom macro definitions in JavaScript will be explained.

### 1 Characters

It is possible to input any UTF-8 character either directly or by character code using one of the following:

- `\symbol{"00A9}`: ©
- `\char"A9`: ©
- `^^A9` or `~~~~00A9`: © or ©

Special characters, like those:

`$ & % # _ { } ~ ^ \`

have to be escaped. More than 200 symbols are accessible through macros. For instance: 30 °C is 86 °F. See also Section [11](#).

### 2 Spaces and Comments

Spaces and comments, of course, work just as they do in L<sup>A</sup>T<sub>E</sub>X. This is an example: Supercalifragilisticexpialidocious It does not matter whether you enter one or several spaces after a word, it will always be typeset as one space—unless you force several spaces, like now. New T<sub>E</sub>Xusers may miss whitespaces after a command. Experienced T<sub>E</sub>X users are T<sub>E</sub>Xperts, and know how to use whitespaces. Longer com-

ments can be embedded in the `comment` environment: This is another example for embedding comments in your document.

### 3 Dashes and Hyphens

L<sup>A</sup>T<sub>E</sub>X knows four kinds of dashes. Access three of them with different numbers of consecutive dashes. The fourth sign is actually not a dash at all—it is the mathematical minus sign:

daughter-in-law, X-rated  
pages 13–67  
yes—or no?  
0, 1 and −1

The names for these dashes are: ‘-’ hyphen, ‘—’ en-dash, ‘—’ em-dash, and ‘−’ minus sign. L<sup>A</sup>T<sub>E</sub>X.js outputs the actual true unicode character for those instead of using the `hyphen` and `minus`.

### 4 Text and Paragraphs, Ligatures

An empty line starts a new paragraph, and so does `\par`.

Like this. A new line usually starts automatically when the previous one is full. However, using `\newline` or `\\`, one can force

to start a new line. Ligatures are supported as well, for instance:

fi, fl, ff, ffi, ffl ... instead of fi, fl, ffl ...

Use `\/` to prevent a ligature.

#### 4.1 Multicolumns

The multi-column layout, using the `multicols` environment, allows easy definition of multiple columns of text—just like in newspapers. The first and mandatory argument specifies the number of columns the text should be divided into. It is often convenient to spread some text over all columns, just before the multicolumn

output. In L<sup>A</sup>T<sub>E</sub>X, this was needed to prevent any page break in between. To achieve this, the `multicols` environment has an optional second argument which can be used for this purpose. For instance, this text you are reading now was started with the argument `\subsection{Multicolumns}`.

### 5 Boxes

L<sup>A</sup>T<sub>E</sub>X.js supports most of the standard L<sup>A</sup>T<sub>E</sub>X boxes.

```
\mbox{text}
```

We already know one of them: it's called `\mbox`. It simply packs up a series of boxes into another one, and can be used to prevent `LATEX.js` from breaking two words. As boxes can be put inside boxes, these horizontal box packers give you ultimate flexibility.

```
\makebox[width][pos]{text}
```

```
\framebox[width][pos]{text}
```

*width* defines the width of the resulting box as seen from the outside. The *pos* parameter takes a one letter value: `center`, `flushleft`, or `flushright`. `spread` is not really working in HTML. The command `\framebox` works exactly the same as `\makebox`, but it draws a box around the text. The following example shows you some things you could do with the `\makebox` and `\framebox` commands.

```
c e n t r a l
```

```
Guess I'm framed now!
```

Bummer, I am too wide

```
never mind, I can source it this?
```

```
\parbox[pos][height][inner-pos]{width}{text}
```

The `\parbox` command produces a box the contents of which are created in paragraph mode. However, only small pieces of text should be used, paragraph-making environments shouldn't be used inside a `\parbox` argument. For larger pieces of text, including ones containing a paragraph-making environment, you should use a `\minipage` environment. By default `LATEX` will position vertically a `\parbox` so its center lines up with the center of the surrounding text line. When the optional position argument is present and equal either to `'t'` or `'b'`, this allows you respectively to align either the top or bottom line in the `\parbox` with the baseline of the surrounding text. You may also specify `'m'` for position to get the default behaviour. The optional *height* argument overrides the natural height of the box. The *inner-pos* argument controls the placement of the text inside the box, as follows; if it is not specified, *pos* is used. `t` text is placed at the top of the box. `c` text is centered in the box. `b` text is placed at the bottom of the box. `s` is not supported in HTML.

The following examples demonstrate simple positioning:

• simple alignment: Some text

parbox default alignment, parbox test text

some text

parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

• alignment with a given height: Some text

parbox default alignment, parbox test text

some text

parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

The following examples demonstrate all explicit *pos/inner-pos* combinations:

□□□

166 / 246

BSN

- center alignment: Some text

parbox default alignment, parbox test text

some text

parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

- top alignment: Some text

parbox default alignment, parbox test text

some text

parbox top alignment, text parbox test text

some text

parbox bottom alignment, text parbox test text

some text.

- bottom alignment: Some text

parbox default alignment, parbox test text

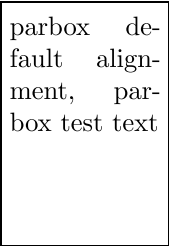
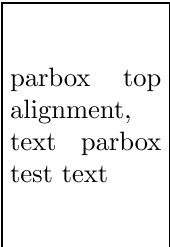
some text

parbox top alignment, text parbox test text

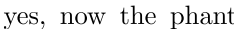
some text

parbox bottom alignment, text parbox test text

some text.

- top/bottom in one line: Some text  some text.
- 

### 5.1 Low-level box-interface

L<sup>A</sup>T<sub>E</sub>X.js supports the following low-level T<sub>E</sub>X commands: `\llap{text}`, `\rlap{text}`, and `\smash{text}`, as well as `\hphantom{text}`, `\vphantom{text}`, and `\phantom{text}`. A phantom looks like this:  yes, now the phantom is `\glap` could be used to put something in the margin. However, there are better alternatives for that. See `\marginpar`.

## 6 Spacing

The following horizontal spaces are supported:

Negative thin space: ||  
 No space (natural): ||  
 Thin space: || or ||  
 Normal space: ||  
 Normal space: ||  
 Non-break space: ||  
 en-space: ||  
 em-space: ||  
 2x em-space: ||  
 3cm horizontal space: | |

## 7 Environments

### 7.1 Lists: Itemize, Enumerate, and Description

The `itemize` environment is suitable for simple lists, the `enumerate` environment for enumerated lists, and the `description` environment for descriptions.

1. You can nest the list environments to your taste:
  - But it might start to look silly.

- With a dash.

2. Therefore remember:

**Stupid** things will not become smart because they are in a list.

**Smart** things, though, can be presented beautifully in a list.

important Technical note: Viewing this in Chrome, however, will show too much vertical space at the end of a nested environment (see above). On top of that, margin collapsing for inline-block boxes is not allowed. Maybe using `dl` elements is too complicated for this and a simple nested `div` should be used instead.

Lists can be deeply nested:

- list text, level one
  - list text, level two
    - \* list text, level three And a new paragraph can be started, too.
      - list text, level four And a new paragraph can be started, too. This is the maximum level.
      - list text, level four
    - \* list text, level three
  - list text, level two
- list text, level one
- list text, level one

## 7.2 Flushleft, Flushright, and Center

The `flushleft` environment:

This text is left-aligned.  $\text{\LaTeX}$  is not trying to make each line the same length.

The `flushright` environment:

This text is right-aligned.  $\text{\LaTeX}$  is not trying to make each line the same length.

And the `center` environment:

At the centre  
of the earth

## 7.3 Quote, Quotation, and Verse

The `quote` environment is useful for quotes, important phrases and examples. A typographical rule of thumb for the

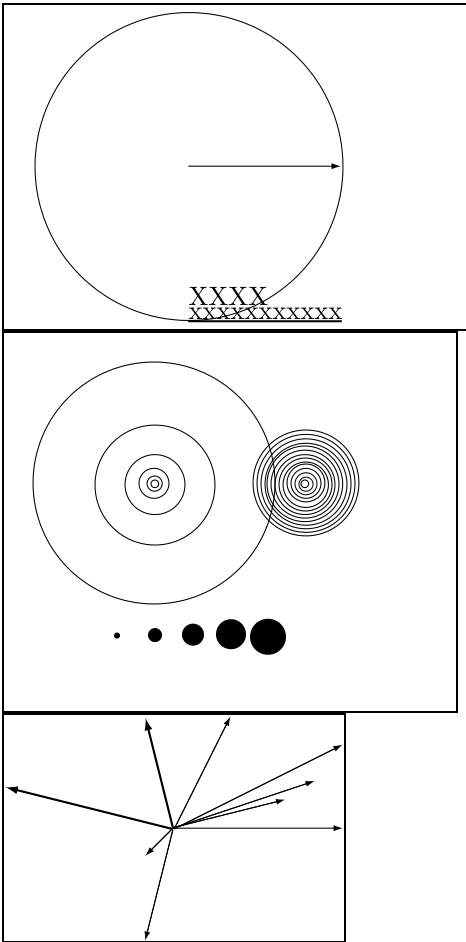
line length is:

On average, no line should be longer than 66 characters.

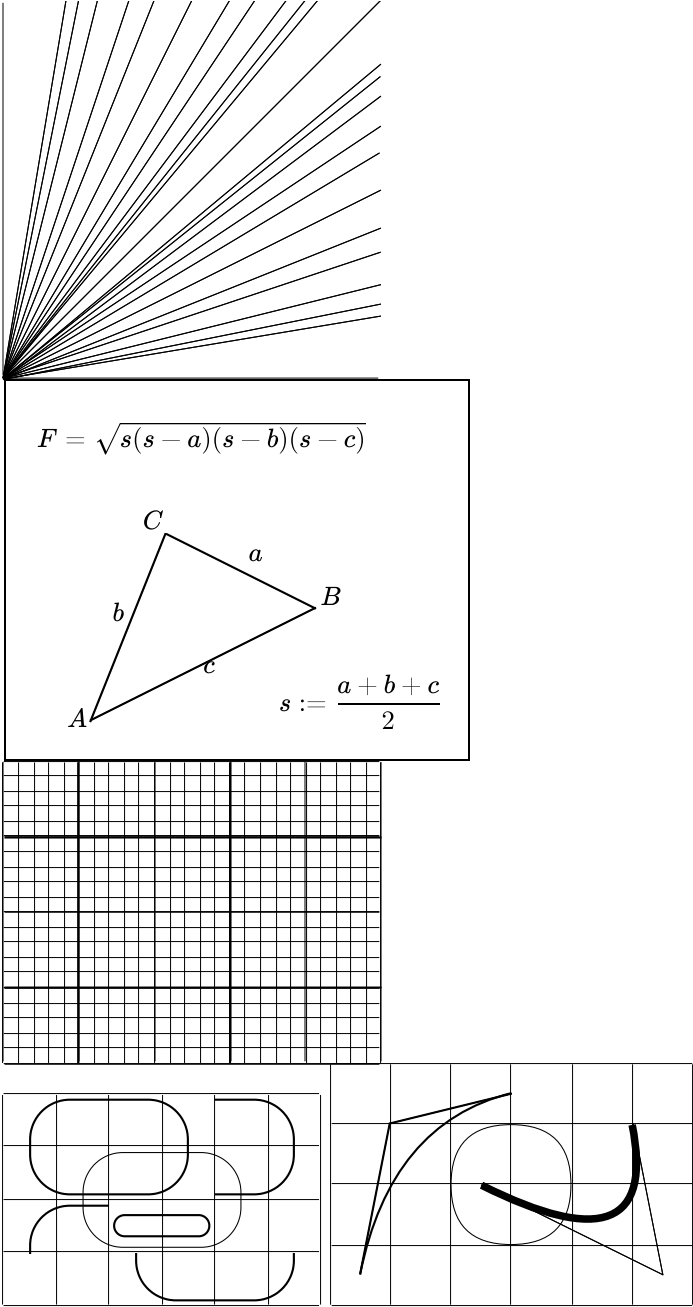
There are two similar environments: the `quotation` and the `verse` environments. The `quotation` environment is useful for longer quotes going over several paragraphs, because it indents the first line of each paragraph. The `verse` environment is useful for poems where the line breaks are important. The lines are separated by issuing a `\\` at the end of a line and an empty line after each verse.

Humpty Dumpty sat on a wall:  
Humpty Dumpty had a great fall.  
All the King's horses and all the King's men  
Couldn't put Humpty together again. —J.W.  
Elliott

7.4 Picture



.....



8   Labels and References

Labels can be attached to parts, chapters, sections, items of enumerations, footnotes, tables and figures. For instance: item [2](#) was important, and regarding fonts, read Section [12.1](#). And below, we can reference item [1\(a\)iB](#) and [1b](#).

- 1. list text, level one
  - (a) list text, level two



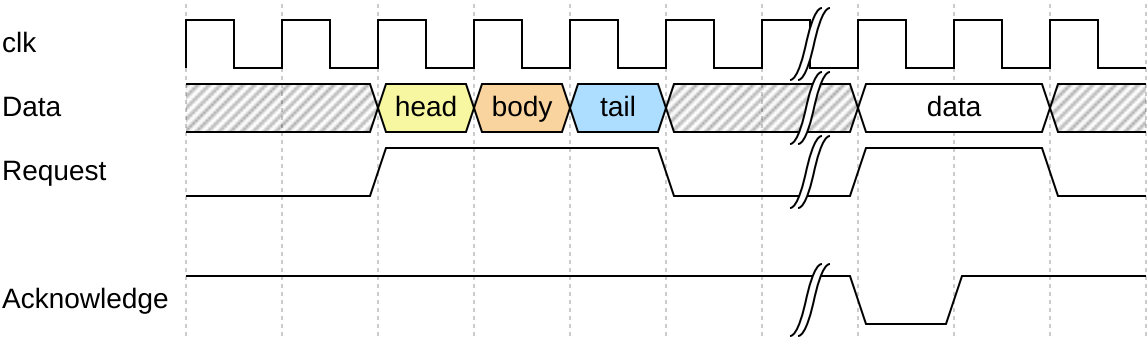
**Remember!** *The MORE fonts **you** use in a document, *the* more  
READABLE and *beautiful* it *become***S**.*

## A Source

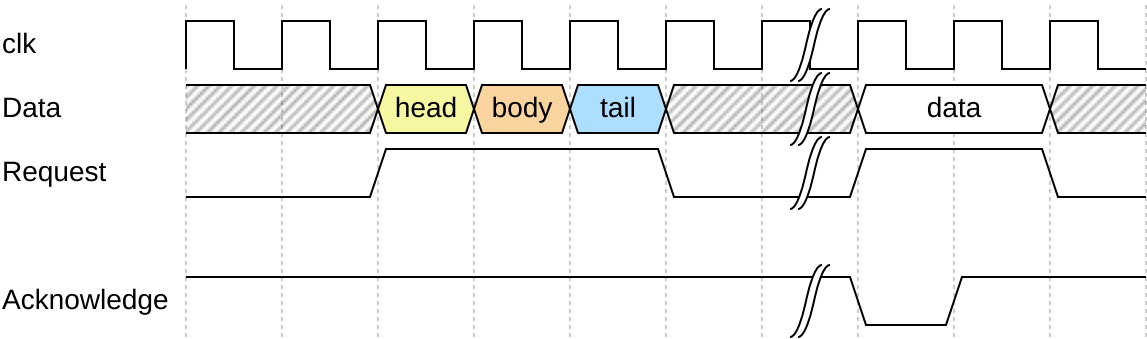
The source of LATEX.js is here on GitHub: <https://github.com/michael-brade/LaTeX.js>

# WaveDrom

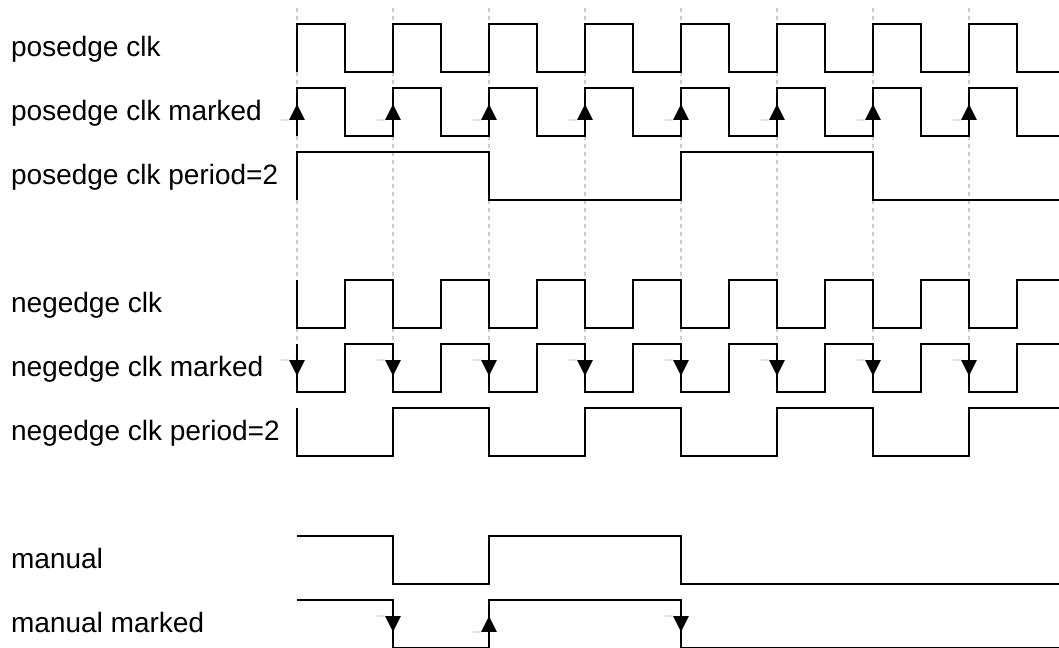
## basic1



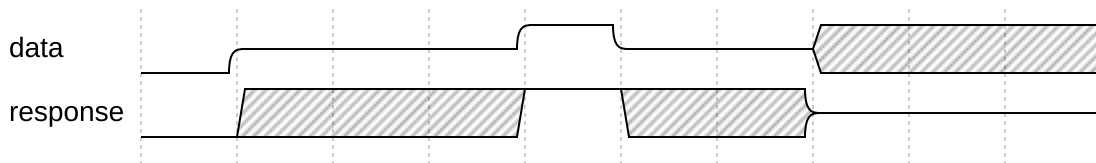
## basic2



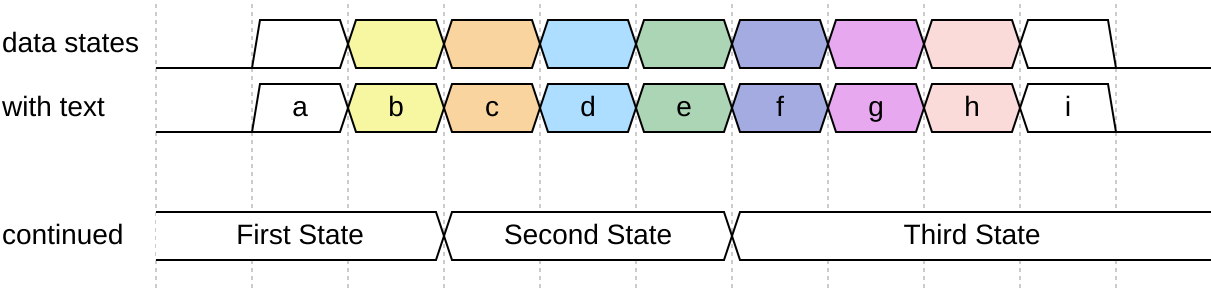
# clock



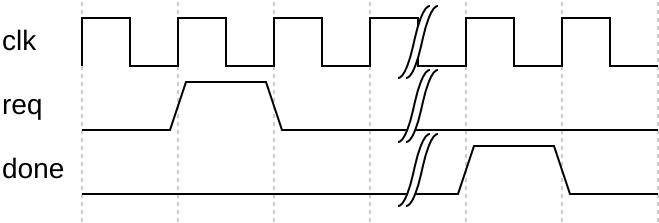
## High Impedance & Undefined



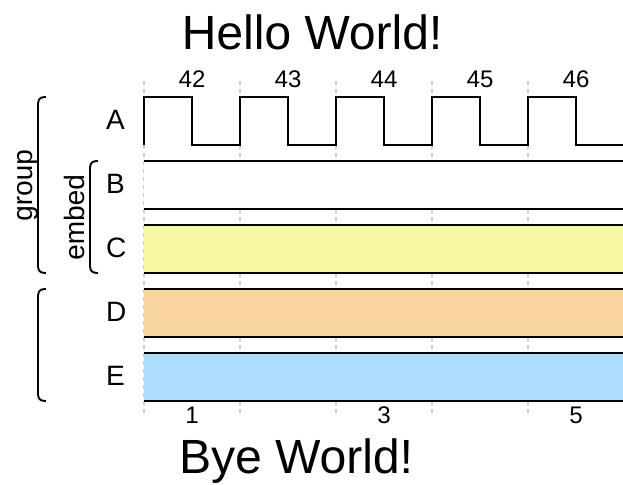
# Data types



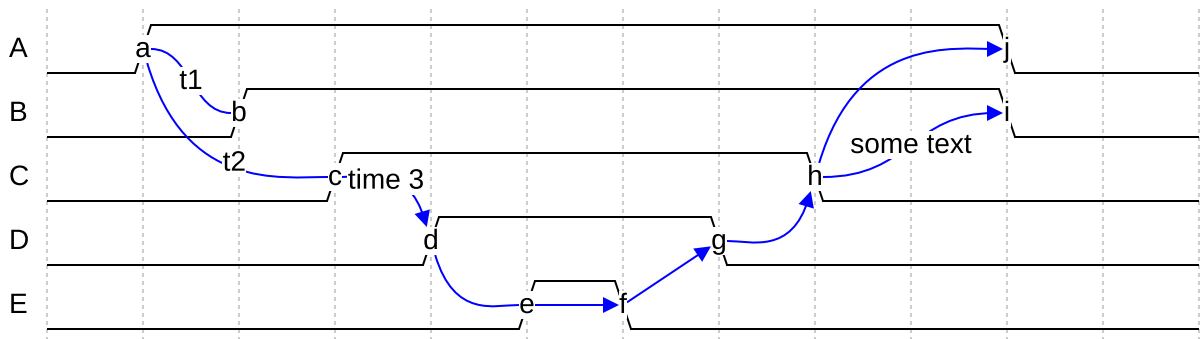
# Graps



# Groups



# Edges



# emoji-cheat-sheet

This cheat sheet is automatically generated from [GitHub Emoji API](#) and [Unicode Full Emoji List](#).

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## Smileys & Emotion

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- [Face Affection](#)
- [Face Tongue](#)
- [Face Hand](#)
- [Face Neutral Skeptical](#)
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- [Face Negative](#)
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Face Smiling

| ico                                                                                | shortcode             | ico                                                                               | shortcode        |
|------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------|------------------|
|   | grinning              |  | smiley           |
|   | smile                 |  | grin             |
|   | laughing<br>satisfied |  | sweat_smile      |
|   | rofl                  |  | joy              |
|   | slightly_smiling_face |  | upside_down_face |
|   | wink                  |  | blush            |
|  | innocent              |                                                                                   |                  |

Face Affection

| ico                                                                                 | shortcode                      | ico                                                                                 | shortcode            |
|-------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------|----------------------|
|  | smiling_face_with_three_hearts |  | heart_eyes           |
|  | star_struck                    |  | kissing_heart        |
|  | kissing                        |  | relaxed              |
|  | kissing_closed_eyes            |  | kissing_smiling_eyes |
|  | smiling_face_with_tear         |                                                                                     |                      |

Face Tongue

| ico                                                                               | shortcode                    | ico                                                                               | shortcode        |
|-----------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------|------------------|
|  | yum                          |  | stuck_out_tongue |
|  | stuck_out_tongue_winking_eye |  | zany_face        |
|  | stuck_out_tongue_closed_eyes |  | money_mouth_face |

Face Hand

| ico                                                                               | shortcode     | ico                                                                               | shortcode       |
|-----------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|-----------------|
|  | hugs          |  | hand_over_mouth |
|  | shushing_face |  | thinking        |

Face Neutral Skeptical

| ico                                                                                 | shortcode         | ico                                                                                 | shortcode      |
|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|----------------|
|   | zipper_mouth_face |   | raised_eyebrow |
|  | neutral_face      |  | expressionless |
|  | no_mouth          |  | face_in_clouds |
|  | smirk             |  | unamused       |
|  | roll_eyes         |  | grimacing      |
|  | face_exhaling     |  | lying_face     |

Face Sleepy

| ico                                                                                 | shortcode | ico                                                                                 | shortcode     |
|-------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|---------------|
|  | relieved  |  | pensive       |
|  | sleepy    |  | drooling_face |
|  | sleeping  |                                                                                     |               |

Face Unwell

| ico                                                                               | shortcode              | ico                                                                               | shortcode             |
|-----------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------|-----------------------|
|  | mask                   |  | face_with_thermometer |
|  | face_with_head_bandage |  | nauseated_face        |
|  | vomiting_face          |  | sneezing_face         |
|  | hot_face               |  | cold_face             |
|  | woozy_face             |  | dizzy_face            |
|  | face_with_spiral_eyes  |  | exploding_head        |

Face Hat

| ico                                                                               | shortcode       | ico                                                                               | shortcode     |
|-----------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|---------------|
|  | cowboy_hat_face |  | partying_face |
|  | disguised_face  |                                                                                   |               |

Face Glasses

| ico                                                                                 | shortcode    | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|-----------|
|  | sunglasses   |  | nerd_face |
|  | monocle_face |                                                                                     |           |

Face Concerned

| ico                                                                                 | shortcode              | ico                                                                                 | shortcode     |
|-------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------|---------------|
|  | confused               |  | worried       |
|  | slightly_frowning_face |  | frowning_face |
|  | open_mouth             |  | hushed        |
|  | astonished             |  | flushed       |

| ico                                                                               | shortcode     | ico                                                                               | shortcode             |
|-----------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|-----------------------|
|  | pleading_face |  | frowning              |
|  | anguished     |  | fearful               |
|  | cold_sweat    |  | disappointed_relieved |
|  | cry           |  | sob                   |
|  | scream        |  | confounded            |
|  | persevere     |  | disappointed          |
|  | sweat         |  | weary                 |
|  | tired_face    |  | yawning_face          |

Face Negative

| ico                                                                                 | shortcode   | ico                                                                                 | shortcode            |
|-------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|----------------------|
|    | triumph     |    | pout<br>rage         |
|    | angry       |    | cursing_face         |
|  | smiling_imp |  | imp                  |
|  | skull       |  | skull_and_crossbones |

Face Costume

| ico                                                                                 | shortcode              | ico                                                                                 | shortcode       |
|-------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------|-----------------|
|  | hankey<br>poop<br>shit |  | clown_face      |
|  | japanese_ogre          |  | japanese_goblin |
|  | ghost                  |  | alien           |
|  | space_invader          |  | robot           |

Cat Face

| ico                                                                               | shortcode   | ico                                                                               | shortcode       |
|-----------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------|-----------------|
|  | smiley_cat  |  | smile_cat       |
|  | joy_cat     |  | heart_eyes_cat  |
|  | smirk_cat   |  | kissing_cat     |
|  | scream_cat  |  | crying_cat_face |
|  | pouting_cat |                                                                                   |                 |

Monkey Face

| ico                                                                               | shortcode     | ico                                                                               | shortcode    |
|-----------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|--------------|
|  | see_no_evil   |  | hear_no_evil |
|  | speak_no_evil |                                                                                   |              |

Heart

| ico                                                                                 | shortcode        | ico                                                                                 | shortcode               |
|-------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|-------------------------|
|  | love_letter      |  | cupid                   |
|  | gift_heart       |  | sparkling_heart         |
|  | heartpulse       |  | heartbeat               |
|  | revolving_hearts |  | two_hearts              |
|  | heart_decoration |  | heavy_heart_exclamation |
|  | broken_heart     |  | heart_on_fire           |
|  | mending_heart    |  | heart                   |
|  | orange_heart     |  | yellow_heart            |
|  | green_heart      |  | blue_heart              |
|  | purple_heart     |  | brown_heart             |
|  | black_heart      |  | white_heart             |

Emotion

| ico                                                                               | shortcode          | ico                                                                               | shortcode          |
|-----------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------|--------------------|
|  | kiss               |  | 100                |
|  | anger              |  | boom<br>collision  |
|  | dizzy              |  | sweat_drops        |
|  | dash               |  | hole               |
|  | speech_balloon     |  | eye_speech_bubble  |
|  | left_speech_bubble |  | right_anger_bubble |
|  | thought_balloon    |  | zzz                |

People & Body

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## Hand Fingers Open

| ico                                                                               | shortcode                        | ico                                                                               | shortcode           |
|-----------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------|---------------------|
|  | wave                             |  | raised_back_of_hand |
|  | raised_hand_with_fingers_splayed |  | hand<br>raised_hand |
|  | vulcan_salute                    |                                                                                   |                     |

## Hand Fingers Partial

| ico                                                                               | shortcode       | ico                                                                               | shortcode        |
|-----------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|------------------|
|  | ok_hand         |  | pinched_fingers  |
|  | pinching_hand   |  | v                |
|  | crossed_fingers |  | love_you_gesture |
|  | metal           |  | call_me_hand     |

## Hand Single Finger

| ico                                                                                 | shortcode  | ico                                                                                 | shortcode           |
|-------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------|---------------------|
|  | point_left |  | point_right         |
|  | point_up_2 |  | fu<br>middle_finger |
|  | point_down |  | point_up            |

## Hand Fingers Closed

| ico                                                                                 | shortcode      | ico                                                                                 | shortcode        |
|-------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|------------------|
|  | +1<br>thumbsup |  | -1<br>thumbsdown |

| ico                                                                               | shortcode           | ico                                                                               | shortcode                           |
|-----------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------|-------------------------------------|
|  | fist<br>fist_raised |  | facepunch<br>fist_oncoming<br>punch |
|  | fist_left           |  | fist_right                          |

Hands

| ico                                                                               | shortcode  | ico                                                                               | shortcode         |
|-----------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------|-------------------|
|  | clap       |  | raised_hands      |
|  | open_hands |  | palms_up_together |
|  | handshake  |  | pray              |

Hand Prop

| ico                                                                                 | shortcode    | ico                                                                                | shortcode |
|-------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------|-----------|
|   | writing_hand |  | nail_care |
|  | selfie       |                                                                                    |           |

Body Parts

| ico                                                                                 | shortcode            | ico                                                                                 | shortcode        |
|-------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------|------------------|
|  | muscle               |  | mechanical_arm   |
|  | mechanical_leg       |  | leg              |
|  | foot                 |  | ear              |
|  | ear_with_hearing_aid |  | nose             |
|  | brain                |  | anatomical_heart |
|  | lungs                |  | tooth            |
|  | bone                 |  | eyes             |

| ico                                                                               | shortcode | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------|-----------|
|  | eye       |  | tongue    |
|  | lips      |                                                                                   |           |

Person

| ico                                                                                 | shortcode         | ico                                                                                 | shortcode                          |
|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|------------------------------------|
|    | baby              |    | child                              |
|    | boy               |    | girl                               |
|    | adult             |    | blond_haired_person                |
|    | man               |    | bearded_person                     |
|    | man_beard         |    | woman_beard                        |
|    | red_haired_man    |    | curly_haired_man                   |
|    | white_haired_man  |    | bald_man                           |
|    | woman             |    | red_haired_woman                   |
|    | person_red_hair   |    | curly_haired_woman                 |
|  | person_curly_hair |  | white_haired_woman                 |
|  | person_white_hair |  | bald_woman                         |
|  | person_bald       |  | blond_haired_woman<br>blonde_woman |
|  | blond_haired_man  |  | older_adult                        |
|  | older_man         |  | older_woman                        |

Person Gesture

| ico                                                                                 | shortcode       | ico                                                                                 | shortcode     |
|-------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------|---------------|
|  | frowning_person |  | frowning_man  |
|  | frowning_woman  |  | pouting_face  |
|  | pouting_man     |  | pouting_woman |

| ico                                                                                | shortcode                                      | ico                                                                                | shortcode                     |
|------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------|
|   | no_good                                        |   | ng_man<br>no_good_man         |
|   | ng_woman<br>no_good_woman                      |   | ok_person                     |
|   | ok_man                                         |   | ok_woman                      |
|   | information_desk_person<br>tipping_hand_person |   | sassy_man<br>tipping_hand_man |
|   | sassy_woman<br>tipping_hand_woman              |   | raising_hand                  |
|   | raising_hand_man                               |   | raising_hand_woman            |
|   | deaf_person                                    |   | deaf_man                      |
|   | deaf_woman                                     |   | bow                           |
|   | bowing_man                                     |   | bowing_woman                  |
|   | facepalm                                       |   | man_facepalming               |
|   | woman_facepalming                              |   | shrug                         |
|  | man_shrugging                                  |  | woman_shrugging               |

Person Role

| ico                                                                                 | shortcode           | ico                                                                                 | shortcode         |
|-------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|-------------------|
|  | health_worker       |  | man_health_worker |
|  | woman_health_worker |  | student           |
|  | man_student         |  | woman_student     |
|  | teacher             |  | man_teacher       |
|  | woman_teacher       |  | judge             |
|  | man_judge           |  | woman_judge       |
|  | farmer              |  | man_farmer        |
|  | woman_farmer        |  | cook              |

| ico                                                                                 | shortcode                 | ico                                                                                 | shortcode               |
|-------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------|-------------------------|
|    | man_cook                  |    | woman_cook              |
|    | mechanic                  |    | man_mechanic            |
|    | woman_mechanic            |    | factory_worker          |
|    | man_factory_worker        |    | woman_factory_worker    |
|    | office_worker             |    | man_office_worker       |
|    | woman_office_worker       |    | scientist               |
|    | man_scientist             |    | woman_scientist         |
|    | technologist              |    | man_technologist        |
|    | woman_technologist        |    | singer                  |
|    | man_singer                |    | woman_singer            |
|    | artist                    |    | man_artist              |
|    | woman_artist              |    | pilot                   |
|    | man_pilot                 |    | woman_pilot             |
|    | astronaut                 |    | man_astronaut           |
|   | woman_astronaut           |   | firefighter             |
|  | man_firefighter           |  | woman_firefighter       |
|  | cop<br>police_officer     |  | policeman               |
|  | policewoman               |  | detective               |
|  | male_detective            |  | female_detective        |
|  | guard                     |  | guardsman               |
|  | guardswoman               |  | ninja                   |
|  | construction_worker       |  | construction_worker_man |
|  | construction_worker_woman |  | prince                  |
|  | princess                  |  | person_with_turban      |
|  | man_with_turban           |  | woman_with_turban       |
|  | man_with_gua_pi_mao       |  | woman_with_headscarf    |

| ico                                                                               | shortcode           | ico                                                                               | shortcode        |
|-----------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------|------------------|
|  | person_in_tuxedo    |  | man_in_tuxedo    |
|  | woman_in_tuxedo     |  | person_with_veil |
|  | man_with_veil       |  | bride_with_veil  |
|  | pregnant_woman      |  | woman_with_veil  |
|  | woman_feeding_baby  |  | breast_feeding   |
|  | person_feeding_baby |                                                                                   | man_feeding_baby |

Person Fantasy

| ico                                                                                 | shortcode        | ico                                                                                 | shortcode          |
|-------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|--------------------|
|    | angel            |    | santa              |
|    | mrs_claus        |    | mx_claus           |
|    | superhero        |    | superhero_man      |
|    | superhero_woman  |    | supervillain       |
|   | supervillain_man |   | supervillain_woman |
|  | mage             |  | mage_man           |
|  | mage_woman       |  | fairy              |
|  | fairy_man        |  | fairy_woman        |
|  | vampire          |  | vampire_man        |
|  | vampire_woman    |  | merperson          |
|  | merman           |  | mermaid            |
|  | elf              |  | elf_man            |
|  | elf_woman        |  | genie              |
|  | genie_man        |  | genie_woman        |
|  | zombie           |  | zombie_man         |
|  | zombie_woman     |                                                                                     |                    |

Person Activity

| ico                                                                                 | shortcode                      | ico                                                                                 | shortcode                   |
|-------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------|-----------------------------|
|    | massage                        |    | massage_man                 |
|    | massage_woman                  |    | haircut                     |
|    | haircut_man                    |    | haircut_woman               |
|    | walking                        |    | walking_man                 |
|    | walking_woman                  |    | standing_person             |
|    | standing_man                   |    | standing_woman              |
|    | kneeling_person                |    | kneeling_man                |
|    | kneeling_woman                 |    | person_with_probing_cane    |
|    | man_with_probing_cane          |    | woman_with_probing_cane     |
|    | person_in_motorized_wheelchair |    | man_in_motorized_wheelchair |
|    | woman_in_motorized_wheelchair  |    | person_in_manual_wheelchair |
|    | man_in_manual_wheelchair       |    | woman_in_manual_wheelchair  |
|   | runner<br>running              |   | running_man                 |
|  | running_woman                  |  | dancer<br>woman_dancing     |
|  | man_dancing                    |  | business_suit_levitating    |
|  | dancers                        |  | dancing_men                 |
|  | dancing_women                  |  | sauna_person                |
|  | sauna_man                      |  | sauna_woman                 |
|  | climbing                       |  | climbing_man                |
|  | climbing_woman                 |                                                                                     |                             |

Person Sport

| ico                                                                                 | shortcode                               | ico                                                                                 | shortcode                           |
|-------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------|
|    | person_fencing                          |    | horse_racing                        |
|    | skier                                   |    | snowboarder                         |
|    | golfing                                 |    | golfing_man                         |
|    | golfing_woman                           |    | surfer                              |
|    | surfing_man                             |    | surfing_woman                       |
|    | rowboat                                 |    | rowing_man                          |
|    | rowing_woman                            |    | swimmer                             |
|    | swimming_man                            |    | swimming_woman                      |
|    | bouncing_ball_person                    |    | basketball_man<br>bouncing_ball_man |
|    | basketball_woman<br>bouncing_ball_woman |    | weight_lifting                      |
|    | weight_lifting_man                      |    | weight_lifting_woman                |
|   | bicyclist                               |   | biking_man                          |
|  | biking_woman                            |  | mountain_bicyclist                  |
|  | mountain_biking_man                     |  | mountain_biking_woman               |
|  | cartwheeling                            |  | man_cartwheeling                    |
|  | woman_cartwheeling                      |  | wrestling                           |
|  | men_wrestling                           |  | women_wrestling                     |
|  | water_polo                              |  | man_playing_water_polo              |
|  | woman_playing_water_polo                |  | handball_person                     |
|  | man_playing_handball                    |  | woman_playing_handball              |
|  | juggling_person                         |  | man_juggling                        |
|  | woman_juggling                          |                                                                                     |                                     |

Person Resting

| ico                                                                               | shortcode            | ico                                                                               | shortcode          |
|-----------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------|--------------------|
|  | lotus_position       |  | lotus_position_man |
|  | lotus_position_woman |  | bath               |
|  | sleeping_bed         |                                                                                   |                    |

Family

| ico                                                                                 | shortcode                    | ico                                                                                 | shortcode                 |
|-------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------|---------------------------|
|    | people_holding_hands         |    | two_women_holding_hands   |
|    | couple                       |    | two_men_holding_hands     |
|    | couplekiss                   |    | couplekiss_man_woman      |
|    | couplekiss_man_man           |    | couplekiss_woman_woman    |
|    | couple_with_heart            |    | couple_with_heart_woman_n |
|    | couple_with_heart_man_man    |    | couple_with_heart_woman_v |
|   | family_man_woman_boy         |   | family_man_woman_girl     |
|  | family_man_woman_girl_boy    |  | family_man_woman_boy_boy  |
|  | family_man_woman_girl_girl   |  | family_man_man_boy        |
|  | family_man_man_girl          |  | family_man_man_girl_boy   |
|  | family_man_man_boy_boy       |  | family_man_man_girl_girl  |
|  | family_woman_woman_boy       |  | family_woman_woman_girl   |
|  | family_woman_woman_girl_boy  |  | family_woman_woman_boy_bc |
|  | family_woman_woman_girl_girl |  | family_man_boy            |
|  | family_man_boy_boy           |  | family_man_girl           |
|  | family_man_girl_boy          |  | family_man_girl_girl      |
|  | family_woman_boy             |  | family_woman_boy_boy      |
|  | family_woman_girl            |  | family_woman_girl_boy     |

| ico                                                                               | shortcode | ico | shortcode |
|-----------------------------------------------------------------------------------|-----------|-----|-----------|
|  | person    |     |           |

Person Symbol

| ico                                                                               | shortcode           | ico                                                                               | shortcode          |
|-----------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------|--------------------|
|  | speaking_head       |  | bust_in_silhouette |
|  | busts_in_silhouette |  | people_hugging     |
|  | family              |  | footprints         |

Animals & Nature

- [Animal Mammal](#)
- [Animal Bird](#)
- [Animal Amphibian](#)
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Animal Mammal

| ico                                                                                 | shortcode   | ico                                                                                 | shortcode   |
|-------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|-------------|
|  | monkey_face |  | monkey      |
|  | gorilla     |  | orangutan   |
|  | dog         |  | dog2        |
|  | guide_dog   |  | service_dog |
|  | poodle      |  | wolf        |
|  | fox_face    |  | raccoon     |

| ico                                                                                 | shortcode       | ico                                                                                 | shortcode          |
|-------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------|--------------------|
|    | cat             |    | cat2               |
|    | black_cat       |    | lion               |
|    | tiger           |    | tiger2             |
|    | leopard         |    | horse              |
|    | racehorse       |    | unicorn            |
|    | zebra           |    | deer               |
|    | bison           |    | cow                |
|    | ox              |    | water_buffalo      |
|    | cow2            |    | pig                |
|    | pig2            |    | boar               |
|    | pig_nose        |    | ram                |
|    | sheep           |    | goat               |
|    | dromedary_camel |    | camel              |
|    | llama           |    | giraffe            |
|   | elephant        |   | mammoth            |
|  | rhinoceros      |  | hippopotamus       |
|  | mouse           |  | mouse2             |
|  | rat             |  | hamster            |
|  | rabbit          |  | rabbit2            |
|  | chipmunk        |  | beaver             |
|  | hedgehog        |  | bat                |
|  | bear            |  | polar_bear         |
|  | koala           |  | panda_face         |
|  | sloth           |  | otter              |
|  | skunk           |  | kangaroo           |
|  | badger          |  | feet<br>paw_prints |

Animal Bird

| ico                                                                               | shortcode  | ico                                                                               | shortcode      |
|-----------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------|----------------|
|  | turkey     |  | chicken        |
|  | rooster    |  | hatching_chick |
|  | baby_chick |  | hatched_chick  |
|  | bird       |  | penguin        |
|  | dove       |  | eagle          |
|  | duck       |  | swan           |
|  | owl        |  | dodo           |
|  | feather    |  | flamingo       |
|  | peacock    |  | parrot         |

Animal Amphibian

| ico                                                                                | shortcode |
|------------------------------------------------------------------------------------|-----------|
|  | frog      |

Animal Reptile

| ico                                                                                 | shortcode   | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|-----------|
|  | crocodile   |  | turtle    |
|  | lizard      |  | snake     |
|  | dragon_face |  | dragon    |
|  | sauropod    |  | t-rex     |

Animal Marine

| ico                                                                               | shortcode          | ico                                                                               | shortcode     |
|-----------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------|---------------|
|  | whale              |  | whale2        |
|  | dolphin<br>flipper |  | seal          |
|  | fish               |  | tropical_fish |
|  | blowfish           |  | shark         |
|  | octopus            |  | shell         |

Animal Bug

| ico                                                                                 | shortcode       | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------|-----------|
|    | snail           |    | butterfly |
|    | bug             |    | ant       |
|    | bee<br>honeybee |    | beetle    |
|  | lady_bettle     |  | cricket   |
|  | cockroach       |  | spider    |
|  | spider_web      |  | scorpion  |
|  | mosquito        |  | fly       |
|  | worm            |  | microbe   |

Plant Flower

| ico                                                                                 | shortcode    | ico                                                                                 | shortcode      |
|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|----------------|
|  | bouquet      |  | cherry_blossom |
|  | white_flower |  | rosette        |
|  | rose         |  | wilted_flower  |

| ico                                                                               | shortcode | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------|-----------|
|  | hibiscus  |  | sunflower |
|  | blossom   |  | tulip     |

Plant Other

| ico                                                                               | shortcode      | ico                                                                               | shortcode        |
|-----------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------|------------------|
|  | seedling       |  | potted_plant     |
|  | evergreen_tree |  | deciduous_tree   |
|  | palm_tree      |  | cactus           |
|  | ear_of_rice    |  | herb             |
|  | shamrock       |  | four_leaf_clover |
|  | maple_leaf     |  | fallen_leaf      |
|  | leaves         |  | mushroom         |

Food & Drink

- [Food Fruit](#)
- [Food Vegetable](#)
- [Food Prepared](#)
- [Food Asian](#)
- [Food Marine](#)
- [Food Sweet](#)
- [Drink](#)
- [Dishware](#)

Food Fruit

| ico                                                                                 | shortcode | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|-----------|
|  | grapes    |  | melon     |

| ico                                                                               | shortcode   | ico                                                                               | shortcode                       |
|-----------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------|---------------------------------|
|  | watermelon  |  | mandarin<br>orange<br>tangerine |
|  | lemon       |  | banana                          |
|  | pineapple   |  | mango                           |
|  | apple       |  | green_apple                     |
|  | pear        |  | peach                           |
|  | cherries    |  | strawberry                      |
|  | blueberries |  | kiwi_fruit                      |
|  | tomato      |  | olive                           |
|  | coconut     |                                                                                   |                                 |

Food Vegetable

| ico                                                                                 | shortcode   | ico                                                                                 | shortcode  |
|-------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|------------|
|   | avocado     |   | eggplant   |
|  | potato      |  | carrot     |
|  | corn        |  | hot_pepper |
|  | bell_pepper |  | cucumber   |
|  | leafy_green |  | broccoli   |
|  | garlic      |  | onion      |
|  | peanuts     |  | chestnut   |

Food Prepared

| ico                                                                                 | shortcode      | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|-----------|
|  | bread          |  | croissant |
|  | baguette_bread |  | flatbread |

| ico                                                                                | shortcode       | ico                                                                                | shortcode           |
|------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------|---------------------|
|   | pretzel         |   | bagel               |
|   | pancakes        |   | waffle              |
|   | cheese          |   | meat_on_bone        |
|   | poultry_leg     |   | cut_of_meat         |
|   | bacon           |   | hamburger           |
|   | fries           |   | pizza               |
|   | hotdog          |   | sandwich            |
|   | taco            |   | burrito             |
|   | tamale          |   | stuffed_flatbread   |
|   | falafel         |   | egg                 |
|   | fried_egg       |   | shallow_pan_of_food |
|   | stew            |   | fondue              |
|   | bowl_with_spoon |   | green_salad         |
|   | popcorn         |   | butter              |
|  | salt            |  | canned_food         |

Food Asian

| ico                                                                                 | shortcode    | ico                                                                                 | shortcode      |
|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|----------------|
|  | bento        |  | rice_cracker   |
|  | rice_ball    |  | rice           |
|  | curry        |  | ramen          |
|  | spaghetti    |  | sweet_potato   |
|  | oden         |  | sushi          |
|  | fried_shrimp |  | fish_cake      |
|  | moon_cake    |  | dango          |
|  | dumpling     |  | fortune_cookie |

| ico                                                                               | shortcode   | ico | shortcode |
|-----------------------------------------------------------------------------------|-------------|-----|-----------|
|  | takeout_box |     |           |

Food Marine

| ico                                                                               | shortcode | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------|-----------|
|  | crab      |  | lobster   |
|  | shrimp    |  | squid     |
|  | oyster    |                                                                                   |           |

Food Sweet

| ico                                                                                 | shortcode | ico                                                                                 | shortcode     |
|-------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|---------------|
|    | icecream  |    | shaved_ice    |
|    | ice_cream |    | doughnut      |
|    | cookie    |    | birthday      |
|  | cake      |  | cupcake       |
|  | pie       |  | chocolate_bar |
|  | candy     |  | lollipop      |
|  | custard   |  | honey_pot     |

Drink

| ico                                                                                 | shortcode   | ico                                                                                 | shortcode      |
|-------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|----------------|
|  | baby_bottle |  | milk_glass     |
|  | coffee      |  | teapot         |
|  | tea         |  | sake           |
|  | champagne   |  | wine_glass     |
|  | cocktail    |  | tropical_drink |

| ico                                                                               | shortcode        | ico                                                                               | shortcode     |
|-----------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------|---------------|
|  | beer             |  | beers         |
|  | clinking_glasses |  | tumbler_glass |
|  | cup_with_straw   |  | bubble_tea    |
|  | beverage_box     |  | mate          |
|  | ice_cube         |                                                                                   |               |

Dishware

| ico                                                                               | shortcode      | ico                                                                               | shortcode          |
|-----------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------|--------------------|
|  | chopsticks     |  | plate_with_cutlery |
|  | fork_and_knife |  | spoon              |
|  | hocho<br>knife |  | amphora            |

Travel & Places

- [Place Map](#)
- [Place Geographic](#)
- [Place Building](#)
- [Place Religious](#)
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Place Map

| ico                                                                               | shortcode    | ico                                                                               | shortcode            |
|-----------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------|----------------------|
|  | earth_africa |  | earth_americas       |
|  | earth_asia   |  | globe_with_meridians |
|  | world_map    |  | japan                |
|  | compass      |                                                                                   |                      |

Place Geographic

| ico                                                                               | shortcode     | ico                                                                               | shortcode      |
|-----------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|----------------|
|  | mountain_snow |  | mountain       |
|  | volcano       |  | mount_fuji     |
|  | camping       |  | beach_umbrella |
|  | desert        |  | desert_island  |
|  | national_park |                                                                                   |                |

Place Building

| ico                                                                                 | shortcode             | ico                                                                                 | shortcode            |
|-------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------|----------------------|
|  | stadium               |  | classical_building   |
|  | building_construction |  | bricks               |
|  | rock                  |  | wood                 |
|  | hut                   |  | houses               |
|  | derelict_house        |  | house                |
|  | house_with_garden     |  | office               |
|  | post_office           |  | european_post_office |
|  | hospital              |  | bank                 |
|  | hotel                 |  | love_hotel           |

| ico                                                                               | shortcode         | ico                                                                               | shortcode       |
|-----------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------|-----------------|
|  | convenience_store |  | school          |
|  | department_store  |  | factory         |
|  | japanese_castle   |  | european_castle |
|  | wedding           |  | tokyo_tower     |
|  | statue_of_liberty |                                                                                   |                 |

Place Religious

| ico                                                                               | shortcode     | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|-----------|
|  | church        |  | mosque    |
|  | hindu_temple  |  | synagogue |
|  | shinto_shrine |  | kaaba     |

Place Other

| ico                                                                                 | shortcode    | ico                                                                                 | shortcode              |
|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|------------------------|
|  | fountain     |  | tent                   |
|  | foggy        |  | night_with_stars       |
|  | cityscape    |  | sunrise_over_mountains |
|  | sunrise      |  | city_sunset            |
|  | city_sunrise |  | bridge_at_night        |
|  | hotsprings   |  | carousel_horse         |
|  | ferris_wheel |  | roller_coaster         |
|  | barber       |  | circus_tent            |

Transport Ground

| ico                                                                                 | shortcode              | ico                                                                                 | shortcode            |
|-------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------|----------------------|
|    | steam_locomotive       |    | railway_car          |
|    | bullettrain_side       |    | bullettrain_front    |
|    | train2                 |    | metro                |
|    | light_rail             |    | station              |
|    | tram                   |    | monorail             |
|    | mountain_railway       |    | train                |
|    | bus                    |    | oncoming_bus         |
|    | trolleybus             |    | minibus              |
|    | ambulance              |    | fire_engine          |
|    | police_car             |    | oncoming_police_car  |
|    | taxi                   |    | oncoming_taxi        |
|    | car                    |    | oncoming_automobile  |
|    | red_car                |                                                                                     |                      |
|  | blue_car               |  | pickup_truck         |
|  | truck                  |  | articulated_lorry    |
|  | tractor                |  | racing_car           |
|  | motorcycle             |  | motor_scooter        |
|  | manual_wheelchair      |  | motorized_wheelchair |
|  | auto_rickshaw          |  | bike                 |
|  | kick_scooter           |  | skateboard           |
|  | roller_skate           |  | busstop              |
|  | motorway               |  | railway_track        |
|  | oil_drum               |  | fuelpump             |
|  | rotating_light         |  | traffic_light        |
|  | vertical_traffic_light |  | stop_sign            |

| ico                                                                               | shortcode    | ico | shortcode |
|-----------------------------------------------------------------------------------|--------------|-----|-----------|
|  | construction |     |           |

Transport Water

| ico                                                                               | shortcode      | ico                                                                               | shortcode        |
|-----------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------|------------------|
|  | anchor         |  | boat<br>sailboat |
|  | canoe          |  | speedboat        |
|  | passenger_ship |  | ferry            |
|  | motor_boat     |  | ship             |

Transport Air

| ico                                                                                 | shortcode            | ico                                                                                 | shortcode          |
|-------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------|--------------------|
|    | airplane             |    | small_airplane     |
|   | flight_departure     |   | flight_arrival     |
|  | parachute            |  | seat               |
|  | helicopter           |  | suspension_railway |
|  | mountain_cableway    |  | aerial_tramway     |
|  | artificial_satellite |  | rocket             |
|  | flying_saucer        |                                                                                     |                    |

Hotel

| ico                                                                                 | shortcode    | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|-----------|
|  | bellhop_bell |  | luggage   |

Time

| ico                                                                                 | shortcode         | ico                                                                                 | shortcode             |
|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|-----------------------|
|    | hourglass         |    | hourglass_flowingsand |
|    | watch             |    | alarm_clock           |
|    | stopwatch         |    | timer_clock           |
|    | mantelpiece_clock |    | clock12               |
|    | clock1230         |    | clock1                |
|    | clock130          |    | clock2                |
|    | clock230          |    | clock3                |
|    | clock330          |    | clock4                |
|    | clock430          |    | clock5                |
|    | clock530          |    | clock6                |
|    | clock630          |    | clock7                |
|    | clock730          |    | clock8                |
|    | clock830          |    | clock9                |
|  | clock930          |  | clock10               |
|  | clock1030         |  | clock11               |
|  | clock1130         |                                                                                     |                       |

Sky & Weather

| ico                                                                                 | shortcode          | ico                                                                                 | shortcode            |
|-------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|----------------------|
|  | new_moon           |  | waxing_crescent_moon |
|  | first_quarter_moon |  | moon                 |
|  | full_moon          |  | waning_gibbous_moon  |
|  | last_quarter_moon  |  | waning_crescent_moon |
|  | crescent_moon      |  | new_moon_with_face   |

| ico                                                                                 | shortcode                    | ico                                                                                 | shortcode                 |
|-------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------|---------------------------|
|    | first_quarter_moon_with_face |    | last_quarter_moon_with_fa |
|    | thermometer                  |    | sunny                     |
|    | full_moon_with_face          |    | sun_with_face             |
|    | ringed_planet                |    | star                      |
|    | star2                        |    | stars                     |
|    | milky_way                    |    | cloud                     |
|    | partly_sunny                 |    | cloud_with_lightning_and_ |
|    | sun_behind_small_cloud       |    | sun_behind_large_cloud    |
|    | sun_behind_rain_cloud        |    | cloud_with_rain           |
|    | cloud_with_snow              |    | cloud_with_lightning      |
|    | tornado                      |    | fog                       |
|    | wind_face                    |    | cyclone                   |
|    | rainbow                      |    | closed_umbrella           |
|    | open_umbrella                |    | umbrella                  |
|   | parasol_on_ground            |   | zap                       |
|  | snowflake                    |  | snowman_with_snow         |
|  | snowman                      |  | comet                     |
|  | fire                         |  | droplet                   |
|  | ice_cube                     |                                                                                     |                           |

## Activities

- [Event](#)
- [Award Medal](#)
- [Sport](#)
- [Game](#)
- ◀ • [Arts & Crafts](#) ▶

Event

| ico                                                                               | shortcode       | ico                                                                               | shortcode      |
|-----------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|----------------|
|  | jack_o_lantern  |  | christmas_tree |
|  | fireworks       |  | sparkler       |
|  | firecracker     |  | sparkles       |
|  | balloon         |  | tada           |
|  | confetti_ball   |  | tanabata_tree  |
|  | bamboo          |  | dolls          |
|  | flags           |  | wind_chime     |
|  | rice_scene      |  | red_envelope   |
|  | ribbon          |  | gift           |
|  | reminder_ribbon |  | tickets        |
|  | ticket          |                                                                                   |                |

Award Medal

| ico                                                                                 | shortcode       | ico                                                                                 | shortcode       |
|-------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------|-----------------|
|  | medal_military  |  | trophy          |
|  | medal_sports    |  | 1st_place_medal |
|  | 2nd_place_medal |  | 3rd_place_medal |

Sport

| ico                                                                                 | shortcode      | ico                                                                                 | shortcode  |
|-------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|------------|
|  | soccer         |  | baseball   |
|  | softball       |  | basketball |
|  | volleyball     |  | football   |
|  | rugby_football |  | tennis     |

| ico                                                                               | shortcode     | ico                                                                               | shortcode               |
|-----------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|-------------------------|
|  | flying_disc   |  | bowling                 |
|  | cricket_game  |  | field_hockey            |
|  | ice_hockey    |  | lacrosse                |
|  | ping_pong     |  | badminton               |
|  | boxing_glove  |  | martial_arts_uniform    |
|  | goal_net      |  | golf                    |
|  | ice_skate     |  | fishing_pole_and_fish   |
|  | diving_mask   |  | running_shirt_with_sash |
|  | ski           |  | sled                    |
|  | curling_stone |                                                                                   |                         |

Game

| ico                                                                                 | shortcode            | ico                                                                                 | shortcode    |
|-------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------|--------------|
|   | dart                 |   | yo_yo        |
|  | kite                 |  | gun          |
|  | 8ball                |  | crystal_ball |
|  | magic_wand           |  | video_game   |
|  | joystick             |  | slot_machine |
|  | game_die             |  | jigsaw       |
|  | teddy_bear           |  | pinata       |
|  | nesting_dolls        |  | spades       |
|  | hearts               |  | diamonds     |
|  | clubs                |  | chess_pawn   |
|  | black_joker          |  | mahjong      |
|  | flower_playing_cards |                                                                                     |              |

## Arts & Crafts

| ico                                                                               | shortcode       | ico                                                                               | shortcode      |
|-----------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|----------------|
|  | performing_arts |  | framed_picture |
|  | art             |  | thread         |
|  | sewing_needle   |  | yarn           |
|  | knot            |                                                                                   |                |

## Objects

- Clothing
- Sound
- Music
- Musical Instrument
- Phone
- Computer
- Light & Video
- Book Paper
- Money
- Mail
- Writing
- Office
- Lock
- Tool
- Science
- Medical
- Household
- Other Object

Clothing

| ico                                                                                 | shortcode            | ico                                                                                 | shortcode          |
|-------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------|--------------------|
|    | eyeglasses           |    | dark_sunglasses    |
|    | goggles              |    | lab_coat           |
|    | safety_vest          |    | necktie            |
|    | shirt<br>tshirt      |    | jeans              |
|    | scarf                |    | gloves             |
|    | coat                 |    | socks              |
|    | dress                |    | kimono             |
|    | sari                 |    | one_piece_swimsuit |
|    | swim_brief           |    | shorts             |
|    | bikini               |    | womans_clothes     |
|    | purse                |    | handbag            |
|    | pouch                |    | shopping           |
|   | school_satchel       |   | thong_sandal       |
|  | mans_shoe<br>shoe    |  | athletic_shoe      |
|  | hiking_boot          |  | flat_shoe          |
|  | high_heel            |  | sandal             |
|  | ballet_shoes         |  | boot               |
|  | crown                |  | womans_hat         |
|  | tophat               |  | mortar_board       |
|  | billed_cap           |  | military_helmet    |
|  | rescue_worker_helmet |  | prayer_beads       |
|  | lipstick             |  | ring               |
|  | gem                  |                                                                                     |                    |

Sound

| ico                                                                               | shortcode                | ico                                                                               | shortcode               |
|-----------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------|-------------------------|
|  | <code>mute</code>        |  | <code>speaker</code>    |
|  | <code>sound</code>       |  | <code>loud_sound</code> |
|  | <code>loudspeaker</code> |  | <code>mega</code>       |
|  | <code>postal_horn</code> |  | <code>bell</code>       |
|  | <code>no_bell</code>     |                                                                                   |                         |

Music

| ico                                                                                | shortcode                  | ico                                                                               | shortcode                      |
|------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------|--------------------------------|
|   | <code>musical_score</code> |  | <code>musical_note</code>      |
|   | <code>notes</code>         |  | <code>studio_microphone</code> |
|   | <code>level_slider</code>  |  | <code>control_knobs</code>     |
|   | <code>microphone</code>    |  | <code>headphones</code>        |
|  | <code>radio</code>         |                                                                                   |                                |

Musical Instrument

| ico                                                                                 | shortcode              | ico                                                                                 | shortcode                     |
|-------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------|-------------------------------|
|  | <code>saxophone</code> |  | <code>accordion</code>        |
|  | <code>guitar</code>    |  | <code>musical_keyboard</code> |
|  | <code>trumpet</code>   |  | <code>violin</code>           |
|  | <code>banjo</code>     |  | <code>drum</code>             |
|  | <code>long_drum</code> |                                                                                     |                               |

Phone

| ico                                                                               | shortcode          | ico                                                                               | shortcode          |
|-----------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------|--------------------|
|  | iphone             |  | calling            |
|  | phone<br>telephone |  | telephone_receiver |
|  | pager              |  | fax                |

Computer

| ico                                                                                 | shortcode      | ico                                                                                | shortcode        |
|-------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|------------------|
|    | battery        |   | electric_plug    |
|    | computer       |   | desktop_computer |
|    | printer        |   | keyboard         |
|    | computer_mouse |   | trackball        |
|    | minidisc       |   | floppy_disk      |
|   | cd             |  | dvd              |
|  | abacus         |                                                                                    |                  |

Light & Video

| ico                                                                                 | shortcode      | ico                                                                                 | shortcode    |
|-------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|--------------|
|  | movie_camera   |  | film_strip   |
|  | film_projector |  | clapper      |
|  | tv             |  | camera       |
|  | camera_flash   |  | video_camera |
|  | vhs            |  | mag          |
|  | mag_right      |  | candle       |
|  | bulb           |  | flashlight   |

| ico                                                                               | shortcode                  | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------|-----------|
|  | izakaya_lantern<br>lantern |  | diya_lamp |

Book Paper

| ico                                                                                | shortcode                      | ico                                                                               | shortcode      |
|------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------|----------------|
|   | notebook_with_decorative_cover |  | closed_book    |
|   | book<br>open_book              |  | green_book     |
|   | blue_book                      |  | orange_book    |
|   | books                          |  | notebook       |
|   | ledger                         |  | page_with_curl |
|   | scroll                         |  | page_facing_up |
|   | newspaper                      |  | newspaper_roll |
|   | bookmark_tabs                  |  | bookmark       |
|  | label                          |                                                                                   |                |

Money

| ico                                                                                 | shortcode        | ico                                                                                 | shortcode   |
|-------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|-------------|
|  | moneybag         |  | coin        |
|  | yen              |  | dollar      |
|  | euro             |  | pound       |
|  | money_with_wings |  | credit_card |
|  | receipt          |  | chart       |

Mail

| ico                                                                               | shortcode            | ico                                                                               | shortcode           |
|-----------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------|---------------------|
|  | envelope             |  | e-mail<br>email     |
|  | incoming_envelope    |  | envelope_with_arrow |
|  | outbox_tray          |  | inbox_tray          |
|  | package              |  | mailbox             |
|  | mailbox_closed       |  | mailbox_with_mail   |
|  | mailbox_with_no_mail |  | postbox             |
|  | ballot_box           |                                                                                   |                     |

Writing

| ico                                                                                 | shortcode      | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|-----------|
|    | pencil2        |    | black_nib |
|   | fountain_pen   |   | pen       |
|  | paintbrush     |  | crayon    |
|  | memo<br>pencil |                                                                                     |           |

Office

| ico                                                                                 | shortcode        | ico                                                                                 | shortcode                |
|-------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|--------------------------|
|  | briefcase        |  | file_folder              |
|  | open_file_folder |  | card_index_dividers      |
|  | date             |  | calendar                 |
|  | spiral_notepad   |  | spiral_calendar          |
|  | card_index       |  | chart_with_upwards_trend |

| ico                                                                               | shortcode                  | ico                                                                               | shortcode      |
|-----------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------|----------------|
|  | chart_with_downwards_trend |  | bar_chart      |
|  | clipboard                  |  | pushpin        |
|  | round_pushpin              |  | paperclip      |
|  | paperclips                 |  | straight_ruler |
|  | triangular_ruler           |  | scissors       |
|  | card_file_box              |  | file_cabinet   |
|  | basket                     |                                                                                   |                |

Lock

| ico                                                                               | shortcode         | ico                                                                               | shortcode            |
|-----------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------|----------------------|
|  | lock              |  | unlock               |
|  | lock_with_ink_pen |  | closed_lock_with_key |
|  | key               |  | old_key              |

Tool

| ico                                                                                 | shortcode         | ico                                                                                 | shortcode       |
|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|-----------------|
|  | hammer            |  | axe             |
|  | pick              |  | hammer_and_pick |
|  | hammer_and_wrench |  | dagger          |
|  | crossed_swords    |  | bomb            |
|  | boomerang         |  | bow_and_arrow   |
|  | shield            |  | carpentry_saw   |
|  | wrench            |  | screwdriver     |
|  | nut_and_bolt      |  | gear            |
|  | clamp             |  | balance_scale   |

| ico                                                                               | shortcode    | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------|-----------|
|  | probing_cane |  | link      |
|  | chains       |  | hook      |
|  | toolbox      |  | magnet    |
|  | ladder       |                                                                                   |           |

Science

| ico                                                                               | shortcode  | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------|-----------|
|  | alembic    |  | test_tube |
|  | petri_dish |  | dna       |
|  | microscope |  | telescope |
|  | satellite  |                                                                                   |           |

Medical

| ico                                                                                 | shortcode   | ico                                                                                 | shortcode        |
|-------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|------------------|
|  | syringe     |  | drop_of_blood    |
|  | pill        |  | adhesive_bandage |
|  | stethoscope |                                                                                     |                  |

Household

| ico                                                                                 | shortcode | ico                                                                                 | shortcode      |
|-------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|----------------|
|  | door      |  | elevator       |
|  | mirror    |  | window         |
|  | bed       |  | couch_and_lamp |
|  | chair     |  | toilet         |
|  | plunger   |  | shower         |

| ico                                                                               | shortcode         | ico                                                                               | shortcode     |
|-----------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------|---------------|
|  | bathtub           |  | mouse_trap    |
|  | razor             |  | lotion_bottle |
|  | safety_pin        |  | broom         |
|  | basket            |  | roll_of_paper |
|  | bucket            |  | soap          |
|  | toothbrush        |  | sponge        |
|  | fire_extinguisher |  | shopping_cart |

Other Object

| ico                                                                                | shortcode    | ico                                                                               | shortcode   |
|------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------|-------------|
|   | smoking      |  | coffin      |
|   | headstone    |  | funeral_urn |
|   | nazar_amulet |  | moyai       |
|  | placard      |                                                                                   |             |

Symbols

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Transport Sign

| ico                                                                               | shortcode     | ico                                                                               | shortcode               |
|-----------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|-------------------------|
|  | atm           |  | put_litter_in_its_place |
|  | potable_water |  | wheelchair              |
|  | mens          |  | womens                  |
|  | restroom      |  | baby_symbol             |
|  | wc            |  | passport_control        |
|  | customs       |  | baggage_claim           |
|  | left_luggage  |                                                                                   |                         |

Warning

| ico                                                                                 | shortcode      | ico                                                                                 | shortcode         |
|-------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|-------------------|
|  | warning        |  | children_crossing |
|  | no_entry       |  | no_entry_sign     |
|  | no_bicycles    |  | no_smoking        |
|  | do_not_litter  |  | non-potable_water |
|  | no_pedestrians |  | no_mobile_phones  |
|  | underage       |  | radioactive       |
|  | biohazard      |                                                                                     |                   |

Arrow

| ico                                                                               | shortcode                 | ico                                                                               | shortcode               |
|-----------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------|-------------------------|
|  | arrow_up                  |  | arrow_upper_right       |
|  | arrow_right               |  | arrow_lower_right       |
|  | arrow_down                |  | arrow_lower_left        |
|  | arrow_left                |  | arrow_upper_left        |
|  | arrow_up_down             |  | left_right_arrow        |
|  | leftwards_arrow_with_hook |  | arrow_right_hook        |
|  | arrow_heading_up          |  | arrow_heading_down      |
|  | arrows_clockwise          |  | arrows_counterclockwise |
|  | back                      |  | end                     |
|  | on                        |  | soon                    |
|  | top                       |                                                                                   |                         |

Religion

| ico                                                                                 | shortcode         | ico                                                                                 | shortcode        |
|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|------------------|
|  | place_of_worship  |  | atom_symbol      |
|  | om                |  | star_of_david    |
|  | wheel_of_dharma   |  | yin_yang         |
|  | latin_cross       |  | orthodox_cross   |
|  | star_and_crescent |  | peace_symbol     |
|  | menorah           |  | six_pointed_star |

Zodiac

| ico                                                                                 | shortcode | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|-----------|
|  | aries     |  | taurus    |

| ico                                                                               | shortcode   | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------|-----------|
|  | gemini      |  | cancer    |
|  | leo         |  | virgo     |
|  | libra       |  | scorpius  |
|  | sagittarius |  | capricorn |
|  | aquarius    |  | pisces    |
|  | ophiuchus   |                                                                                   |           |

Av Symbol

| ico                                                                                 | shortcode                 | ico                                                                                 | shortcode             |
|-------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------|-----------------------|
|    | twisted_rightwards_arrows |    | repeat                |
|    | repeat_one                |    | arrow_forward         |
|    | fast_forward              |    | next_track_button     |
|    | play_or_pause_button      |    | arrow_backward        |
|   | rewind                    |   | previous_track_button |
|  | arrow_up_small            |  | arrow_double_up       |
|  | arrow_down_small          |  | arrow_double_down     |
|  | pause_button              |  | stop_button           |
|  | record_button             |  | eject_button          |
|  | cinema                    |  | low_brightness        |
|  | high_brightness           |  | signal_strength       |
|  | vibration_mode            |  | mobile_phone_off      |

Gender

| ico                                                                                 | shortcode   | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|-----------|
|  | female_sign |  | male_sign |

| ico                                                                               | shortcode          | ico | shortcode |
|-----------------------------------------------------------------------------------|--------------------|-----|-----------|
|  | transgender_symbol |     |           |

Math

| ico                                                                               | shortcode              | ico                                                                               | shortcode           |
|-----------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------|---------------------|
|  | heavy_multiplication_x |  | heavy_plus_sign     |
|  | heavy_minus_sign       |  | heavy_division_sign |
|  | infinity               |                                                                                   |                     |

Punctuation

| ico                                                                                 | shortcode        | ico                                                                               | shortcode                             |
|-------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------|---------------------------------------|
|    | bangbang         |  | interrobang                           |
|    | question         |  | grey_question                         |
|    | grey_exclamation |  | exclamation<br>heavy_exclamation_mark |
|  | wavy_dash        |                                                                                   |                                       |

Currency

| ico                                                                                 | shortcode         | ico                                                                                 | shortcode         |
|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|-------------------|
|  | currency_exchange |  | heavy_dollar_sign |

Other Symbol

| ico                                                                                 | shortcode      | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|-----------|
|  | medical_symbol |  | recycle   |
|  | fleur_de_lis   |  | trident   |

| ico                                                                               | shortcode                | ico                                                                               | shortcode                   |
|-----------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------|-----------------------------|
|  | name_badge               |  | beginner                    |
|  | o                        |  | white_check_mark            |
|  | ballot_box_with_check    |  | heavy_check_mark            |
|  | x                        |  | negative_squared_cross_mark |
|  | curly_loop               |  | loop                        |
|  | part_alternation_mark    |  | eight_spoked_asterisk       |
|  | eight_pointed_black_star |  | sparkle                     |
|  | copyright                |  | registered                  |
|  | TM                       |                                                                                   |                             |

Keycap

| ico                                                                                 | shortcode  | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------|-----------|
|    | hash       |    | asterisk  |
|   | zero       |   | one       |
|  | two        |  | three     |
|  | four       |  | five      |
|  | six        |  | seven     |
|  | eight      |  | nine      |
|  | keycap_ten |                                                                                     |           |

Alphanum

| ico                                                                                 | shortcode    | ico                                                                                 | shortcode |
|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|-----------|
|  | capital_abcd |  | abcd      |
|  | 1234         |  | symbols   |
|  | abc          |  | a         |

| ico                                                                                 | shortcode | ico                                                                                 | shortcode           |
|-------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|---------------------|
|    | ab        |    | b                   |
|    | cl        |    | cool                |
|    | free      |    | information_source  |
|    | id        |    | m                   |
|    | new       |    | ng                  |
|    | o2        |    | ok                  |
|    | parking   |    | sos                 |
|    | up        |    | vs                  |
|    | koko      |    | sa                  |
|    | u6708     |    | u6709               |
|    | u6307     |    | ideograph_advantage |
|    | u5272     |    | u7121               |
|    | u7981     |    | accept              |
|    | u7533     |    | u5408               |
|   | u7a7a     |   | congratulations     |
|  | secret    |  | u55b6               |
|  | u6e80     |                                                                                     |                     |

Geometric

| ico                                                                                 | shortcode         | ico                                                                                 | shortcode     |
|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|---------------|
|  | red_circle        |  | orange_circle |
|  | yellow_circle     |  | green_circle  |
|  | large_blue_circle |  | purple_circle |
|  | brown_circle      |  | black_circle  |
|  | white_circle      |  | red_square    |
|  | orange_square     |  | yellow_square |

| ico                                                                               | shortcode                       | ico                                                                               | shortcode              |
|-----------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------|------------------------|
|  | green_square                    |  | blue_square            |
|  | purple_square                   |  | brown_square           |
|  | black_large_square              |  | white_large_square     |
|  | black_medium_square             |  | white_medium_square    |
|  | black_medium_small_square       |  | white_medium_small_sq  |
|  | black_small_square              |  | white_small_square     |
|  | large_orange_diamond            |  | large_blue_diamond     |
|  | small_orange_diamond            |  | small_blue_diamond     |
|  | small_red_triangle              |  | small_red_triangle_dov |
|  | diamond_shape_with_a_dot_inside |  | radio_button           |
|  | white_square_button             |  | black_square_button    |

Flags

- [Flag](#)
- [Country Flag](#)
- [Subdivision Flag](#)

Flag

| ico                                                                                 | shortcode        | ico                                                                                 | shortcode               |
|-------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|-------------------------|
|  | checkered_flag   |  | triangular_flag_on_post |
|  | crossed_flags    |  | black_flag              |
|  | white_flag       |  | rainbow_flag            |
|  | transgender_flag |  | pirate_flag             |

Country Flag

| ico                                                                                 | shortcode             | ico                                                                                  | shortcode          |
|-------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------|--------------------|
|    | ascension_island      |    | andorra            |
|    | united_arab_emirates  |    | afghanistan        |
|    | antigua_barbuda       |    | anguilla           |
|    | albania               |    | armenia            |
|    | angola                |    | antarctica         |
|    | argentina             |    | american_samoa     |
|    | austria               |    | australia          |
|    | aruba                 |    | aland_islands      |
|    | azerbaijan            |    | bosnia_herzegovina |
|    | barbados              |    | bangladesh         |
|    | belgium               |    | burkina_faso       |
|    | bulgaria              |    | bahrain            |
|    | burundi               |    | benin              |
|  | st_barthelemy         |  | bermuda            |
|  | brunei                |  | bolivia            |
|  | caribbean_netherlands |  | brazil             |
|  | bahamas               |  | bhutan             |
|  | bouvet_island         |  | botswana           |
|  | belarus               |  | belize             |
|  | canada                |  | cocos_islands      |
|  | congo_kinshasa        |  | central_african_r  |
|  | congo_brazzaville     |  | switzerland        |
|  | cote_divoire          |  | cook_islands       |
|  | chile                 |  | cameroon           |
|  | cn                    |  | colombia           |

| ico                                                                                 | shortcode                            | ico                                                                                  | shortcode            |
|-------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------|----------------------|
|    | clipperton_island                    |    | costa_rica           |
|    | cuba                                 |    | cape_verde           |
|    | curacao                              |    | christmas_island     |
|    | cyprus                               |    | czech_republic       |
|    | de                                   |    | diego_garcia         |
|    | djibouti                             |    | denmark              |
|    | dominica                             |    | dominican_republic   |
|    | algeria                              |    | ceuta_melilla        |
|    | ecuador                              |    | estonia              |
|    | egypt                                |    | western_sahara       |
|    | eritrea                              |    | es                   |
|    | ethiopia                             |    | eu<br>european_union |
|    | finland                              |    | fiji                 |
|    | falkland_islands                     |    | micronesia           |
|   | faroe_islands                        |   | fr                   |
|  | gabon                                |  | gb<br>uk             |
|  | grenada                              |  | georgia              |
|  | french_guiana                        |  | guernsey             |
|  | ghana                                |  | gibraltar            |
|  | greenland                            |  | gambia               |
|  | guinea                               |  | guadeloupe           |
|  | equatorial_guinea                    |  | greece               |
|  | south_georgia_south_sandwich_islands |  | guatemala            |
|  | guam                                 |  | guinea_bissau        |
|  | guyana                               |  | hong_kong            |

| ico                                                                                 | shortcode              | ico                                                                                  | shortcode         |
|-------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------|-------------------|
|    | heard_mcdonald_islands |    | honduras          |
|    | croatia                |    | haiti             |
|    | hungary                |    | canary_islands    |
|    | indonesia              |    | ireland           |
|    | israel                 |    | isle_of_man       |
|    | india                  |    | british_indian_oc |
|    | iraq                   |    | iran              |
|    | iceland                |    | it                |
|    | jersey                 |    | jamaica           |
|    | jordan                 |    | jp                |
|    | kenya                  |    | kyrgyzstan        |
|    | cambodia               |    | kiribati          |
|    | comoros                |    | st_kitts_nevis    |
|    | north_korea            |    | kr                |
|   | kuwait                 |   | cayman_islands    |
|  | kazakhstan             |  | laos              |
|  | lebanon                |  | st_lucia          |
|  | liechtenstein          |  | sri_lanka         |
|  | liberia                |  | lesotho           |
|  | lithuania              |  | luxembourg        |
|  | latvia                 |  | libya             |
|  | morocco                |  | monaco            |
|  | moldova                |  | montenegro        |
|  | st_martin              |  | madagascar        |
|  | marshall_islands       |  | macedonia         |
|  | mali                   |  | myanmar           |
|  | mongolia               |  | macau             |

| ico                                                                                 | shortcode                | ico                                                                                  | shortcode          |
|-------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------|--------------------|
|    | northern_mariana_islands |    | martinique         |
|    | mauritania               |    | montserrat         |
|    | malta                    |    | mauritius          |
|    | maldives                 |    | malawi             |
|    | mexico                   |    | malaysia           |
|    | mozambique               |    | namibia            |
|    | new_caledonia            |    | niger              |
|    | norfolk_island           |    | nigeria            |
|    | nicaragua                |    | netherlands        |
|    | norway                   |    | nepal              |
|    | nauru                    |    | niue               |
|    | new_zealand              |    | oman               |
|    | panama                   |    | peru               |
|    | french_polynesia         |    | papua_new_guinea   |
|   | philippines              |   | pakistan           |
|  | poland                   |  | st_pierre_miquelon |
|  | pitcairn_islands         |  | puerto_rico        |
|  | palestinian_territories  |  | portugal           |
|  | palau                    |  | paraguay           |
|  | qatar                    |  | reunion            |
|  | romania                  |  | serbia             |
|  | ru                       |  | rwanda             |
|  | saudi_arabia             |  | solomon_islands    |
|  | seychelles               |  | sudan              |
|  | sweden                   |  | singapore          |
|  | st_helena                |  | slovenia           |
|  | svalbard_jan_mayen       |  | slovakia           |

| ico                                                                                 | shortcode                   | ico                                                                                  | shortcode         |
|-------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------|-------------------|
|    | sierra_leone                |    | san_marino        |
|    | senegal                     |    | somalia           |
|    | suriname                    |    | south_sudan       |
|    | sao_tome_principe           |    | el_salvador       |
|    | sint_maarten                |    | syria             |
|    | swaziland                   |    | tristan_da_cunha  |
|    | turks_caicos_islands        |    | chad              |
|    | french_southern_territories |    | togo              |
|    | thailand                    |    | tajikistan        |
|    | tokelau                     |    | timor_leste       |
|    | turkmenistan                |    | tunisia           |
|    | tonga                       |    | tr                |
|    | trinidad_tobago             |    | tuvalu            |
|   | taiwan                      |   | tanzania          |
|  | ukraine                     |  | uganda            |
|  | us_outlying_islands         |  | united_nations    |
|  | us                          |  | uruguay           |
|  | uzbekistan                  |  | vatican_city      |
|  | st_vincent_grenadines       |  | venezuela         |
|  | british_virgin_islands      |  | us_virgin_islands |
|  | vietnam                     |  | vanuatu           |
|  | wallis_futuna               |  | samoa             |
|  | kosovo                      |  | yemen             |
|  | mayotte                     |  | south_africa      |
|  | zambia                      |  | zimbabwe          |

Subdivision Flag

| ico                                                                               | shortcode | ico                                                                               | shortcode |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------|-----------|
|  | england   |  | scotland  |
|  | wales     |                                                                                   |           |

# typst.

## typst demo.

▼ demo0\_0.typ

t

```
// demo0
#import "@preview/cetz:0.3.4": canvas, draw
#import draw: line, content, rect, circle, bezier, group
#set page(width: auto, height: auto, margin: 15pt)
#canvas({
  // Core constants and reusable values
  let plot_width = 24
  let timeline_width = 16
  let time_tick_spacing = 1.0
  let step_width = time_tick_spacing
  let step_gap = 0.1
  let box_width_factor = 0.9
  let rect_height = 0.3
  let border_radius = 0.05
  let gap = 0.15
  let strategy_y_positions = (11, 6.5, 2.0) // Further compressed y-
positions
  // Colors - more pastel version for better text readability
  // Grays
  let dark_gray = rgb("#5D6B7A")
  let cpu_color = rgb("#8899AA")
  let gpu_color = rgb("#6A7A8A")
  let section_bg = rgb("#F7FAFC")
  // Define a consistent CPU light gray color for all strategies
  let cpu_light_gray = cpu_color.lighten(70%)
  // Blues with pastel tones
  let light_blue = rgb("#BBDEFB")
  let mid_blue = rgb("#90CAF9")
  let dark_blue = rgb("#64B5F6")
  // Greens with pastel tones
  let light_green = rgb("#C8E6C9")
  let mid_green = rgb("#A5D6A7")
  let dark_green = rgb("#81C784")
  // Oranges with pastel tones
  let light_orange = rgb("#FFE0B2")
  let mid_orange = rgb("#FFCC80")
  let dark_orange = rgb("#FFB74D")
  // Reds with pastel tones
  let light_red = rgb("#FFCDD2")
  let mid_red = rgb("#EF9A9A")
  let dark_red = rgb("#E57373")
  // Helper functions
  let draw_sim_block(x_pos, y_pos, width, color, label, task_label: "",
```

```

opacity: 100%) = {
  rect(
    (x_pos, y_pos - rect_height / 2),
    (x_pos + width, y_pos + rect_height / 2),
    fill: color.transparentize(100% - opacity),
    stroke: color.darken(10%),
    radius: border_radius,
    name: "block-" + label,
  )
  if task_label != "" {
    content((x_pos + width / 2, y_pos), text(size: 8pt)[#task_label],
name: "label-" + label)
  }
}
let draw_structure_rect(x_pos, y_pos, width, color, label, struct_name)
= {
  rect(
    (x_pos, y_pos),
    (x_pos + 0.95 * width, y_pos + 0.1),
    fill: color.transparentize(20%),
    stroke: none,
    name: "block-" + struct_name,
  )
  content((x_pos + width / 2, y_pos + 0.15), text(size: 8pt)[#label],
name: "label-" + struct_name, anchor: "south")
}
let draw_util_bar(x_pos, y_pos, percentage, label, is_bad: false) = {
  // Bar showing CPU or GPU utilization - consolidated function
  let width = 2.8
  let height = 0.4
  // Background bar
  rect(
    (x_pos, y_pos - height / 2),
    (x_pos + width, y_pos + height / 2),
    fill: rgb("#E2E8F0"),
    stroke: 0.5pt,
    radius: 0.1,
    name: "bar-bg-" + label,
  )
  // Filled portion with appropriate color
  let fill_width = width * percentage / 100
  // Determine color based on percentage and is_bad flag
  let fill_color = if is_bad {
    light_red.darken(20%) // Use darker red for explicitly marked "bad"
  }
  utilization
}

```

```

    } else if percentage < 30 {
        light_red
    } else if percentage < 70 {
        light_orange
    } else {
        light_green
    }
    rect(
        (x_pos, y_pos - height / 2),
        (x_pos + fill_width, y_pos + height / 2),
        fill: fill_color,
        stroke: none,
        radius: (north-west: 0.1, south-west: 0.1),
        name: "bar-fill-" + label,
    )
    // Label
    content(
        (rel: (0.03, 0), to: "bar-bg-" + label),
        text(size: 8pt, weight: "bold")[#label #percentage%],
        name: "bar-label-" + label,
        anchor: "center",
    )
}
// Title, subtitle and legend
let title_y = 15.0
content(
    (plot_width / 2, title_y),
    text(weight: "bold", size: 16pt)[GPU Batching Strategies for Atomistic
Simulations],
    name: "main-title",
)
content(
    (rel: (0, -1), to: "main-title"),
    text(size: 12pt)[Comparison of Unbatched vs. BinningAutoBatcher vs.
InFlightAutoBatcher],
    name: "subtitle",
)
// Add section backgrounds and scenario setups
let strategies = (
    (strategy_y_positions.at(0), "Unbatched\nSimulations", (80, 5)),
    (strategy_y_positions.at(1), "Binning\nAutoBatcher", (40, 60)),
    (strategy_y_positions.at(2), "InFlight\nAutoBatcher", (60, 90)),
)
for (idx, (y_pos, label, (cpu_util, gpu_util))) in
strategies.enumerate() {

```

```

// Background
rect(
  (0.5, y_pos - 2.0),
  (plot_width - 0.5, y_pos + 2.0),
  fill: section_bg,
  stroke: none,
  radius: border_radius * 3,
  name: "section-bg-" + str(idx),
)
// Scenario label
content((2, y_pos + 1.0), text(weight: "bold", size: 12pt)[#label],
name: "scenario-label-" + str(idx))
// Utilization meters
draw_util_bar(.8, y_pos - 0.1, cpu_util, "CPU Utilization")
// GPU utilization - mark as bad for unbatched simulations (idx == 0)
draw_util_bar(.8, y_pos - 1, gpu_util, "GPU Utilization", is_bad: idx
== 0)
// CPU/GPU labels
content(
  (4.6, y_pos + 1.3),
  text(fill: cpu_color, size: 10pt, weight: "bold")[CPU],
  anchor: "east",
  name: "cpu-label-" + str(idx),
)
content(
  (4.6, y_pos - 0.5),
  text(fill: gpu_color, size: 10pt, weight: "bold")[GPU],
  anchor: "east",
  name: "gpu-label-" + str(idx),
)
// Timeline axis with ticks
line(
  (4.8, y_pos - 1.5),
  (4.8 + timeline_width, y_pos - 1.5),
  stroke: 0.8pt,
  mark: (end: "stealth", fill: black, scale: 0.5),
  name: "timeline-" + str(idx),
)
for t in range(1, 17) {
  let x_pos = 4 + t * time_tick_spacing
  if x_pos <= 4 + timeline_width {
    line(
      (x_pos, y_pos - 1.5 - 0.05),
      (x_pos, y_pos - 1.5 + 0.05),
      stroke: 0.8pt,

```

```

        name: "tick-" + str(idx) + "-" + str(t),
    )
    content(
        (rel: (0, -0.2), to: "tick-" + str(idx) + "-" + str(t)),
        text(size: 8pt)[t=#t],
        anchor: "north",
        name: "tick-label-" + str(idx) + "-" + str(t),
    )
}
}
// CPU/GPU separator
let separator_y = if idx == 0 { y_pos + 0.6 } else { y_pos + 0.95 }
line(
    (4, separator_y),
    (4 + timeline_width, separator_y),
    stroke: (dash: "dotted", paint: dark_gray.lighten(30%)),
    name: "separator-" + str(idx),
)
}
// 1. Unbatched Simulations
let unbatched_cpu_y = strategy_y_positions.at(0) + 1.4
let unbatched_gpu_y = strategy_y_positions.at(0) - 0.5
// Structure rectangles with consistent pattern
let unbatched_structures = (
    (5.0, 2.0, mid_blue, "Structure 1"),
    (7.0, 2.3, mid_green, "Structure 2"),
    (9.3, 1.8, mid_orange, "Structure 3"),
    (11.1, 2.1, mid_red, "Structure 4"),
    (13.2, 1.8, dark_green, "Structure 5"),
    (15.0, 2.0, dark_green.darken(10%), "Structure 6"),
    (17.0, 1.6, dark_green.darken(15%), "Structure 7"),
    (18.6, 1.5, dark_green.darken(20%), "Structure 8"),
)
for (idx, (x_pos, width, color, label)) in
unbatched_structures.enumerate() {
    draw_structure_rect(x_pos, unbatched_cpu_y, width, color, label,
"unbatched-" + str(idx + 1) + "-cpu")
}
    content((21.8, unbatched_cpu_y - 1.9), text(size: 10pt, weight: "bold")
[... continues], anchor: "center")
// CPU operation blocks in a loop
for x in range(22) {
    let x_offset = 5.25 + x * 0.7
    if x_offset < 20 {
        draw_sim_block(

```

```

        x_offset,
        unbatched_cpu_y - 0.4,
        0.4,
        cpu_light_gray,
        "unbatched-cpu-op-" + str(x),
        task_label: "",
        opacity: 90%,
    )
}
}
// GPU blocks with structure IDs and positions
let gpu_blocks = (
    (1, dark_blue, (5.5, 6.5)),
    (2, dark_green, (7.5, 8.5)),
    (3, dark_orange, (9.8, 10.6)),
    (4, dark_red, (11.5, 12.5)),
    (5, dark_green, (13.8, 14.6, 15.4)),
    (6, dark_green.darken(10%), (16.2, 16.8, 17.4, 18.0, 18.6)),
    (7, dark_green.darken(15%), (19.2, 19.8)),
)
for (struct_num, color, positions) in gpu_blocks {
    for (idx, pos) in positions.enumerate() {
        draw_sim_block(
            pos,
            unbatched_gpu_y,
            0.3 * box_width_factor,
            color,
            "unbatched-" + str(struct_num) + "-gpu-" + str(idx + 1),
            task_label: "S" + str(struct_num),
        )
    }
}
// 2. BinningAutoBatcher
let binning_cpu_y = strategy_y_positions.at(1) + 1.3
let binning_gpu_y = strategy_y_positions.at(1) - 0.85
// CPU processing blocks - simplified to only show prep batch operations
let cpu_blocks = (
    (5.0, "Prep batch"),
    (10.4, "Prep batch"),
)
for (idx, (x_pos, label)) in cpu_blocks.enumerate() {
    draw_sim_block(x_pos, binning_cpu_y, 1.3, cpu_light_gray, "binning-op-"
+ str(idx), task_label: label)
}
// Batch visualization setup

```

```

let s_colors = (
    dark_blue,
    dark_green,
    dark_orange,
    dark_red,
    dark_blue.darken(10%),
)
let s_labels = ("S1", "S2", "S3", "S4", "S5")
// Bin parameters and active patterns
let bins = (
    (
        // Bin 1: 6 steps with completion patterns
        0,
        false,
        (
            (1, 1, 1, 1, 1), // Step 0: all active
            (1, 1, 1, 1, 1), // Step 1: all active
            (0, 1, 1, 1, 1), // Step 2: structure 1 completes earlier
            (0, 0, 1, 1, 1), // Step 3: structures 1,2 complete
            (0, 0, 0, 1, 0), // Step 4: structures 1,2,3,5 complete
            (0, 0, 0, 1, 0), // Step 5: only 4 remains
        ),
    ),
    (
        // Bin 2: 6 steps with different completion pattern
        6,
        true,
        (
            (1, 1, 1, 1, 1), // Step 0: all active
            (1, 1, 1, 1, 1), // Step 1: structure 3 completes early
            (0, 1, 0, 1, 1), // Step 2: structures 1,3 complete
            (0, 1, 0, 0, 1), // Step 3: only 4,5 remain
            (0, 1, 0, 0, 0), // Step 4: only 5 remains
            (0, 1, 0, 0, 0), // Step 5: all complete
        ),
    ),
)
// Draw both bins with common function
for (start_step, is_second_bin, patterns) in bins {
    for step in range(patterns.len()) {
        let x_pos = 5.0 + (start_step + step) * (step_width - step_gap)
        let box_width = (step_width - step_gap) * box_width_factor
        let box_x_start = x_pos + ((step_width - step_gap) - box_width) / 2
        // Draw each structure slot
        for i in range(5) {

```

```

    let block_y = binning_gpu_y - 0.3 + i * (rect_height + gap)
    let binning_block_name = "binning-block-" + str(start_step) + "-"
+ str(step) + "-" + str(i)
    if patterns.at(step).at(i) == 1 {
        let color = s_colors.at(i)
        let label = if is_second_bin { "S" + str(i + 6) } else {
s_labels.at(i) }
        rect(
            (box_x_start, block_y - rect_height / 2),
            (box_x_start + box_width, block_y + rect_height / 2),
            fill: color,
            stroke: color.darken(20%),
            radius: border_radius,
            name: binning_block_name,
        )
        content(
            (box_x_start + box_width / 2, block_y),
            text(size: 7pt)[#label],
            anchor: "center",
            name: binning_block_name + "-label",
        )
    } else {
        // Empty placeholder
        rect(
            (box_x_start, block_y - rect_height / 2),
            (box_x_start + box_width, block_y + rect_height / 2),
            fill: light_red.transparentize(90%),
            stroke: (dash: "dotted", paint:
light_red.transparentize(30%)),
            radius: border_radius,
            name: binning_block_name + "-empty",
        )
    }
}
// Add completion label for the last step
if step == patterns.len() - 1 {
    content(
        (box_x_start - 0.25, binning_gpu_y - 0.45),
        text(size: 8pt, fill: dark_gray, style: "italic")[Batch #if
is_second_bin [2] else [1] Complete],
        anchor: "center",
        name: "batch-" + str(if is_second_bin { 2 } else { 1 }) + "-
complete-label",
    )
}
}

```

```

    }
  }
  for (x_pos, percentage) in ((5.6, 100), (8.0, 60), (9.8, 30), (11, 100),
(13.5, 60), (15.5, 20)) {
    content(
      (x_pos, binning_gpu_y + 2.7),
      text(size: 8pt, fill: dark_gray)[#percentage% GPU],
      anchor: "center",
    )
  }
  // Explanatory arrows and labels - removing the isolated red arrow
  content(
    (10.5, binning_gpu_y - 1.5),
    text(size: 7pt, fill: dark_red, style: "italic")["Must wait for batch
to complete, resulting in underutilized GPU"],
    anchor: "center",
    name: "must-wait",
  )
  let batch_end_x = 5.0 + 6 * (step_width - step_gap)
  let timeline_y = strategy_y_positions.at(1) - 1.5
  line(
    "must-wait.north",
    (batch_end_x, timeline_y),
    stroke: (dash: "dotted", paint: dark_red),
    mark: (end: "stealth", fill: dark_red, scale: 0.4, offset: 0.05),
    name: "must-wait-arrow",
  )
  // 3. In-flightAutoBatcher
  let inflight_cpu_y = strategy_y_positions.at(2) + 1.3
  let inflight_gpu_y = strategy_y_positions.at(2) - 0.9
  // Structure colors and labels
  let all_colors = (
    dark_blue,
    dark_green,
    dark_orange,
    dark_red,
    dark_blue.darken(10%),
    dark_green.darken(10%),
    dark_orange.darken(10%),
    dark_red.darken(10%),
    dark_blue.darken(20%),
    dark_green.darken(20%),
  )
  let all_labels = ("S1", "S2", "S3", "S4", "S5", "S6", "S7", "S8", "S9",
"S10")

```

```

// Structure placement by time step - fix structure continuity
let structures_by_step = (
  (0, 1, 2, 3, 4), // t=1: S1-S5
  (0, 1, 2, 3, 4), // t=2: S1-S5 (S1 continues for longer)
  (0, 6, 2, 3, 4), // t=3: S1 continues, S6 replaces S2, S3-S5 continue
  (5, 6, 7, 3, 4), // t=4: S5, S6-S8, S4 continue (S1 completed, S5
swapped in)
  (5, 6, 7, 8, 4), // t=5: S5-S9, S4 (S3 completed, S8 swapped in)
  (5, 6, 7, 8, 9), // t=6: S5-S10 (S4 completed, S9 swapped in)
  (5, 6, 7, 8, 9), // t=7: still fully filled
  (5, 6, 7, -1, -1), // t=8: final step, partially filled (about 50%) -
using S6 instead of isolated S4
)
// Place CPU operations equidistant, one for each time step
// Create the same number of prep boxes as structure steps (8 steps)
// Draw CPU operations - one per time step
for step in range(structures_by_step.len()) {
  let x_pos = 5.0 + step * (step_width - step_gap)
  let box_width = (step_width - step_gap) * box_width_factor * 0.7 //
Make boxes 10% wider (70% instead of 60%)
  // Left align by using 10% of remaining space as left margin instead
of centering
  let box_x_start = x_pos + ((step_width - step_gap) - box_width) * 0.1
  // Use the same generic gray for all prep blocks
  draw_sim_block(
    box_x_start,
    inflight_cpu_y,
    box_width,
    cpu_light_gray,
    "inflight-prep-" + str(step),
    task_label: "Prep",
  )
}
// Draw all structures by step
for step in range(structures_by_step.len()) {
  let x_pos = 5.0 + step * (step_width - step_gap)
  let box_width = (step_width - step_gap) * box_width_factor
  let box_x_start = x_pos + ((step_width - step_gap) - box_width) / 2
  // Create a batch name for this step
  let batch_name = "inflight-batch-" + str(step)
  // Draw each structure position
  for pos in range(5) {
    let struct_idx = structures_by_step.at(step).at(pos)
    let block_y = inflight_gpu_y - 0.3 + pos * (rect_height + gap)
    let block_name = "inflight-block-" + str(step) + "-" + str(pos)

```

```

    if struct_idx != -1 {
        let color = all_colors.at(struct_idx)
        let label = all_labels.at(struct_idx)
        // Structure box
        rect(
            (box_x_start, block_y - rect_height / 2),
            (box_x_start + box_width, block_y + rect_height / 2),
            fill: color,
            stroke: color.darken(20%),
            radius: border_radius,
            name: block_name,
        )
        content(
            (box_x_start + box_width / 2, block_y),
            text(size: 7pt)[#label],
            name: block_name + "-label",
            anchor: "center",
        )
        // Swap-in indicator
        if step > 0 {
            let prev_structures = structures_by_step.at(step - 1)
            let prev_block_name = "inflight-block-" + str(step - 1) + "-" +
str(pos)
            if pos < prev_structures.len() and prev_structures.at(pos) !=
struct_idx {
                // Simple diagonal line for swap-in indicator
                line(
                    (rel: (0, 0), to: block_name + ".south-west"), // End at the
block edge
                    (rel: (-0.25, -0.3), to: block_name + ".south-west"), //
Start from outside
                    stroke: (dash: "dotted", paint: dark_green),
                    mark: (start: "stealth", fill: dark_green, scale: 0.3),
                    name: block_name + "-swap-indicator",
                )
            }
        }
    } else {
        // Empty placeholders with dotted lines to indicate unused GPU
cycles
        rect(
            (box_x_start, block_y - rect_height / 2),
            (box_x_start + box_width, block_y + rect_height / 2),
            fill: light_red.transparentize(90%),
            stroke: (dash: "dotted", paint: light_red.transparentize(30%)),

```

```

        radius: border_radius,
        name: block_name + "-empty",
    )
}
}
// Mark the last batch specially
if step == structures_by_step.len() - 1 {
    // Store a reference to the last batch for later reference
    let final_block_name = "inflight-block-" + str(step) + "-0"
    // Instead of using absolute positioning, use relative positioning
    // based on the named elements we just created
    content(
        (rel: (box_width + 1.7, 0.2), to: final_block_name),
        text(size: 8pt, fill: dark_gray, style: "italic")[Final batch
partially filled],
        anchor: "center",
        name: "final-batch-label",
    )
}
}
// First label with frame
content(
    (plot_width / 2, -0.8),
    text(
        size: 9pt,
        weight: "bold",
        fill: dark_green,
    ) [In-flight batching achieves highest GPU utilization and maximizes
predictions per unit time],
    frame: "rect",
    fill: light_green.transparentize(70%),
    stroke: dark_green,
    padding: 3pt,
    radius: border_radius,
)
// Summary notes
content(
    (plot_width / 2, -3),
    box(width: 50em)[
        *Unbatched:* Each simulation runs sequentially with most
calculations on CPU and minimal GPU utilization\
        *Binning:* Fixed-size batches improve GPU utilization but can't
adapt to varying simulation completion times\
        *In-flight:* Dynamic reallocation eliminates GPU idle time by
immediately adding new structures when others complete. Color changes

```

```
indicate in-flight structure replacement.
],
frame: "rect",
fill: rgb("#F7FAFC"),
stroke: 0.5pt,
padding: (10pt, 10pt, 0pt),
radius: border_radius,
)
})
```

GPU Batching Strategies for Atomistic Simulations

Comparison of Unbatched vs. BinningAutoBatcher vs. InFlightAutoBatcher



**Unbatched:** Each simulation runs sequentially with most calculations on CPU and minimal GPU utilization  
**Binning:** Fixed-size batches improve GPU utilization but can't adapt to varying simulation completion times  
**In-flight:** Dynamic reallocation eliminates GPU idle time by immediately adding new structures when others complete. Color changes indicate in-flight structure replacement.